European Commission

Economic
and Financial Affairs

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EUROPEAN
ECONOMY

DISCUSSION PAPER 234 | NOVEMBER 2025

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Authorised for publication by Géraldine Mahieu, Director for Investment, Growth and Structural Reforms.

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Luxembourg: Publications Office of the European Union, 2025

PDF ISBN 978-92-68-31838-6 ISSN 2443-8022 doi:10.2765/6056667 KC-01-25-065-EN-N

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Abstract

US trade policy has taken a sharp protectionist turn under the second Trump administration, with the aim of boosting domestic manufacturing production, reducing the US trade deficit, and raising budgetary revenues "paid by foreigners." This paper assesses the macroeconomic consequences of recent US tariff announcements, based on quantitative simulations by the European Commission's multi-region New Keynesian DSGE model, QUEST. Results indicate that, rather than aiding domestic production, tariff hikes weaken the US economy. While tariffs shift demand from imports towards US-produced goods, they also act as an adverse supply shock. In addition, the equilibrium terms-of-trade appreciation crowds out exports, and monetary tightening in response to inflationary pressures hurts domestic demand as well. Although tariff revenues generate additional fiscal space for the US government, only around a quarter of the burden falls on foreigners in the form of a US terms-of-trade gain. Finally, tariff hikes reduce US trade deficits only temporarily. The effects on EU GDP are moderately negative, driven mainly by weaker exports to the US. At the same time European exporters gain market share in third countries at the expense of less competitive American firms. US tariffs on other countries lead to trade diversion, slightly deepening the short-term economic losses in Europe, but reversing later on. A general tit-for-tat retaliation would deepen the negative impacts in both the US and the EU. Beyond the direct effects of tariffs, rising uncertainty and a loss of investor confidence in the US economy further aggravates the adverse economic consequences by tightening financing conditions. Sensitivity analyses highlight the role of tariff persistence, the currency of trade invoicing and the monetary policy response.

JEL Classification: E62, F13, F41, F42, F47.

Keywords: tariffs, trade war, international macroeconomics, DSGE modelling.

Acknowledgements: I thank Philipp Pfeiffer, Björn Döhring and János Varga for their useful comments, as well as for Rafal Raciborski and Adrian Ifrim for their insightful reviews, and to colleagues from DG TRADE, the ECB and IMF modelling teams for fruitful discussions and exchanges.

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1 INTRODUCTION

US trade policy has taken a sharp protectionist turn under the second Trump administration. The effective tariff rate levied on US imports has risen by around 15 percentage points (The Budget Lab Yale, 2025*a*), the steepest increase since the 1930s. The macroeconomic effects of such a radical policy shift are likely to be significant not only domestically but, given the considerable size of the US economy, could also markedly reshape global trade flows.

Tariffs are traditionally regarded as harmful since, by distorting patterns of comparative advantage, they impose efficiency costs and reduce the "gains from trade". For large enough economies, these effects might be counterbalanced by purchasing power gains stemming from a stronger terms-of-trade, as tariffs shift demand from imports towards domestic goods. However, tariffs can generate additional short-term costs by disrupting global supply chains and triggering recessionary pressures, which weigh on economic activity and real incomes. The effect of tariffs on welfare are influenced by the balance of these forces.

That said, the US government is not focused on the abstract notion of welfare, but hopes to achieve a range of concrete policy goals: boosting domestic manufacturing production, reducing the US trade deficit, and raising budgetary revenues which it claims would be "paid by foreigners." These expectations rest on the idea that, by raising the relative price of foreign goods to US consumers, tariffs can improve external competitiveness much like a currency devaluation. Whether a sharp hike in tariffs can accomplish these objectives is unclear, since several offsetting mechanisms may limit or even reverse their intended effects. Properly assessing the macroeconomic consequences of tariffs therefore requires a structured approach to evaluate the interaction between the various transmission channels.

This paper contributes to this effort by quantitatively estimating the macroeconomic effects of US tariff hikes (as announced up to 2 Apr 2025), using QUEST, a rich general equilibrium model maintained by the European Commission. Static models of international trade typically focus on the long-run effects of tariffs, such as the costs stemming from a less efficient exploitation of comparative advantages. In contrast, dynamic New Keynesian models like ours allow to analyse the short-run cyclical implications of tariff-induced demand shifts in the presence of nominal rigidities. By incorporating trade in intermediate inputs, the model can capture linkages through cross-border supply chains that tend to amplify the effects of trade barriers. The multi-region setup also enables us to assess spillovers to trading partners like the EU, and trace trade diversion effects through third countries.

The results indicate that, rather than aiding domestic production, a unilateral increase in import tariffs would weaken the US economy and lower real GDP. Although higher tariffs do shift US demand from imports towards domestically produced goods, they also act as an adverse supply shock. On the one hand, they raise production costs by taxing imported inputs; on the other hand, they hurt labour supply by eroding the purchasing power of wages. As excess demand for domestic products puts a strain on the economy's resources, the US terms-of-trade (ToT) appreciates. In addition to limiting the tariff-induced shift away from imports, this also makes US exporters less competitive abroad, thereby crowding out exports. Furthermore, monetary tightening in response to inflationary pressures raises real interest rates and constrains domestic demand as well.

At the same time, the stronger US terms-of-trade lifts the purchasing power of incomes earned from domestic production (real gross disposable income, GDI), creating a terms-of-trade gain for the American economy as a whole. As a result, tariffs raise the after-tax import costs of the US private sector by less than they raise the tax revenue collected by the government. This difference represents a real transfer from abroad to the US economy, i.e. the US ToT gain is the way in which the rest of the world is made to pay the tariffs – however, it covers only around a quarter of the increase in tariff revenues. Therefore, while tariffs can generate additional fiscal space for the US government, most of their burden falls on domestic firms and households, who face lower real incomes through higher consumer prices or reduced profit margins – at least until the extra tariff revenues are handed back to them through tax cuts or transfers. Moreover, despite the ToT gain, falling production levels amid a recession (i.e. lower real GDP) eventually reduce real GDI as well.

As for the external balance, tariff hikes reduce US trade deficits only temporarily. While they lower demand for imports, the resulting terms-of-trade appreciation crowds out exports as well. Therefore, despite gross trade flows declining significantly, the effect on the net balance is smaller. This is a (partial) manifestation of the Lerner symmetry, which points out that a tax on imports behaves similarly to a tax on exports. To the extent net export volumes do rise, it is less the direct consequence of tariff-induced expenditure switching, but instead follows more from weaker domestic demand which mitigates import leakage. On top of the volume component, the ToT gain contributes positively to the trade balance, as it reduces the import bill and boosts export revenues through changes in relative prices. However, in the medium-run these effects fade, as tariffs have no lasting influence on saving or investment rates, the fundamental drivers behind the US external balance.

A unilateral increase in US tariffs would lower EU real GDP moderately, driven mainly by weaker exports to the US. Although European exporters suffer from the shrinking American market *size*, they also gain market *share* in third countries at the expense of American exporters who become less competitive due to a stronger dollar and more costly imported inputs. The flipside of this increased European competitiveness is a terms-of-trade loss, eroding the purchasing power of EU incomes beyond the fall in real GDP. The fact that other countries are also targeted by US tariffs initially weighs on the EU economy due to trade diversion, as third-country products that were shut out of US markets become relatively cheaper in search of new consumers. However, this also prevents EU exporters from losing market share inside the US, with the latter beneficial effect eventually prevailing.

Stylised scenarios, featuring tit-for-tat retaliations by trading partners, indicate that these would deepen GDP losses in the US as well as in the retaliating countries. At the same time, one-for-one retaliation would eliminate the EU's ToT loss, reflecting how the burden of US tariffs is shifted back entirely to US firms and households in the form of further hits to their real incomes.

The reaction of financial markets following the April 2 tariff announcements (e.g. the unusual combination of a weakening dollar and rising US Treasury yields) seems to indicate that a loss of confidence in the US economy and rising uncertainty are highly relevant forces in the current context – in addition to the tariff hikes themselves. In simulations including a stylised increase in US risk premiums, interest rates would rise further and the nominal dollar exchange rate would *depreciate*, more than offsetting the appreciation pressure implied by unilateral US tariff hikes alone. While this would lead to a more gradual ToT appreciation, supporting US net export volumes, the higher interest rates induced by rising risk premiums would dampen domestic capital accumulation and consumption demand further – on balance deepening the decline in US GDP.

The main qualitative results hold even under alternative calibrations, but with important quantitative consequences. In our main scenario we simulated highly persistent, but not fully permanent tariff hikes, in light of the uncertainties regarding US trade policies beyond the term of the current administration. Instead, if tariffs are percieved to be permanent, the US terms-of-trade would appreciate twice as much, supporting the real income of Americans and making foreigners bear roughly half the burden of US tariffs. In addition, as tariffs are not expected to decline over time, they would no longer act as an intertemporal tax on consumption, thereby mitigating the fall in US real GDP. In another scenario, we explore Dominant Currency

Pricing (DCP) whereby pre-tariff import prices are sticky in US dollar terms – instead of in the exporter's currency. This makes the terms-of-trade more rigid, hindering international relative price adjustments, leading to a sharper decline in global trade and GDP.

Under nominal rigidities, monetary policy is an important determinant of the transmission of tariff shocks, which present central bankers with trade-offs. If the Federal Reserve were to adopt a relatively looser monetary stance (compared to the baseline assumption of a CPI-inflation targeting interest rate rule), this would support US domestic demand, leading to better GDP outcomes, but at the cost of higher inflation and a smaller ToT gain – while the spillovers to European economic activity would be positive. In comparison, alternative assumptions about how US fiscal policy uses the extra tariff revenues seem less consequential. In our baseline scenario tariff revenues contribute to public debt reduction. Instead, if they are spent in a budget neutral way, it could dampen GDP losses, but only to the extent that they are targeted in a progressive fashion at liquidity-constrained households (boosting aggregate demand) or are used to cut distortionary taxes (enhancing supply-side efficiency).

Related literature. The idea that tariffs can stimulate the economy via improving external competitiveness (beggar-thy-neighbor) goes back as far as Keynes (1931), who proposed that import tariffs in combination with export subsidies could have the same impact as a currency devaluation. But importantly, Keynes suggested tariffs as a macroeconomic stabilisation tool in the context of a depressed economy operating below capacity, when monetary and fiscal policies were constrained, limited by the fixed exchange rate of the gold standard. In contrast, Mundell (1961) showed that, when exchange rates are flexible and the economy is at full employment, the tariff-induced strain on the economy's resources would lead to a currency appreciation in general equilibrium, offsetting the direct stimulative effects by crowding out exports, and rendering the policy neutral.[1] This echoes the symmetry result of Lerner (1936) stating that a tax on imports is equivalent to a tax on exports.

In dynamic models with rational expectations and nominal rigidities in domestic wage and price setting, Dornbusch (1987), Krugman (1982), and Eichengreen (1981) show that tariffs can even be contractionary. In such a setting, tariffs become a cost-push shock, having adverse effect on labour supply as workers aim to regain the purchasing power of their wages. Even with neutral monetary policy (targeting nominal incomes), this implies lower real output. Tighter monetary policy that cares about tariff-induced inflationary pressures, could lead to larger terms-of-trade appreciation and be an additional contractionary force.

More recent New Keynesian models convey similar messages. Auclert, Rognlie and Straub (2025) demonstrate how the demand-boosting effects of tariffs via expenditure switching from imports can be counterbalanced by a short-term decline in consumption driven by intertemporal substitution in response to *temporay* tariffs, as well as by a decline in less competitive exports that contain imported intermediate inputs. They argue that with reasonable elasticities tariffs are contractionary under neutral monetary policy. In turn, Monacelli (2025) and Auray, Devereux and Eyquem (2025*a*) emphasise the role of monetary policy: a tighter monetary stance under inflation targeting exacerbates the contractionary effects, while recession could only be prevented at the cost of higher inflation, highlighting the inherently stagflationary nature of tariffs and the trade-offs faced by policymakers. Jeanne and Son (2024) illustrates how the long-run effect of *permanent* tariff hikes (i.e. with no more intertemporal substitution effects either from monetary tightening or from an expected decline in tariffs), is likely to be contractionary independently of trade elasticities. In this case labour supply is hurt, as the adverse substitution effects from lower real wages are

[1]This neutrality result hinges on the absence of the Laursen-Metzler effect, i.e. that a stronger terms-of-trade would not lead to higher saving rates. Otherwise, as argued by Mundell (1961), tariffs can even be contractionary.

no longer counterbalanced by beneficial income effects from weaker consumption. These adverse supply side effects determine the long-run GDP outcome, while the strength of the demand boost via expenditure switching just determines the required size of the offsetting terms-of-trade appreciation.[2]

Regarding optimal monetary policy, Werning, Lorenzoni and Guerrieri (2025) characterise tariff hikes as cost-push shocks that create a trade-off between inflation and output stabilisation, and show that the central bank should temporarily accommodate the shock, i.e. tolerate higher short-term inflation in exchange for mitigating output losses. Similarly, Monacelli (2025) shows optimal policy to be more expansionary than domestic PPI inflation targeting (that would replicate the flexible-price allocation in his model without imported inputs), keeping output above its distorted natural level. Bianchi and Coulibaly (2025) arrives to the same result, arguing that the reason is a *fiscal externality*: since households do not internalise the tax revenue from tariffs, they reduce their import demand too sharply which must be counterbalanced by stimulus. In contrast, in a model with firm dynamics, Bergin and Corsetti (2023) show the optimal (cooperative) monetary reaction to be contractionary relative to the flexible-price allocation (for a country imposing unilateral tariffs). Auray, Devereux and Eyquem (2025*b*) and Auray, Devereux and Eyquem (2024) explore the interaction between trade and monetary policies, showing how different monetary strategies can change the intensity of trade wars.

A recent literature pioneered by Farhi, Gopinath and Itskhoki (2014) examines import tariffs in the context of *fiscal devaluations*, revenue-neutral changes in tax policies that aim to replicate the competitiveness-enhancing effects of nominal exchange rate devaluation. In addition to the original Keynesian idea of combining import tariffs with export subsidies, these could also mean reducing payroll taxes financed by an increase in VAT rates – often discussed as a policy option for less competitive euro area members in need of current account adjustment, but without the option to devalue their currency or to endure austerity-driven internal devaluation.[3] Under the Lerner symmetry, these policies should have no stimulative effect due to offsetting exchange rate movements in equilibrium. In a fully-fledged New Keynesian DSGE model, Lindé and Pescatori (2019) explore conditions for deviating from this benchmark neutrality result, such as exchange rate rigidity or financial market completeness (see also Costinot and Werning (2019) and Barbiero et al. (2019)). Similarly, Erceg, Prestipino and Raffo (2023) find that fiscal devaluations involving tariffs could provide some stimulus under fixed exchange rates, but their effect diminishes under flexible exchange rates. Vogel (2017) compares different fiscal devaluation strategies in a small open economy version of QUEST, and finds that under fixed exchange rates import tariffs with export subsidies can mitigate the output losses after a domestic demand contraction, and aid current account rebalancing.

Looking at adverse supply side effects, Barattieri, Cacciatore and Ghironi (2021) build a New Keynesian model with firm heterogeneity, endogenous selection into trade and capital accumulation to demonstrate how tariffs lower aggragate productivity by reallocating market share toward less efficient domestic producers, which

[2] Appendix B.2 compares the effects of permanent tariff hikes in the flexible price model versions of Monacelli (2025) and Jeanne and Son (2024), generalising them to include imported intermediate inputs in production. There are no intertemporal substitution effects either from monetary tightening or from an expected decline in tariffs, and flexible prices allow us to isolate the supply-side channels. The only difference between the models is financial market completeness, which is shown to matter crucially for the GDP effects of tariffs. Whether allowing for the terms-of-trade gain to be consumed (or given to foreigners as insurance payment), these assumptions affect the path of consumption, and thereby labour supply, materially. The exercise can also illustrate the effects of tariff persistence under incomplete markets: perfect risk-sharing under complete markets (Monacelli, 2025) approximates the response to temporary tariffs under incomplete markets (Auclert, Rognlie and Straub, 2025).

[3] In addition, border adjustment of corporate income taxes (C-BAT) have been discussed in the US, allowing firms to deduct export sales from their taxable profits, but not the cost of imported inputs. Essentially, border adjustment makes goods taxed based on the location of final consumption rather than of production.

make tariffs recessionary even under fixed exchange rates. In addition, tariffs hurt investment by raising the price of import-intensive capital goods, amplifying the negative supply-side consequences in the long run, as also pointed out by Baqaee and Malmberg (2025). Furthermore, Kalemli-Özcan, Soylu and Yildirim (2025) demonstrates the role of international production networks, showing how tariffs lead to endogenous fragmentation of global supply chains.

On the empirical side, Barattieri, Cacciatore and Ghironi (2021) estimate panel VARs showing how trade barriers cause output to fall and inflation to rise. Furceri et al. (2022) present similar evidence from local projections, associating tariff hikes also with real appreciation and insignificant trade balance changes. Documenting the effects of the 2018 US tariff measures, Amiti, Redding and Weinstein (2019) estimate that most of their burden fell on US households, which is also in line with the findings of Fajgelbaum et al. (2020), showing how the US-China trade war lowered US aggregate real income.[4]

The new US tariff announcements have prompted a series of studies analysing their economic impact. Quantitative model simulations in rich policy models, such as in Jouvanceau et al. (2025) and Mckibbin, Hogan and Noland (2024), find recessionary effects, similarly to our results. Other papers focus on the long-run effect on the trade deficit (Costinot and Werning, 2025) or analyse welfare implications and optimal tariff policy (Itskhoki and Mukhin, 2025; Ignatenko et al., 2025; Dávila et al., 2025).

[4] See also Fajgelbaum and Khandelwal (2022), Amiti, Redding and Weinstein (2020) and Flaaen and Pierce (2024).

2 MODEL DESCRIPTION

This paper explores the macroeconomic effects of tariff hike scenarios through the lenses of the European Commission's macroeconomic model, QUEST (Burgert et al., 2020).

QUEST is a New Keynesian open economy dynamic stochastic general equilibrium (DSGE) model, which incorporates the main features relevant for understanding tariffs. Among these are the imperfect substitutability of goods produced in different regions, and sluggish adjustment of import volumes in response to relative price changes. Trade flows and nominal exchange rates are modelled bilaterally, via integrated international goods and capital markets. The dynamic model structure enables us to trace both short and long-run effects. The model distinguishes between a tradable and a non-tradable sector. While it lacks the sectoral detail of typical trade Computable General Equilibrium (CGE) models and therefore cannot assess impacts on specific manufacturing sectors and services, it includes trade in intermediate inputs for both the tradable and non-tradable sectors. This captures linkages through cross-border value chains, which is crucial as trade in intermediate inputs amplifies the effects of trade barriers. Imported goods (hit by tariffs) therefore appear not only in the consumption basket of households but are also used in the production process.

The rest of the model features are standard in the New Keynesian literature. There are two types of infinitely-lived households: liquidity-constrained and unconstrained Ricardian agents. Both consume goods and supply labour (via a labour union setting sticky wages), while Ricardians also invest into domestic physical capital, government bonds and international bonds, and also receive firm profits. On the production side, monopolistically competitive firms operate in tradeable and non-tradeable sectors whose price setting is subject to nominal rigidities. Monetary policy controls the short term nominal interest rate to stabilise inflation, while fiscal policy finances various transfers and public spending by levying taxes (including tariffs) on domestic agents and issuing nominal government bonds.

The analysis applies a 6-region model setup covering the US, the EU, China, Canada, Mexico and the rest of the world (RoW), with trade linkages calibrated based on the FIGARO input-output database.

For the sake of brevity, a full detailed description will not be repeated here, highlighting only key model features relevant to the subsequent discussion. The interested reader is referred to Burgert et al. (2020) for more details.

2.1. GOODS MARKET STRUCTURE

Each region is home to a domestic tradable sector and a non-tradable sector $j \in \{TD, NT\}$, whose gross output Y_t^j is used for various expenditure types, such as (private and public) final consumption $C_t + G_t + IG_t$, investment goods I_t^j, intermediate inputs INT_t^j or exports X_t.

For these various purposes, the output of different sectors is bundled together with imports according to a nested, 3-layered CES structure, where components are imperfect substitutes of each other:

- final and intermediate goods for uses $H \in \{C, G, IG, I^{NT}, I^{TD}, INT^{NT}, INT^{TD}, X\}$ are bundles of tradables (T) and non-tradable goods (NT)
- the tradable bundle (T) consist of domestically produced (TD) and imported (M) goods
- imported goods (M) needed for the above come from various trading partners $f \in \mathcal{F}$

Demand for imports and domestic goods (delayed substitution). Solving the expenditure minimisation problems in the first two CES layers yields demand functions for the two types of domestic goods (NT, TD) as well as for imported goods (M), given relative prices and total demand for each expenditure type H_t (except for exports).[5]

$$H_t = \left[\left(1 - s_H^T\right)^{\frac{1}{\sigma_{TNT}}} \left(H_t^{NT}\right)^{\frac{\sigma_{TNT}-1}{\sigma_{TNT}}} + \left(s_H^T\right)^{\frac{1}{\sigma_{TNT}}} \left(H_t^T\right)^{\frac{\sigma_{TNT}-1}{\sigma_{TNT}}} \right]^{\frac{\sigma_{TNT}}{\sigma_{TNT}-1}} \tag{2.1}$$

$$H_t^{NT} = \left(1 - s_H^T\right) \left(\frac{P_t^{NT}}{P_t^H}\right)^{-\sigma_{TNT}} H_t \tag{2.2}$$

$$H_t^T = s_H^T \left(\frac{P_t^T}{P_t^H}\right)^{-\sigma_{TNT}} H_t \tag{2.3}$$

$$H_t^T = \left[\left(1 - s^M\right)^{\frac{1}{\sigma_m}} \left(\widetilde{H}_t^{TD}\right)^{\frac{\sigma_m-1}{\sigma_m}} + \left(s^M\right)^{\frac{1}{\sigma_m}} \left(\widetilde{H}_t^{M}\right)^{\frac{\sigma_m-1}{\sigma_m}} \right]^{\frac{\sigma_m}{\sigma_m-1}} \tag{2.4}$$

$$\underbrace{\frac{H_t^{TD}}{H_t^T}}_{h_t^{TD}} = \left[\underbrace{\left(1 - s^M\right) \left(\frac{P_t^{TD}}{\widetilde{\mathcal{P}}_t^T}\right)^{-\sigma_m}}_{\widetilde{H}_t^{TD}/H_t^T} \right]^{1-\rho^{im}} \left[\underbrace{\frac{H_{t-1}^{TD}}{H_{t-1}^T}}_{h_{t-1}^{TD}} \right]^{\rho^{im}} \tag{2.5}$$

$$\underbrace{\frac{H_t^{M}}{H_t^T}}_{h_t^{M}} = \left[\underbrace{s^M \left(\frac{P_t^{M}}{\widetilde{\mathcal{P}}_t^T}\right)^{-\sigma_m}}_{\widetilde{H}_t^{M}/H_t^T} \right]^{1-\rho^{im}} \left[\underbrace{\frac{H_{t-1}^{M}}{H_{t-1}^T}}_{h_{t-1}^{M}} \right]^{\rho^{im}} \tag{2.6}$$

where $H_t \in \{C_t, G_t, IG_t, I_t^{NT}, I_t^{TD}, INT_t^{NT}, INT_t^{TD}\}$, with the (use-specific) steady state shares governed by s_H^T and s^M, while σ_{TNT} and σ_m denote elasticities of substitution. In the second CES layer $\widetilde{H}_t^{TD}, \widetilde{H}_t^{M}$ denote "frictionless" demand for domestically produced tradable (TD) and imported (M) goods, which tell us what the *optimal* split between those components would be. This also makes use of a shadow price, defined as $\widetilde{\mathcal{P}}_t^T \equiv \left[\left(1 - s^M\right) \left(P_t^{TD}\right)^{1-\sigma_m} + \left(s^M\right) \left(P_t^{M}\right)^{1-\sigma_m} \right]^{\frac{1}{1-\sigma_m}}$.

However, (2.5)-(2.6) deviate from the above "frictionless" case by introducing ***delayed import substitution*** to make this split initially less elastic, i.e. making imports and domestic goods less substitutable in the short run than implied by the long-run import elasticity σ_m. This is implemented by applying and ad-hoc real rigidity, governed by $\rho^{im} \neq 0$, to the *actual* shares of domestically produced and imported goods h_t^{TD}, h_t^{M}, making them more persistent than they would be in the frictionless case.[6] The sluggish import substitution gives rise to a J-curve-type behaviour for the trade balance, where relative price movements can initially dominate more rigid volume adjustments.

Import demand from trading partners (delayed substitution). In the third CES layer, total imports $M_t = \sum_H H_t^M$, which are the sum of import demands for each type of expenditure H_t^M as specified by (2.6),

[5]The exports bundle consists only of domestically produced tradable (TD) goods, with no non-tradables or *direct* import content, therefore no demand function is necessary for its components. But exports still have *indirect* import content, as domestically produced TD goods (their only component) use imported intermediate inputs during the production process. See the detailed supply-and-use I/O structure in Table Appendix A.1

[6]Importantly, h_t^{TD} and h_t^M are the same for $\forall\, H \in \{C, G, IG, I^{NT}, I^{TD}, INT^{NT}, INT^{TD}\}$. This is due to the same recursive structure, that involves the same starting shares in steady state ($h_0^M = s^M, h_0^{TD} = 1 - s^M$) and same relative prices $\frac{P_t^M}{\widetilde{\mathcal{P}}_t^T}$ and $\frac{P_t^{TD}}{\widetilde{\mathcal{P}}_t^T}$ across $\forall\, H$.

are split between various trading partners $f \in \mathcal{F}$.[7]

$$M_t = \left[\sum_f \left(s^f\right)^{\frac{1}{\sigma_x}} \left(\widetilde{M}_t^f\right)^{\frac{\sigma_x - 1}{\sigma_x}} \right]^{\frac{\sigma_x}{\sigma_x - 1}}$$

$$\frac{M_t^f}{M_t} = \underbrace{\left[s^f \left(\frac{P_t^{M,f}}{\widetilde{\mathcal{P}}_t^M} \right)^{-\sigma_x} \right]}_{\widetilde{M}_t^f / M_t}^{1-\rho^{ex}} \left[\frac{M_{t-1}^f}{M_{t-1}} \right]^{\rho^{ex}} \tag{2.7}$$

where s^f denotes the share of trading partner f in the total import basket of the domestic economy, and σ_x captures the long-run elasticity of substitution between different trading partners (export elasticity). The "frictionless" optimal split between trading partners would be $\widetilde{M}_t^f$, making use of the shadow price definition $\widetilde{\mathcal{P}}_t^M \equiv \left[\sum_f s^f \left(P_t^{M,f} \right)^{1-\sigma_x} \right]^{\frac{1}{1-\sigma_x}}$. As before, (2.7) deviates from this frictionless split by applying an ad hoc rigidity via $\rho^{ex} \neq 0$, effectively lowering short-run elasticities relative to σ_x, and ***delaying substitution within the import basket.***

2.2. PRODUCTION WITH INTERMEDIATE INPUTS

Each domestic sector $j \in \{NT, TD\}$ consists of two types of firms. Perfectly competitive **manufacturers** take prices as given and produce a *homogenous* sectoral good (determining gross output Y_t^j). A continuum of monopolistically competitive **retailers** then create a *differentiated variety* of the manufacturer's output, having pricing power subject to nominal rigidities (determining sticky sectoral output prices P_t^{NT} and P_t^{TD}). A third type of **export retailers** then differentiate TD goods further (determining sticky export prices P_t^X).

Production technology. Manufacturing firms have access to a nested CES production technology that combines value-added VA_t^j and intermediate inputs INT_t^j to produce gross sectoral output Y_t^j:

$$Y_t^j = \left[\left(1 - s_{in}^j\right)^{\frac{1}{\sigma_{in}}} \left(VA_t^j\right)^{\frac{\sigma_{in}-1}{\sigma_{in}}} + \left(s_{in}^j\right)^{\frac{1}{\sigma_{in}}} \left(INT_t^j\right)^{\frac{\sigma_{in}-1}{\sigma_{in}}} \right]^{\frac{\sigma_{in}}{\sigma_{in}-1}} \tag{2.8}$$

where s_{in}^j and σ_{in} are, respectively, the steady-state share of intermediates in gross output and the elasticity of substitution between intermediates and value-added.

In turn, VA_t^j is produced with a Cobb-Douglas production technology, combining labour L_t^j, sector-specific private capital K_t^j, and public capital KG_t:

$$VA_t^j = A_t^j \left[\left(u_t^j K_t^j\right)^{1-\alpha} \left(L_t^j - fcl\right)^{\alpha} \left(KG_t\right)^{\alpha_g} - fcy \right] \tag{2.9}$$

where A_t^j, u_t^j, fcl, and fcy denote total factor productivity (TFP), capacity utilisation, overhead labour and fixed costs, respectively.

After solving the profit maximisation problem, the demand for intermediate inputs used by manufacturers in sector j is derived as:

$$INT_t^j = s_{in}^j \left(\frac{P_t^{INT,j}}{\eta_t^j P_t^j} \right)^{-\sigma_{in}} Y_t^j \tag{2.10}$$

[7] Note that it is not each H_t^M that is split separately between imports from different trading partners, but rather *total* imports M_t.

where η_t^j denotes the relative price at which manufacturers sell their output to retailers (in effect, the real marginal costs of retailers). The price of intermediate goods $P_t^{INT,j}$ is pinned down by the price index (2.14), after substituting in $H = INT^j$. Recall, just like final good bundles, intermediate inputs are a composite of different types of domestic sectoral and imported goods. Plug in $H = INT^j$ to the CES baskets (2.1) and (2.4):

$$INT_t^j = \left[\left(1 - s_{INT,j}^T\right)^{\frac{1}{\sigma_{TNT}}} \left(INT_t^{NT,j}\right)^{\frac{\sigma_{TNT}-1}{\sigma_{TNT}}} + \left(s_{INT,j}^T\right)^{\frac{1}{\sigma_{TNT}}} \left(INT_t^{T,j}\right)^{\frac{\sigma_{TNT}-1}{\sigma_{TNT}}}\right]^{\frac{\sigma_{TNT}}{\sigma_{TNT}-1}}$$

$$INT_t^{T,j} = \left[\left(1 - s^M\right)^{\frac{1}{\sigma_m}} \left(\widetilde{INT}_t^{TD,j}\right)^{\frac{\sigma_m-1}{\sigma_m}} + \left(s^M\right)^{\frac{1}{\sigma_m}} \left(\widetilde{INT}_t^{M,j}\right)^{\frac{\sigma_m-1}{\sigma_m}}\right]^{\frac{\sigma_m}{\sigma_m-1}}$$

where $INT_t^{k,j}$ denotes the volume of type $k \in \{NT, TD, M\}$ goods needed for assembling intermediate input composites used for production in sector $j \in \{NT, TD\}$. The demand for these components are pinned down by (2.2), (2.3), (2.5) and (2.6), after plugging in $H = INT^j$. Via imported inputs, domestically produced gross output also has import content.

2.3. PRICE SETTING

Domestic firms. Retailers set the prices of gross domestic sectoral output P_t^{NT} and P_t^{TD} which are subject to Rotemberg style nominal rigidities and are governed by standard New Keynesian Phillips Curves (NKPC). As a result, output price inflation reacts sluggishly to deviations in real marginal costs η_t^j, that in turn are driven by wage, capital and intermediate input costs (via (2.8), (2.9)).

$$\pi_t^j \approx E_t \left\{\Lambda_{t,t+1}^j \frac{Y_{t+1}^j}{Y_t^j} \pi_{t+1}^j\right\} + \frac{\sigma_j}{\gamma_j}\left[\eta_t^j - \frac{\sigma_j - 1}{\sigma_j}\right] \qquad \text{for } j \in \{NT, TD\} \tag{2.11}$$

where γ_j captures nominal rigidities, σ_j governs retailer's market power, and $\pi_t^j = \frac{P_t^j}{P_{t-1}^j} - 1$ denotes sectoral output price inflation, while $\Lambda_{t,t+1}^j$ is the stochastic discount factor of firm-owning households (expressed in terms of the sectoral good).

Export prices P_t^X depend on P_t^{TD} (since exports consist only of *further* differentiated *TD* goods), but with an additional layer of price stickiness leading to temporary deviations between the two, affecting the markup of exporters. The exporter NKPC is:

$$\pi_t^X \approx E_t \left\{\Lambda_{t,t+1}^X \frac{X_{t+1}}{X_t} \pi_{t+1}^X\right\} + \frac{\sigma_{px}}{\gamma_{px}}\left[\frac{P_t^{TD}}{P_t^X} - \frac{\sigma_{px} - 1}{\sigma_{px}}(1 + \tau^x)\right] \tag{2.12}$$

where γ_{px} captures nominal rigidities for exporters (*on top* of retailers), σ_{px} governs their market power, and $\pi_t^X = P_t^X / P_{t-1}^X - 1$ denotes export price inflation, while $\Lambda_{t,t+1}^X$ is the stochastic discount factor of firm-owning households (expressed in terms of exports). The steady state desired markup is set to 0 by an appropriate subsidy τ^x, such that $\eta^x = \frac{P^{TD}}{P^X} = 1$.

Import prices. Bilateral import prices from region f, $P_t^{M,f}$ (faced by domestic importers, expressed in domestic currency) depend on the prices set by foreign exporters in foreign currency $P_t^{X,f}$, on the bilateral nominal exchange rate $\frac{\mathcal{E}_t}{\mathcal{E}_t^f}$, and on tariffs levied on imports from that country $\tau_t^{tariff,f}$.

$$P_t^{M,f} = \left(1 + \tau_t^{tariff,f}\right) \underbrace{\frac{\mathcal{E}_t}{\mathcal{E}_t^f} P_t^{X,f}}_{\equiv P_t^{M1,f}} \tag{2.13}$$

where $P_t^{M1,f}$ is the bilateral *import price net of tariffs*, i.e. the price at the border paid to the foreign trading partner f, expressed in domestic currency.

As shown by (2.12), in our baseline setup of ***producer currency pricing (PCP)*** prices are sticky in the currency of the exporter (in $P_t^{X,f}$), and on top of this there is full exchange rate and tariff pass-through to import prices. Section 5.2 explores other alternatives such as dominant currency pricing (DCP), where it is the pre-tariff US dollar price $\frac{P_t^{M1,f}}{\mathcal{E}_t} = \frac{P_t^{X,f}}{\mathcal{E}_t^f}$ that is rigid (see also Appendix A.7).[8]

Price indices. Aggregate price indices for the 3 layers of CES bundles P_t^H, P_t^T, P_t^M are defined implicitly, to ensure that nominal expenditures add up to the total value of the bundle:

$$P_t^H \equiv P_t^{NT}\frac{H_t^{NT}}{H_t} + P_t^T\frac{H_t^T}{H_t} \tag{2.14}$$

$$P_t^T \equiv P_t^{TD}\underbrace{\frac{H_t^{TD}}{H_t^T}}_{h_t^{TD}} + P_t^M\underbrace{\frac{H_t^M}{H_t^T}}_{h_t^M} \tag{2.15}$$

$$P_t^M \equiv \sum_f P_t^{M,f}\frac{M_t^f}{M_t} \tag{2.16}$$

where recall that h_t^{TD} and h_t^M are the same for $\forall\, H \in \{C, G, IG, I^{NT}, I^{TD}, INT^{NT}, INT^{TD}\}$. In addition, one can show that in the absence of delayed import substitution $\rho^{im} = \rho^{ex} = 0$, the actual price indices for tradables and imports coincide with the shadow price definitions $P_t^T = \widetilde{\mathcal{P}}_t^T$ and $P_t^M = \widetilde{\mathcal{P}}_t^M$. Note that P_t^M (faced by domestic importers) already includes tariffs.

Terms-of-trade. The terms-of-trade (ToT) is defined is the ratio of export prices P_t^X and the import price index net of tariffs, P_t^{M1}:

$$ToT_t \equiv \frac{P_t^X}{P_t^{M1}} \tag{2.17}$$

where P_t^{M1} is defined implicitly as:

$$P_t^{M1} \equiv \sum_f \underbrace{\frac{\mathcal{E}_t}{\mathcal{E}_t^f}P_t^{X,f}}_{P_t^{M1,f}}\frac{M_t^f}{M_t} \tag{2.18}$$

which makes ToT_t a kind of real effective exchange rate measure, expressing the relative price of domestic and foreign *exported* goods (as opposed to the whole consumption basket), depending on nominal exchange rates as well as on domestic and foreign price levels. In other words, the ToT shows how much a country can import in exchange for its exports (with a higher ToT making a country richer), but also how expensive its exporters are in world markets (with higher/stronger ToT hurting competitiveness).

[8] Another alternative would be local currency pricing (LCP, or "pricing to market") where the rigidity is in the currency of the importer $P_t^{M,f}$. This would mean that fluctuations in nominal exchange rates and tariffs more directly and quickly affect the prices and markups of exporters. Note that this effect is there also under PCP, just indirectly and more sluggishly: as quick FX- or tariff-induced fluctuations in import prices lead to changing demand for their products, exporters are prompted to change export prices once nominal rigidities allow.

Effective tariff rate. The *effective* tariff rate τ_t^{tariff} expresses the total tariff revenues collected as a share of the pre-tax import bill:

$$\tau_t^{tariff} \equiv \frac{tariffrev_t}{P_t^{M1}\, M_t} \tag{2.19}$$

where

$$tariffrev_t = \sum_f \tau_t^{tariff,f} \, \frac{\mathcal{E}_t}{\mathcal{E}_t^f} \, P_t^{X,f} \, M_t^f \tag{2.20}$$

which shows that, to the extent *bilateral* tariff rates are heterogeneous, even if they remain unchanged, composition effects stemming from substitution within the import basket (i.e. changes in M_t^f/M_t) can lead to fluctuations in the *effective* tariff rate.[9]

Combining (2.13), (2.16), (2.18) with (2.19) and (2.20), it follows that the following relationship holds for the effective price indices:

$$P_t^M = \left(1 + \tau_t^{tariff}\right) P_t^{M1} \tag{2.21}$$

2.4. RESOURCE CONSTRAINTS AND TOTAL IMPORTS

The resource constraints (market clearing) for type $k \in \{NT, TD\}$ goods require that the gross output of each sector Y_t^k is exhausted by all the various uses of that output, which includes being a component H_t^k in composite bundles for final consumption, investment and intermediate goods $H \in \{C, G, IG, I^{NT}, I^{TD}, INT^{NT}, INT^{TD}\}$, as well as exports X_t and adjustment costs adj_t^k.

$$\begin{aligned} Y_t^k &= \sum_H H_t^k \;+\; \mathbb{I}_{k=TD}\, X_t \;+\; adj_t^k \;= && \text{for } k \in \{NT, TD\} \\ &= C_t^k + G_t^k + IG_t^k + \sum_{j \in \{NT,TD\}} I_t^{k,j} + \sum_{j \in \{NT,TD\}} INT_t^{k,j} \;+\; \mathbb{I}_{k=TD}\, X_t \;+\; adj_t^k && \end{aligned} \tag{2.22}$$

where recall that exports X_t consist only of domestically produced tradeable $k = TD$ goods.

Total imports are equal to the sum of imported goods used in all the various expenditure bundles:

$$\begin{aligned} M_t &= \sum_H H_t^M = \\ &= C_t^M + G_t^M + IG_t^M + \sum_{j \in \{NT,TD\}} I_t^{M,j} + \sum_{j \in \{NT,TD\}} INT_t^{M,j} \end{aligned} \tag{2.23}$$

Demand for the components H_t^k for $k \in \{NT, TD, M\}$ are governed by (2.2), (2.3), (2.5) and (2.6).

For a representation of the above described goods market structure as supply and use and input-output tables, see Annex A.1. This makes it clear, as (2.23) also shows, that imports are used both *directly* for final consumption, and for intermediate inputs during the production of domestic goods in (2.8), which means they show up in final consumption *indirectly* as well, via its domestic components. Thereby, as tariffs disturb import prices, and alter import demand, this can have rich cascading effects through the domestic economy.

[9] However, up to first order, these composition effects are zero as long as the steady state features symmetric tariffs.

2.5. EXTERNAL BALANCE

Total exports of the domestic economy are determined by the import demand of all foreign regions for the domestic region's (Home) goods (trade market clearing):

$$X_t = \sum_f \nu^f M_t^{H,f} \tag{2.24}$$

where foreigners' import demand for Home goods $M_t^{H,f}$ is analogous to domestic (Home) import demand for region f goods $M_t^{f,H}$ as in (2.7) – the superscript H is just suppressed in the latter for simpler notation. ν^f denotes the relative size of region f to the domestic economy.

Trade balance. The trade balance of the domestic economy is defined as net exports in (nominal) value terms:

$$TB_t \equiv P_t^X X_t - P_t^{M1} M_t \tag{2.25}$$

where the import bill is measured at *net-of-tariff* prices P_t^{M1}, i.e. prices at the border which are paid to foreign trading partners.

2.6. GDP DEFINITION WITH TARIFFS AND INTERMEDIATE INPUTS

Nominal GDP. Nominal GDP is defined from the production side as the sum of value added from all sectors (gross value added, GVA), plus taxes (such as VAT and tariffs) less subsidies on products.[10]

$$\begin{aligned} nGDP_t^{prod} \equiv & \underbrace{\sum_{j\in\{NT,TD\}} \left[P_t^j Y_t^j - P_t^{INT,j} INT_t^j\right] \;+\; (P_t^X - P_t^{TD})\, X_t}_{GVA_t} \;+ \\ & +\; \tau_t^{vat} P_t^C C_t \;+\; \tau_t^{tariff} P_t^{M1} M_t \end{aligned} \tag{2.26}$$

As shown in Section A.2, (2.26) is equivalent to GDP definitions from the expenditure or income sides:

$$\begin{aligned} nGDP_t^{exp} \equiv & \left(1+\tau_t^{vat}\right) P_t^C C_t \;+\; P_t^C \left[G_t + IG_t\right] + \sum_{j\in\{NT,TD\}} P_t^{I,j} I_t^j \;+ \\ & +\; \underbrace{\left(P_t^X X_t \;-\; P_t^{M1} M_t\right)}_{TB_t} + \sum_{j\in\{NT,TD\}} P_t^j adj_t^j \end{aligned} \tag{2.27}$$

$$\begin{aligned} nGDP_t^{inc} \equiv & \sum_{j\in\{NT,TD\}} \left[\left(1+\tau_t^{ssc}\right) W_t L_t^j \;+\; i_t^{K,j} P_t^{I,j} K_t^j \;+\; profit_t^j\right] \;+ \\ & +\; profit_t^X \;+\; \tau_t^{vat} P_t^C C_t \;+\; \tau_t^{tariff} P_t^{M1} M_t \end{aligned} \tag{2.28}$$

Note that tariffs are implicitly included in the expenditure side definition, since price indices for final expenditures (e.g. P_t^C, $P_t^{I,j}$) are already inclusive of tariffs via (2.14), while the import bill (which is deducted) is measured at prices net of tariffs (P_t^{M1} in (2.18)).

[10] The latter adjustment is in line with current statistical standards, which measure GDP at *market prices* paid by final users (as opposed to at *basic prices*, like GVA). This distinction is crucial for our analysis which focuses on the effect of tariffs, which can therefore directly affect the GDP-deflator (see Appendices A.6.3 and A.6.1).

Real GDP and GDP-deflator. Real GDP is defined as a constant price volume index (i.e. measuring output at fixed steady state prices), based on expenditure side nominal GDP (2.27):

$$rGDP_t \equiv \left(1+\tau^{vat}\right) C_t + G_t + IG_t + \sum_{j\in\{NT,TD\}} I_t^j + \left(\frac{1}{\eta^x} X_t - \frac{1}{1+\tau^{tariff}} M_t\right) \tag{2.29}$$

where most steady state prices are normalised to 1, including the pre-VAT consumption price index $P^C = 1$ and the *post-tariff* import price index $P^M = 1$, which implies via (2.21) that the steady state *pre-tariff* import price (which is the relevant measure in the expenditure side GDP definition) is $P^{M1} = \frac{1}{1+\tau^{tariff}}$. Export prices $P^X = \frac{P^{TD}}{\eta^x} = \frac{1}{\eta^x}$ via (2.12).

The GDP deflator is implicitly determined as the ratio of nominal and real GDP:

$$P_t^{GDP} = \frac{nGDP_t}{rGDP_t} \tag{2.30}$$

Real GDI and terms-of-trade gain. Real gross domestic income ($rGDI_t$) is defined as the sum of real GDP and the terms-of-trade gain $\widehat{TB}_t^{ToT}$. Real GDI measures the ***purchasing power of incomes*** earned from domestic production, i.e. it expresses nominal GDP (incomes) in terms of domestic expenditures (what incomes are spent on):

$$\begin{aligned} rGDI_t \equiv rGDP_t \; + & \underbrace{\left(\frac{P_t^{GDP}}{P_t^D} - 1\right) rGDP_t}_{\widehat{TB}_t^{ToT}} = \\ = & \frac{P_t^{GDP}}{P_t^D} \, rGDP_t \; = \; \frac{nGDP_t}{P_t^D} \end{aligned} \tag{2.31}$$

where $P_t^D \equiv \frac{P_t^C\left[(1+\tau_t^{vat})C_t+G_t+IG_t\right]+\sum_j P_t^{I,j} I_t^j}{(1+\tau^{vat})C_t+G_t+IG_t+\sum_j I_t^j}$ is the implicit deflator for domestic expenditures.

By changing the relative price of exports and imports, a fluctuating terms-of-trade $ToT_t = \frac{P_t^X}{P_t^{M1}}$ also drives a wedge between the GDP deflator P_t^{GDP} (the price of what the domestic economy *produces*, including exports) and the domestic expenditure deflator P_t^D (the price of what the domestic economy *spends* on, including imports). This wedge governs the terms-of-trade gain $\widehat{TB}_t^{ToT}$, which captures how much more the economy as a whole can spend (without having to borrow from abroad) purely due to relative price changes, even if the volume of its production remained the same. In other words, the terms-of-trade gain reflects the divergence between the real income of an economy (real GDI) and its real production (real GDP).

The terms-of-trade gain also has a major impact on the evolution of the trade balance. As derived in Appendix A.3, a change in the value of the trade balance can be decomposed into contributions by trade volume changes, *average* price level growth, and *relative* price changes (i.e. the ToT-gain):

$$\underbrace{TB_t - TB}_{\widehat{TB}_t} = \underbrace{\left(\widehat{X}_t - \widehat{M}_t\right)}_{\widehat{TB}_t^{volume}} + \underbrace{\left(1 - \frac{1}{P_t^D}\right) TB_t}_{\widehat{TB}_t^{price}} + \underbrace{\left[\left(\frac{P_t^X}{P_t^D} - \frac{1}{\eta^x}\right) X_t - \left(\frac{P_t^{M1}}{P_t^D} - \frac{1}{1+\tau^{tariff}}\right) M_t\right]}_{\widehat{TB}_t^{ToT}} \tag{2.32}$$

where $\widehat{X}_t \equiv \frac{1}{\eta^x}(X_t - X)$ and $\widehat{M}_t \equiv \frac{1}{1+\tau^{tariff}}(M_t - M)$, and the "average" price level is chosen to be P_t^D. As shown in Appendix A.3, the term $\widehat{TB}_t^{ToT}$ is exactly the same in (2.32) as in (2.31), but in this context it illustrates how the value of the trade balance would change purely due to fluctuations in relative prices, even if the volume of goods crossing borders remained the same. Put differently, the ToT gain shows how much more goods the economy can import in exchange for a given amount of exports, without affecting the trade balance (i.e. without having to borrow more from abroad).

2.7. CALIBRATION

Most behavioural parameters and dynamics adjustment costs in the model are kept at their standard calibration, which are based on model estimations (Ratto, Roeger and in 't Veld, 2009; Albonico et al., 2019). Trade linkages and our stylised global supply chains with traded intermediate inputs are calibrated based on the FIGARO input-output database.

Table 2.1: Trade elasticity parameters

Parameter	Description	Value
σ_m	Import elasticity (long-run)	1.5
σ_x	Export elasticity (long-run)	3.0
ρ^{im}	Delay in import substitution (quarterly)	0.90
ρ^{ex}	Delay in substitution across trading partners (quarterly)	0.95

For the analysis of tariffs, trade elasticities are of crucial importance. Our baseline values are shown in Table 2.1, which are in line with what the literature uses in similarly structured DSGE models. Auclert et al. (2024) adopts a symmetric $\sigma_m = \sigma_x = 4$, but with very rigid delayed substitution ($\rho^{im} = 0.976$ quarterly), citing Boehm, Levchenko and Pandalai-Nayar (2023). In turn, Acharya and Challe (2025) have $\sigma_m = 1.5, \sigma_x = 4$, citing Egorov and Mukhin (2023), but with no delayed substitution feature. In a simpler New Keynesian model without intermediate inputs Jeanne and Son (2024) use $\sigma_m = 1, \sigma_x = 3$, citing Feenstra et al. (2018). Section 5.5 conducts a sensitivity analysis.

3 UNILATERAL US TARIFF HIKES

Simulation scenario. The main scenario aims to capture US announcements up to and including 2 Apr 2025 (dubbed "Liberation Day"), featuring sizeable *unilateral* tariff hikes on US imports of goods from virtually all trading partners, resulting in a roughly 20 percentage point increase in the effective US tariff rate.[11]

In order to isolate the effect of US tariffs hikes themselves, this scenario features no escalation in trade tensions, i.e. no retaliation by trading partners (an alternative trade war scenario with stylised tit-for-tat retaliations is explored in Section 4.1). In addition, this scenario abstracts from the strong financial market reaction whereby investors started to demand higher risk premia on dollar-denominated assets, arguably triggered by the policy uncertainty surrounding the sweeping US tariff announcements (an alternative scenario including additional risk premium shocks is simulated in Section 4.2).

The simulated increase in tariffs starts in 2025 and is highly persistent but, given the uncertainties about the long-term outlook, not fully permanent (Section 5.1 presents sensitivity analysis with permanent tariffs).[12] Finally, the main scenario features no further fiscal policy action on the part of the US government, assuming that tariff revenues contribute towards public debt reduction (alternative fiscal policies are explored in Section 5.4).

3.1. THE TRANSMISSION OF TARIFF SHOCKS IN THE US

Direct effects of tariffs. Tariffs are a tax on imports that make foreign goods more expensive for US households and firms. The change in relative prices induces some substitution away from imports, which boosts aggregate demand for US-produced goods. This ***expenditure switching channel*** operates both via the consumption basket, as American households shift their demand from imported goods towards domestic ones, as well as on the production side, as US firms substitute imported intermediate inputs to relatively cheaper domestic value added (to produce *given* gross output).

This demand-boosting effect of tariffs is partially weakened in our model by several channels:

a) delayed import substitution: substitution between imports and domestic goods is sluggish in the short run, as captured by $\rho^{im} > 0$ in (2.5)-(2.6), making volume adjustments more rigid in the short run than implied by the long-run import elasticity σ_m;

b) liquidity-constrained households: as tariffs erode the purchasing power of incomes, this weakens the demand of constrained households not just for imports but also for domestically produced goods (***real income channel***);[13]

c) non-permanent tariffs: the expectation of declining tariffs makes them act as an intertemporal tax, which encourages postponing consumption, similarly to an increase in real interest rates (***intertemporal substitution channel***).

[11] More details about the tariff hike assumptions, and their cross-country patterns, can be found in Appendix D.

[12] Truly permanent tariffs would be expected to last forever, which is a consequential assumption in a model where forward-looking investors have perfect foresight. Uncertainty about trade policy beyond the term of the current administration could warrant the assumption of non-fully permanent shocks, with a quarterly autoregressive coefficient of 0.975. That said, our sensitivity analysis indicates that most qualitative messages of our analysis survive even with permanent shocks (for details see Section 5.1).

[13] Non-constrained "Ricardian" households anticipate tariff revenues to be (eventually) passed on to them in the form of tax cuts or transfers in the future, so their lifetime disposable income is not much affected, and they can smooth out temporary real income fluctuations. In contrast, liquidity-constrained households can only consume their current real income, which is lowered by the tariffs as long as tariff revenues are not *immediately* transferred back to households, or if these transfers are not distributed across households proportionally to tariff-induced purchasing power losses.

At the same time, tariffs also hurt aggregate supply by making domestic production more costly (***production cost channel***). On the one hand, higher tariffs erode the purchasing power of wages (to the extent households consume imported goods), which lowers labour supply as the opportunity cost of leisure falls (substitution effect).[14] On the other hand, by taxing imported intermediate inputs, tariffs raise domestic production costs further, amplifying the adverse effects on the supply of domestic gross output.

In summary, while higher tariffs shift US demand from imports towards domestic production, they also act as an adverse supply shock. This creates a demand-supply imbalance (see Figure 3.1, or Appendices B.1-B.2 for a stylised graphical illustration of these channels).

Indirect general equilibrium effects. As tariffs shift demand from imports towards domestically produced goods, this puts strain on the US economy's resources, leading to a 5.6% **terms-of-trade (ToT) appreciation**. The ToT-appreciation is reflected partly in higher inflationary pressures for domestic output prices, and partly in a stronger nominal exchange rate for the US dollar that is induced by monetary tightening in response to higher inflation. To a smaller extent, weakened US import demand puts downward pressure on the export prices of US trading partners which also contributes somewhat to the US ToT-appreciation.[15]

By making domestic goods more expensive relative to foreign ones, a stronger terms-of-trade offsets roughly a quarter of the direct effect of tariffs on relative prices, dampening the rise in *after-tariff* import prices faced by US consumers. Thereby, the ToT-appreciation is one of the general equilibrium mechanisms to bring demand back in line with (adversely affected) supply capacities. It operates via similar channels as the direct tariff effects:

a) ***expenditure switching (competitiveness):*** a stronger ToT makes US-produced goods less competitive both at home (against imports) and abroad (exports), constraining aggregate demand for American products;

b) ***real income channel (ToT-gain):*** by making imports cheaper, a stronger ToT raises the purchasing power of domestic incomes, generating a terms-of-trade gain that makes the American economy richer as a whole. This ToT-gain boosts the consumption demand of US households not just for imports but for domestically produced goods as well (which works against the expenditure switching channel).[16]

c) ***production cost channel:*** a stronger ToT raises the purchasing power of wages, which boosts labour supply by raising the opportunity cost of leisure (substitution effect). In addition, a stronger ToT makes

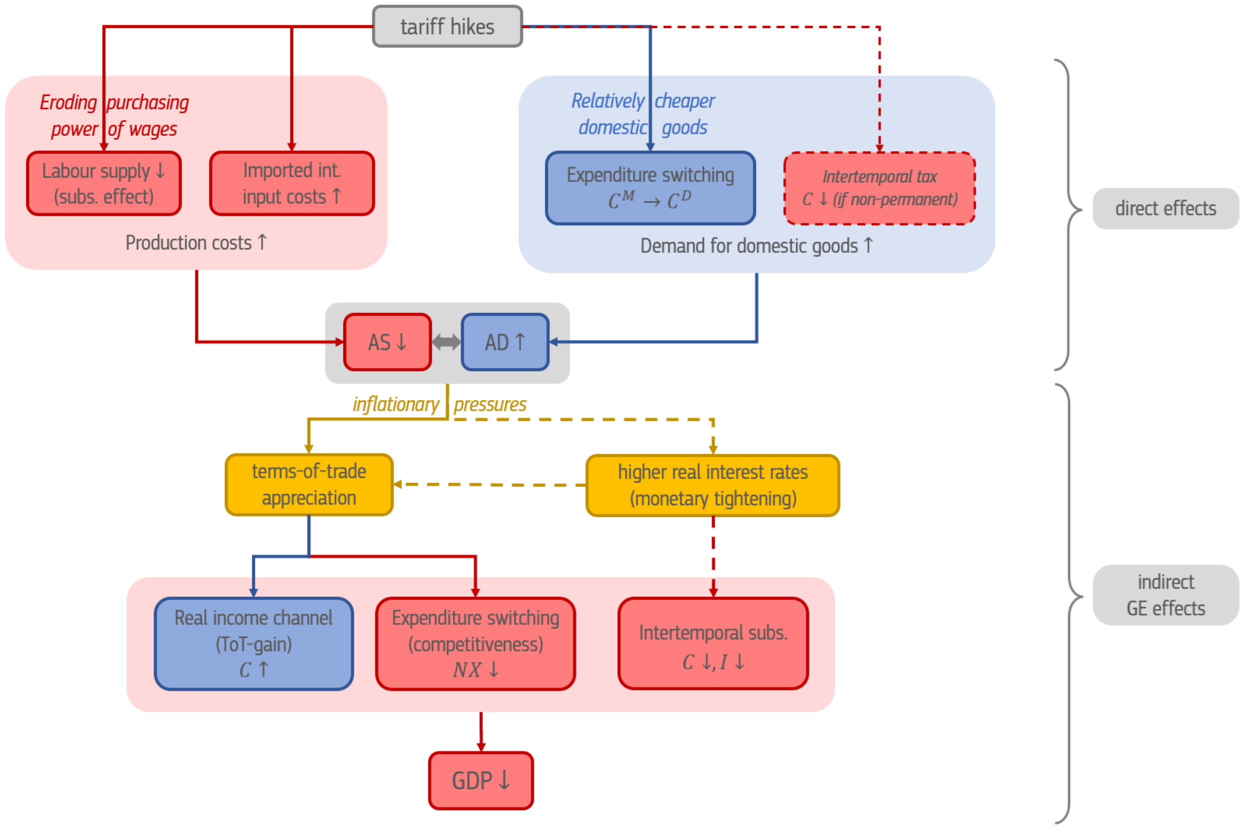
[14] In the case of tariffs, the opposing income effect from lower real wages – which would typically *increase* labour supply, as poorer households can afford less leisure – is offset by a lifetime income boost from higher expected net government transfers, funded by tariff revenues. That said, this income effect is missing only for unconstrained Ricardian households who can internalise the government budget constraint, while for liquidity-constrained agents the effect is there, mitigating the overall hit to labour supply (unless tariff revenues are immediately handed back to them in the form of transfers as in Section 5.4). In addition to falling real wages, another source of tariff-induced beneficial income effects on labour supply stems from intertemporal substitution if non-permanent tariffs are expected to decline over time: this makes Ricardian households postpone their consumption, thereby raising their labour supply (see (AS-CM) or (AS-gdp-CM), where perfect international risk sharing under complete markets approximates temporary shocks under incomplete markets).

[15] From the perspective of currency markets, tariffs lower US import demand which in turn reduces the net international supply of US dollars. With unchanged foreign capital flows (demand for US dollars), this puts appreciation pressure on the dollar exchange rate to bring the trade balance and US net borrowing back in line with foreign saving decisions. Of course, changing exchange rates also influence foreign saving decisions themselves.

[16] Notice that, unlike with the direct effect of tariffs, the real income channel via the ToT-gain operates not just for liquidity-constrained but also for Ricardian households, as the stronger ToT represents a lifetime income boost, a real transfer from foreigners that will not have to be "paid back". This channel is more pronounced the more persistent the terms-of-trade appreciation is, or the higher the share of liquidity-constrained households is.

imported inputs cheaper, further lowering production costs and boosting aggregate supply. At the same time however, by raising real incomes (both via real wages, and via savings on input costs) a stronger ToT affords households to consume not just more goods, but also more leisure, hurting labour supply (income effect). These forces should roughly offset each other as long as the intertemporal elasticity of substitution is around unity, making aggregate supply less responsive to terms-of-trade adjustments (see stylised illustration in Appendices B.1-B.2).

Figure 3.1: A stylised illustration of tariff transmission channels

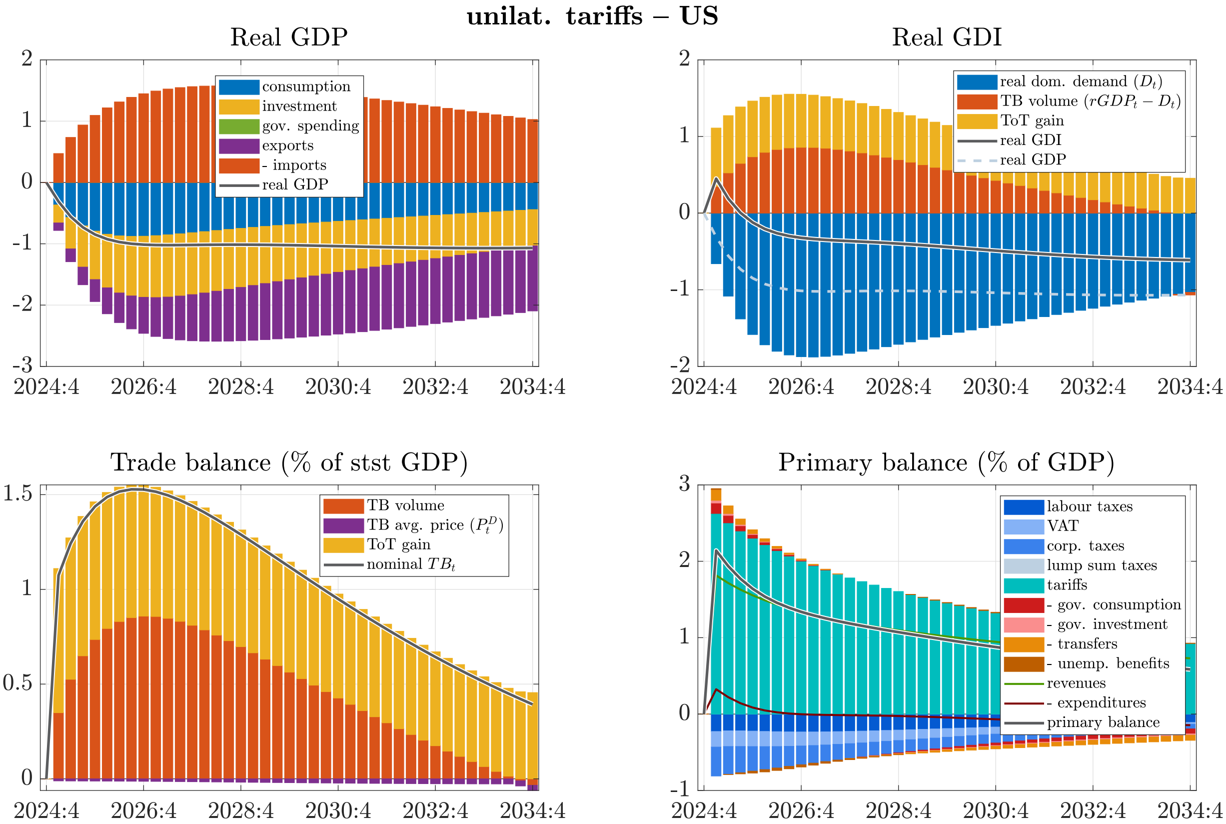


Note: Red (blue) nodes and arrows indicate contractionary (expansionary) channels, while yellow ones capture general equilibrium price adjustments. Dashed lines are operational only in certain cases: a) if tariffs are not expected to stay permanently; b) if monetary policy tightens in response to inflationary pressures. For simplicity, direct channels relating only to liquidity constrained households (real income channel on demand, and income effect on labour supply) are omitted, equivalently to the case when tariff revenues are handed back to them *immiediately*, proportionally to their tariff-induced purchasing power losses. The indirect effect on production costs from terms-of-trade changes is also omitted, given this is roughly zero if income and substitution effects on labour supply cancel out. For more details see Figure B.2. *Source:* European Commission services..

In addition to the terms-of-trade appreciation, **higher real interest rates** are another mechanism to restore equilibrium in the goods market. Under nominal rigidities central banks have some control over real interest rates (as well as over real exchange rate such as the terms-of-trade), which makes monetary policy an important determinant of the transmission mechanism for tariffs (as also highlighted by Monacelli (2025)). Our main scenario assumes the Federal Reserve follows in interest rate rule targeting CPI inflation, and therefore raises real interest rates in response to the inflationary pressures induced by tariffs. In turn, tighter monetary conditions (in addition to also contributing to the terms-of-trade appreciation via a stronger nominal dollar exchange rate) weigh on consumption and investment demand in the short term by encouraging saving (***intertemporal substitution channel***). Alternative monetary policy regimes are explored in Section 5.3.

GDP. On balance, the US economy is hurt by the tariff hikes, as falling exports and weaker domestic demand drive GDP 0.6-1.0% lower (see top left panel of Figure 3.2). Although tariffs shift the composition of demand away from imports and towards US production, the offsetting mechanisms discussed above reduce

Figure 3.2: Macroeconomic effects in the US – unilateral tariffs



Note: The top panels report real GDP (GDI) in %-deviation from baseline, with bars reflecting the contribution of expenditure volumes (to real GDP), and of the terms-of-trade gain (to real GDI), based on (2.29) and (A.26). Real GDI is the sum of real GDP and the terms-of-trade (ToT) gain. The bottom left panel shows the nominal trade balance (in pp-deviation from baseline, as a share of steady state GDP), with bars capturing contributions by trade volumes, average price level change and terms-of-trade (relative price) changes, based on (A.9). The bottom right panel shows the pp-deviation in the primary budget balance-to-GDP ratio, with contributions from various revenue and spending items. *Source:* European Commission services..

aggregate demand sufficiently so that the adverse supply side effects would prevail, and US economic activity would decline. A stronger terms-of-trade, in addition to limiting the shift away from imports, also makes US exporters less competitive abroad, thereby crowding out exports. In addition, higher real interest rates, as monetary policy reacts to higher inflation, reduce domestic demand for investment and consumption.[17] At the same time, the terms-of-trade gain from a stronger dollar provides some cushion for the fall in consumption demand, as it raises the purchasing power of domestic incomes – however, this is not sufficient to offset the negative effects from elsewhere.

3.2. WHO PAYS FOR THE TARIFFS?

Tariff revenues and government budget. While tariff hikes might not manage to stimulate US GDP, they can create additional fiscal space for the US government. Despite the tax base (i.e. the value of imports) declining in response to trade restrictions, higher tariff rates still raise additional tariff revenues, worth over 2% of GDP initially (see bottom right panel of Figure 3.2). The primary budget balance increases by somewhat less, given that other taxes (on consumption, and on labor and corporate income) endogenously decline as a share of GDP.[18] Since in this scenario no further fiscal policy action is assumed, these higher primary

[17] Recall, that this negative intertemporal substitution effect on consumption demand is amplified by the non-permanent nature of the tariff hikes. For a permanent tariff simulation see Section 5.1.

[18] This happens *despite* income tax rates remaining unchanged. With proportional income taxation and unchanged tax rates one might expect income tax revenues to stay constant as a share of GDP. However, as shown by (2.28), GDP is the sum of market incomes *and* taxes on products. And in this case tariff hikes cause product taxes to take up a larger share of GDP, necessarily lowering the

balances contribute towards reducing public debt (alternative fiscal setups are explored in Section 5.4, and in Figure C.2).

These tariff revenues represent additional resources for the US government, but this does not mean they also represent the same amount of additional resources for the US economy as a whole (i.e. that "foreigners pay for it"). Tariffs are technically paid by US firms and households who import foreign goods, so each dollar of extra tax revenue for the government *ceteris paribus* reduces the disposable income of the US private sector by the same amount, leading to just within-country redistribution. But of course, pre-tax private incomes also change in general equilibrium, which will determine how tariff revenues are paid for in an economic sense (tax incidence), and how much foreigners contribute to this.

Real income and terms-of-trade gain. The appreciating US terms-of-trade raises the purchasing power of incomes earned from domestic production, by making imports cheaper relative to US-produced goods. As shown by the top right panel of Figure 3.2, this raises the *real gross disposable income (real GDI)* of the US economy above its production volumes (real GDP), with the wedge between the two referred to as the terms-of-trade gain. The ToT-gain captures how much more the US economy as whole can spend beyond what is justified by its production, purely due to relative price changes vis-a-vis foreigners. In this sense, it represents a change in American incomes that is "paid by foreigners": via a stronger ToT the US extracts purchasing power gains from its trading partners.

Initially, the ToT-gain accrues mostly to the US government budget (beyond US exporters) in the form of additional tariff revenue, from where it can eventually be passed on to households as transfers or tax cuts. Due to the stronger ToT, tariffs raise the after-tax import costs of the US private sector by *less* than they raise the tax revenue collected by the government. Put differently, the real disposable income of American importers declines by less than the increase in fiscal resources (which will eventually raise the disposable income of US households), with the difference reflecting the part of tariff revenues that is paid by foreigners. This mechanism lies behind the familiar textbook illustration of the welfare effects of tariffs, shown in Appendix B.3 and Figure B.10.

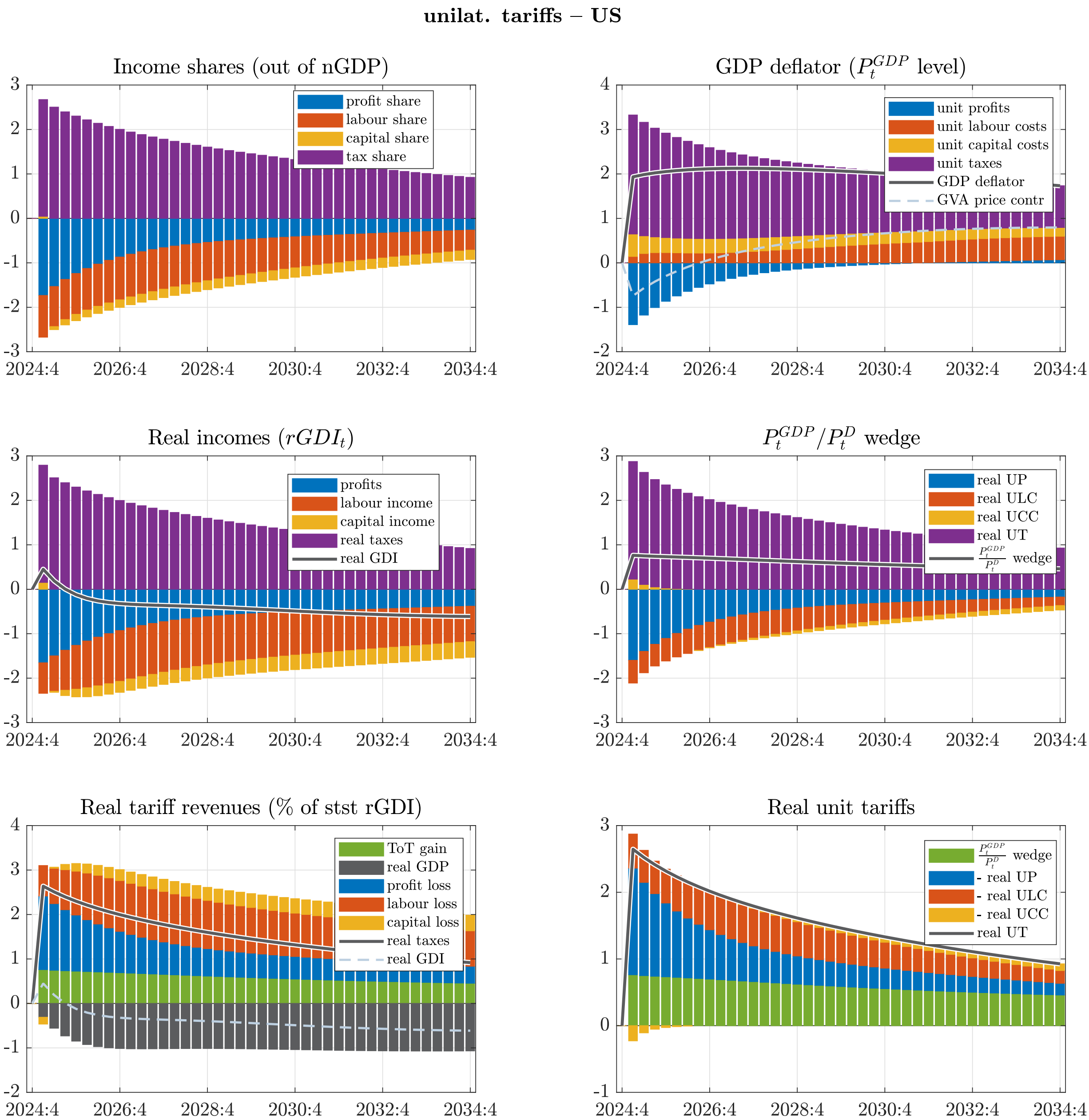
Figure 3.3 maps how the burden of US tariffs is distributed across various economic actors by looking at decompositions of US real GDI (with the underlying equations spelled out in Appendix A.5.2). Similarly to the income-side nominal GDP definition (2.28), real GDI can be written as the sum of real profits, labour and capital incomes, as well as real taxes on products – using the price index of domestic expenditures P_t^D to deflate each component. As the middle-left panel shows, the rise in real tariff revenues (purple bars) is offset by a fall in real private sector incomes for workers, capital and firm owners alike (blue, red, yellow bars), with the balance reflecting the change in the aggregate real GDI of the US economy as a whole. But as we have seen before, this same change in real GDI is the result of not just the ToT-gain, but also of falling real GDP. The part of the decline in US private real incomes that is due to falling production levels (real GDP) also entails less working hours or lower capital utilisation, and therefore shouldn't strictly be thought of as "paying for" higher tariff revenues in a direct sense.[19]

To control for changing production levels, in the middle right panel each component is divided by real GDP, arriving to real unit production costs (UP, ULC, UCC) and real unit taxes (UT). In line with the above-explained intution, this shows that the decline in real private incomes that is *not* due to falling production volumes (blue, red and yellow bars) is smaller than the increase in real unit tariffs (purple bars). The balance is the

share of labour and corporate incomes (see also the top left panel of Figure 3.3).

[19]Put differently, American workers have less money to spend not just because they have to pay more taxes to the government (or because ToT-changes altered the purchasing power of their wages), but also because they work less.

Figure 3.3: US real income decomposition – unilateral tariffs



Note: Panels show the decomposition of income between production factors and product taxes. Left columns focus on income *levels* (except for income *shares*), while right columns adjust for production volumes (dividing through by *$rGDP_t$*) to display *unit prices*. The first row looks at GDP, while the second row looks at real GDI (deflated by P_t^D). The third row is just a rearranged version of the second row such that bars add up to tariffs, showing how those are "paid for". In the bottom left panel, real GDI is split to real GDP and the terms-of-trade gain. Results are in terms of percentage point deviation from baseline. Underlying equations are (A.28), (A.34), (A.29), (A.32), (A.31), (A.33). Tariffs raise the GDP-deflator above the contribution by GVA prices. *Source:* European Commission services..

portion of tariff revenues that is covered by the ToT-gain, which is captured by the wedge between the GDP deflator P_t^{GDP} and the deflator for domestic expenditures P_t^D.[20] In other words, this is the part of tariffs paid by foreigners as they suffer real income losses from a stronger US terms-of-trade.

The bottom right panel of Figure 3.3 is a rearranged version of the middle right panel, where the contributions sum to real unit tariffs. This shows clearly that most of the increase in US tariffs is paid by domestic agents

[20] Note that both P_t^{GDP} and P_t^D are raised directly by tariffs, but their ratio moves together with the terms-of-trade. See (A.6) and (A.9). A stronger terms-of-trade pushes the price of domestic GVA (i.e. P_t^{GDP} cleaned of tariffs) upwards, while it *lowers* the pre-tariff price of expenditures that include imports.

as they suffer losses in real unit profits and labour incomes (blue and red bars), while only around a quarter of the tariff burden is paid by foreigners via a stronger US ToT eroding their purchasing power (green bars).

Therefore, while the terms-of-trade gain represents a very real transfer from abroad to the US economy as a whole, this transfer is not sufficient to cover all the increase in tariff revenues, which therefore requires domestic firms and households to endure lower real incomes, at least until the additional tariff revenues are handed back to them in the form of tax cuts or transfers. Recall, however, that the situation is even worse than that because US real GDI does not even grow by the full amount of the terms-of-trade gain: due to the adverse effects of tariffs on production, US real GDP falls so much that, despite a positive ToT-gain, it leads to a *lower* real GDI after the first few quarters (see top right panel of Figure 3.2). In other words, the real incomes of the US private sector decline not only because the stronger terms-of-trade does not fully offset the effect of higher tariffs on their purchasing power, but also because they produce less output with the American economy falling into a recession.

Inflation decomposition. The decline in US real unit profits and real unit labour costs is in part due to tariffs raising the price of expenditures P_t^D that erodes the purchasing power of *given* nominal profits and wages. Meanwhile, the rest of the change is due to nominal income prices (i.e. the nominal price of domestic gross value added, the GVA-deflator) themselves being affected. The top right panel of Figure 3.3 zooms into the these nominal changes.

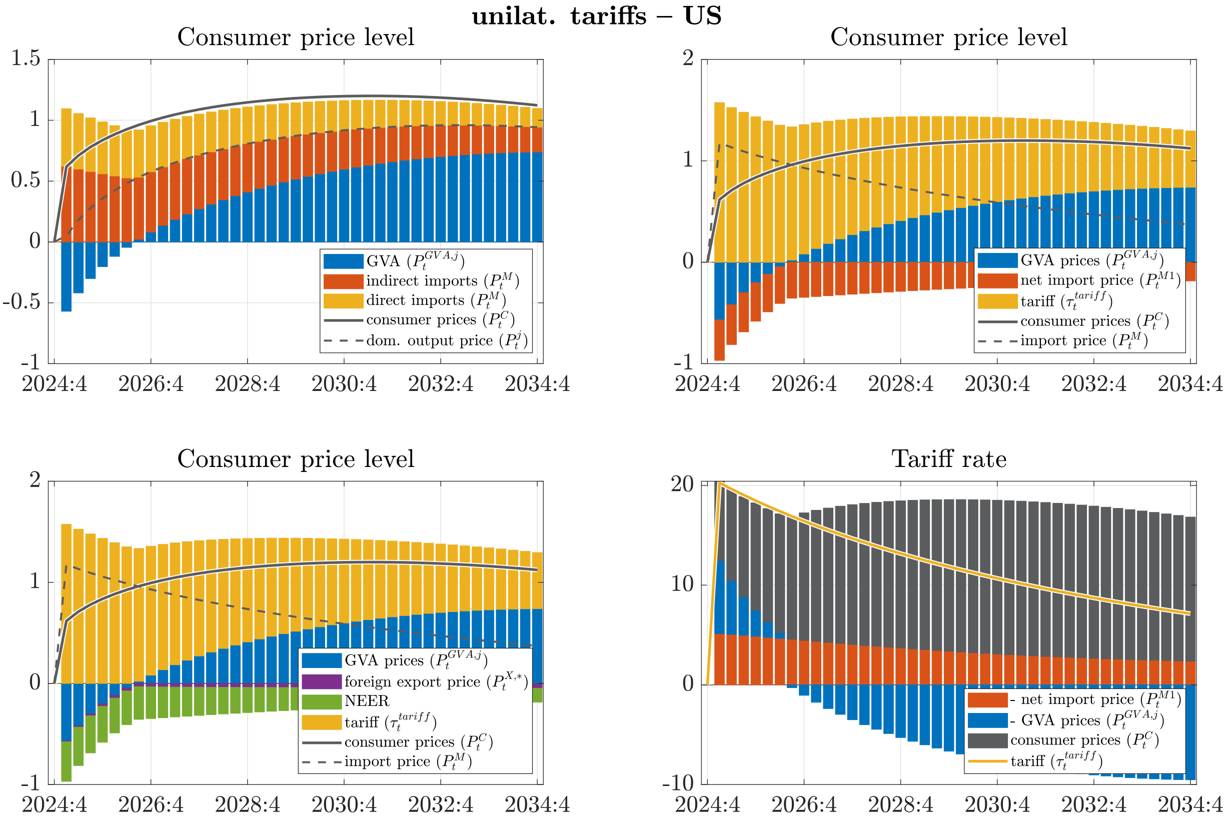
This shows that in response to tariff hikes, the unit profits of American firms decline (blue bars). This is mostly due to nominal rigidities, which prevent firms from fully passing on the increase in their production costs to their sticky final goods prices. As a result of this non-perfect pass-through, higher costs are absorbed by falling markups. In turn, production costs rise both because tariffs make imported intermediate inputs more expensive, and because domestic workers and capital owners demand higher wages and capital rental rates to try to compensate for their own tariff-induced purchasing power losses. The latter are themselves part of domestic value added, so higher unit labour costs and unit capital costs contribute positively to the GVA-deflator (red and yellow bars), even though the weakening economic activity exerts some downward price pressure. On balance, the GVA deflator initially declines, with falling unit profits dominating its dynamics, but as firms rebuild their markups and second round effects gain ground through rising unit labour costs, the nominal price of GVA gradually increases above its baseline level.

The top right panel of Figure 3.3 also illustrates how tariffs drive a wedge between the GDP deflator and GVA prices. This is made clear by the nominal GDP definition (2.26) which includes product taxes (such as tariffs) on top of domestic gross value added (Appendix A.6.3 provides further details). Therefore, despite GVA prices *declining* initially, the GDP deflator still increases as it is pulled upwards by higher unit tariffs.

An alternative approach to answer who bears the burden of tariffs is explored in Figure 3.4, which provides a decomposition of US consumer prices. Higher *after-tariff* import prices P_t^M raise consumer prices both directly, via imports of final consumption goods (yellow bars in the top left panel), as well as indirectly, through imported intermediate inputs used in domestically-produced consumption goods (red bars). This effect on the CPI is initially mitigated by lower domestic GVA prices, but as second-round effects gradually kick in and raise the GVA-deflator, consumer prices are pushed upwards also through their domestic value added content (blue bars).

The contribution to CPI from higher *after-tariff* import prices P_t^M can alternatively be broken down to the mechanical effect of the tariff hikes τ_t^{tariff} (yellow bars in the top right panel) and to declining *pre-tariff* import prices P_t^{M1} (red bars). The latter captures the already-discussed terms-of-trade offset, as a stronger ToT lowers pre-tariff import prices, thereby dampening the effect of tariffs on import prices faced by US

Figure 3.4: US CPI decomposition – unilateral tariffs



Note: The solid lines on the first three panels report the *level* of consumer prices P_t^C in %-deviation from baseline, with the bars reflecting *contributions* to this deviation by domestic value added and import content in the consumption basket. Underlying equations are (A.21), (A.22) and (A.23). The bottom right panel is a rearranged, rescaled version of the top right panel, showing the percentage point deviation in effective tariff rates τ_t^{tariff}, and how that is "paid for". Contributions by domestic output prices P_t^j are the sum of GVA and indirect import contribution terms, while contributions by import prices P_t^M are the sum of indirect and direct import terms, or of net import price P_t^{M1} and tariff rate τ_t^{tariff} terms. *Source:* European Commission services.

consumers (dashed line) by roughly a quarter. Most of this ToT-offset is driven by a stronger nominal USD exchange rate (green bars in the bottom left panel), as opposed to outright cuts in the export prices of trading partners in their own currency (purple bars).[21] But by having the price of their US-bound products fall in USD terms, foreigners bear part of the burden.

Through the lenses of the consumption basket, the full burden of effective tariff rate hikes is distributed as shown in the bottom right panel of Figure 3.4, born partly by

i) US consumers whose purchasing power is eroded by higher consumer prices (grey bars);

ii) US workers and firm owners in their capacity as domestic value added producers who see nominal GVA prices initially decline (mostly due to higher taxes on imported inputs eating into unit profits) (blue bars);

iii) foreigners who receive lower prices for their products in USD terms due to a stronger dollar (red bars).

Nominal GVA prices eventually do pick up (negative blue bars), but this also pushes consumer prices higher through their domestic value added content (grey bars). So the overall burden faced by US households is just reshuffled, and is manifested less through their declining nominal incomes and more through higher

[21] This is the result of the Producer Currency Pricing (PCP) assumption, making export prices sticky in the exporter's currency. The above decomposition would somewhat change under Dominant Currency Pricing (DCP), as explored in Section 5.2 and in Figure C.15, where export prices are sticky in USD.

consumption prices eroding their real income. As captured by the fall in net import prices due to the terms-of-trade offset (positive red bars), foreigners pay for roughly a quarter of the tariff hikes, in line with the previous analysis of $\frac{P_t^{GDP}}{P_t^D}$ decomposition.

Note that without the terms-of-trade offset, foreigners would not pay for any of the tariffs. Then it would just be a pure within-country redistribution between US households in their different capacities as taxpayers (who can expect tax cuts from higher budget revenues), consumers and workers/firm owners. And for roughly three-quarters of the tariff hikes, this is indeed the case.

3.3. EXTERNAL BALANCE AND TARIFFS

The tariff hikes reduce US trade deficits only temporarily. As discussed above, while tariffs shift demand away from imports towards US-produced goods, this puts strain on the US economy's resources, and the resulting terms-of-trade appreciation also crowds out exports by making them less competitive abroad. Therefore, the tariff hike reduces *gross* trade flows, lowering *both* imports and exports, with only a small effect on the net balance. This is a (partial) manifestation of the ***Lerner symmetry***, which points out that a tax on imports behaves similarly to a tax on exports, via general equilibrium price adjustments. In our model exports are taxed even in a more direct sense, since the imported intermediate inputs used during their production are also subject to tariffs.

That said, initially, import volumes decline more (by 6-11%) than export volumes (by 2-6%), implying a positive contribution to the trade balance (see bottom left panel of Figure 3.2, red bars). But this is less the result of expenditure *switching* induced by tariffs and instead follows more from the fall in overall consumption levels which mitigates the import leakage (expenditure *changing*). Lower consumption, in turn, is caused by adverse intertemporal substitution effects due to the non-permanent nature of tariff hikes and higher real interest rates.[22] On top of the volume component, the terms-of-trade gain raises the nominal trade balance further (yellow bars), as it reduces the import bill and boosts export revenues through changes in relative prices. Altogether, the tariff hikes raise the US trade balance by 1.3-1.5% of GDP in the first years. However, in the medium run, these effects fade as the fundamental drivers behind the US external balance – saving relative to investment – are not significantly altered by tariffs.[23]

The temporarily higher US trade balance also drives the current account balance higher, which in turn must be a reflection of a temporarily higher saving-investment differential in the US. This stems from lower domestic demand, i.e. spending on consumption and investment falling more than national income (blue bars below solid line in the top right panel of Figure 3.2). Another way to break down the higher current account balance is as higher government budget balances, driven by tariff revenues of around 2% of GDP, while the net lending

[22]This is confirmed in the alternative scenario with permanent tariffs (see Figure C.10, and Section 5.1), where the volume contribution to the trade balance would be *negative* (!), i.e. import volumes would fall less than export volumes. Without the adverse intertemporal substitution effects on consumption, and with the beneficial effects of a larger ToT-gain, US consumption would *increase*, supporting import volumes through a higher import leakage (expenditure *changing*).

[23]To see why tariffs are unlikely to have lasting effects on the trade balance, consider a simple static framework where the trade balance is just the difference between real income from production and consumption. For a higher trade balance, the saving rate in the economy should go up, but it is unclear why tariffs would achieve this on their own. Tariffs are not a tax on overall consumption, just on imports: they shift the *composition* of goods within the consumption basket, rather than reducing its *overall size* relative to real income by pushing up saving rates. Similarly, tariffs are not a subsidy to overall production, just to import-competing domestic producers, while exporters are crowded out – moreover, adverse supply side effects actually *hurt* overall production. The national saving rate could only go up if the government decides to save the tariff revenues (i.e. lower the fiscal deficit) on behalf of liquidity-constrained households who would have to cut their overall consumption (since they cannot smooth out the tariff-induced drops in their disposable income by anticipating future transfers or tax cuts).

balance of the private sector initially declines by a smaller amount due to the ToT-gain.

Effects on the US – taking stock. In light of the above analysis, US tariff hikes are unlikely to achieve most of their desired policy goals.

- Instead of **reviving US-based production (manufacturing jobs)**, tariffs would depress US real GDP, acting as an adverse supply shock and raising production costs (Section 3.1).
- Tariffs indeed **raise budgetary revenues**, but most of it is paid by American households, with foreigners bearing only a quarter of the burden, in the form of a US ToT-gain (Section 3.2).
- Instead of **eliminating the US trade deficit**, tariffs would raise the trade balance only temporarily and to a limited extent, as the long run drivers behind the US saving-investment imbalance remain unaffected (Section 3.3).

3.4. EFFECTS ON THE EU AND TRADE DIVERSION

Via trade linkages, unilateral US tariff hikes also affect the EU economy. The US is one of the EU's largest export markets, with its US-bound exports comprising nearly 3% of EU GDP. As American firms and households respond to tariffs by importing less, including from Europe, this lowers EU exports by 1.1-1.5%, which weigh on European economic activity (see top left panel of Figure 3.5, purple bars). The flipside of a stronger dollar is a weaker (more competitive) terms-of-trade rate for Europe. This does not only cushion the fall in EU exports but also reins in European import demand (red bars), inducing substitution towards domestically produced goods that mitigates the fall in GDP. At the same time, a depreciating ToT also erodes the purchasing power of European incomes that weighs on consumption demand (blue bars). On balance, EU GDP would be lowered moderately, by around 0.2%.

The terms-of-trade loss makes the EU poorer as a whole. As shown on the top right panel of Figure 3.5, the ToT-loss (yellow bars) pushes real gross disposable income lower than the fall in real GDP alone would imply, capturing how Europeans bear the burden of US tariffs.[24] As a result, EU real GDI declines by 0.4%.

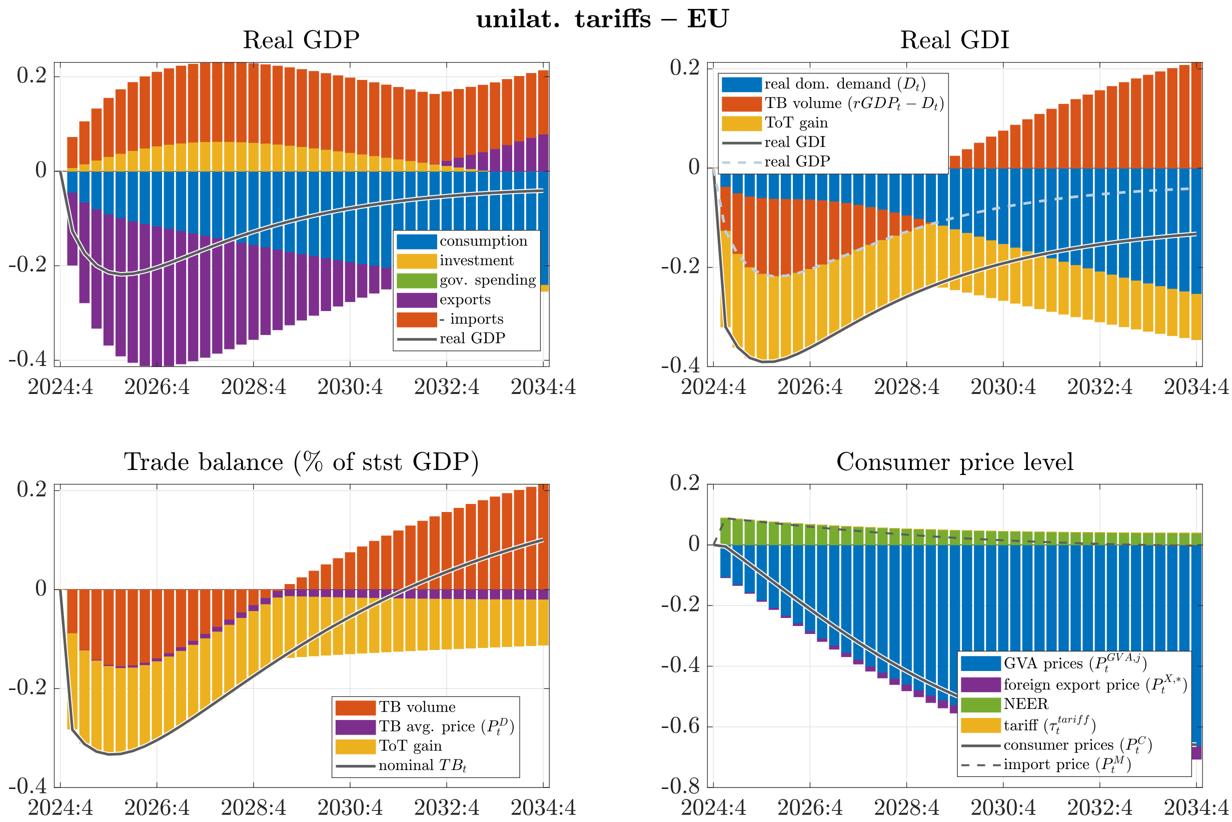
The EU's trade balance would decrease by about 0.3% of GDP in response to the US tariff hikes (see bottom left panel of Figure 3.5). In part, this is driven by an initially negative volume contribution (red bars), as exports fall more than imports. However, in the medium run exports gradually recover as they start responding to the more competitive terms-of-trade. In addition to the volume component, the EU trade balance is further weighed down by the ToT-loss (yellow bars), as a weaker euro raises the import bill.

Inflation in the EU declines only mildly, by around 0.1 percentage points (bottom right panel of Figure 3.5). Recessionary forces lower domestic inflationary pressures in Europe, pulling down GVA prices (blue bars). However, the euro exchange rate depreciation partly counteracts this effect via rising import prices (green bars). Responding to lower inflation and slower growth, monetary easing in Europe would provide some support to domestic demand.

Trade diversion – third country effects. Using a 6-region model allows us to zoom in behind the aggregate change in EU exports, and map how unilateral US tariff hikes affect global trade patterns (see Figure 3.6). The decline in EU exports is driven mainly by the shrinking American market *size*, as US firms

[24] Initially, most of the ToT-loss is reflected in lower real unit profits of European firms, since a weaker euro raises the cost of their imported intermediate inputs, which they cannot immediately pass on to their sticky final output prices. As markups are gradually rebuilt, however, the ToT-loss shifts to lower real unit labour costs, making European workers bear the burden of purchasing power losses induced by US tariffs (see middle left panel of Figure C.1).

Figure 3.5: Macroeconomic effects in the EU-27 – unilateral tariffs



Note: The top panels report real GDP (GDI) in %-deviation from baseline, with bars reflecting the contribution of expenditure volumes (to real GDP), and of the terms-of-trade gain (to real GDI), based on (2.29) and (A.26). Real GDI is the sum of real GDP and the terms-of-trade (ToT) gain. The bottom left panel shows the nominal trade balance (in pp-deviation from baseline, as a share of steady state GDP), with bars capturing contributions by trade volumes, average price level change and terms-of-trade (relative price) changes, based on (A.9). The bottom right panel shows the *level* of consumer prices P_t^C in %-deviation from baseline, with the bars reflecting *contributions* to this deviation by domestic value added and import content in the consumption basket, based on (A.23). *Source:* European Commission services.

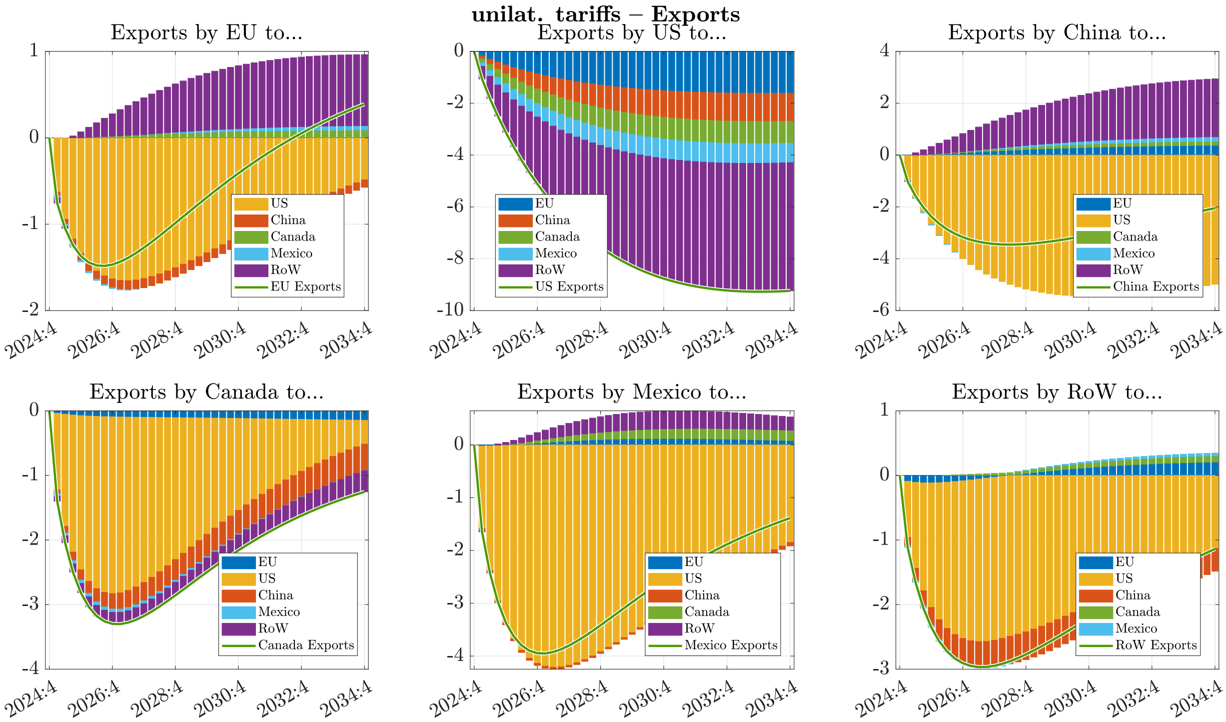
and households respond to tariffs by importing less overall, including from Europe (yellow bars in the top left panel). However, this effect is somewhat mitigated via two channels. First, as we have seen above, Europe's bilateral ToT vis-a-vis the US depreciates, improving the competitiveness of EU exporters. On the one hand, this mitigates the losses in the US market (i.e. the yellow bars would be larger without this). On the other hand, EU exporters gain market *share* in third countries at the expense of American exporters who become less competitive due to a stronger dollar and more costly imported inputs (purple bars).

Second, to the extent that the US targets China with asymmetrically higher tariffs, EU exporters would also gain market share in the US, at the expense of Chinese exporters whose products face steeper levies (smaller yellow bars than under symmetry). At the same time, this asymmetry also implies a larger ToT depreciation for China than for the EU, eroding the competitiveness of EU exporters in the Chinese market (see red bars), and slightly raising EU imports from China (blue bars in the top right panel).

Figure 3.6 also highlights the importance of exposure to the US market: despite a similar rise in tariff rates, the decline in US-bound exports is significantly smaller for the EU than for Mexico or Canada, who are more reliant on the US as an export destination.

In order to isolate how the trade diversion stemming from US tariffs levied on third countries affects Europe, Figure 3.7 splits the total effects into a part caused by tariffs directly targeting the EU (blue bars), and the spillovers to Europe due to US tariffs targeting *other* countries (red and yellow bars). Trade restrictions imposed on their exports by the US generate a depreciation pressure on the terms-of-trade of third

Figure 3.6: Trade patterns: export decomposition by destination markets – unilateral tariffs



Note: Solid lines report export volumes in %-deviation from baseline, with the *contribution* of various destination markets (EU, US, China, Canada, Mexico, Rest of the World) represented by bars, based on (2.24). *Source:* European Commission services.

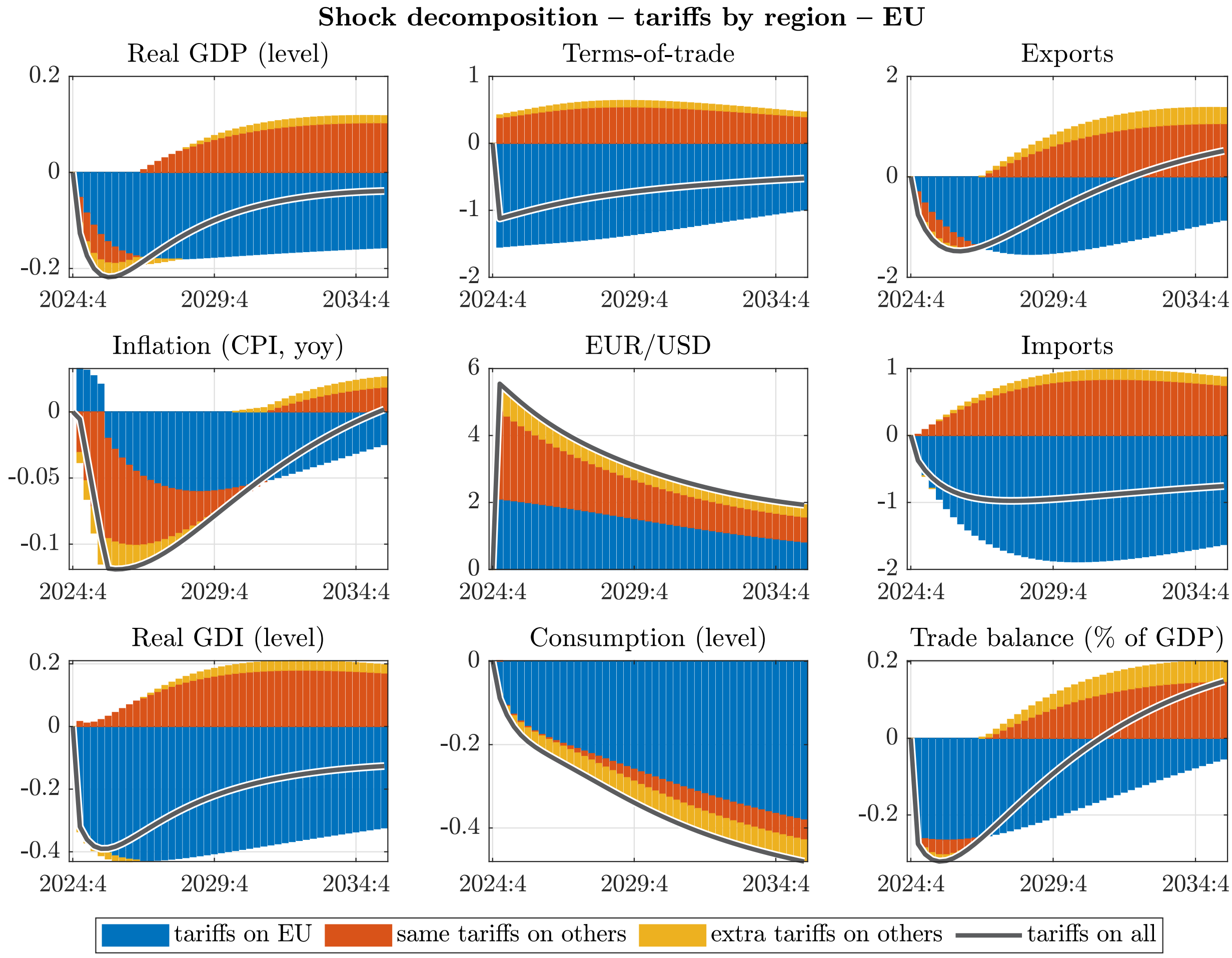
countries, the flipside of which is a relatively stronger effective ToT for Europe.[25] This implies the EU loses competitiveness against these other countries, as their products that were shut out of US markets become cheaper in search of new consumers. As a result, EU imports increase, and initially, European exporters are also hurt. In the first years, this weighs on the trade balance and also aggravates the fall in real GDP.

At the same time, the appreciating terms-of-trade generates a ToT-gain for Europe that raises real incomes. By putting downward pressure on import prices in the EU, a stronger real exchange rate contributes to lower CPI inflation, raising the purchasing power of European consumers. As the chart shows, in the absence of trade diversion EU inflation would even briefly increase as import prices would be higher (blue bars).

Finally, as delayed substitution gradually gets under way, European exporters gain market share within the US market, at the expense of third country exporters who are targeted with tariffs (while tariffs on EU goods are ignored in the scenario captured by red bars). As a result, EU exports pick up in the medium run, which eventually also supports EU GDP, making the trade diversion effects ultimately beneficial for the EU economy. Further details about the European spillovers resulting from US tariffs targeting non-EU countries *only* (i.e. the scenario captured by the sum of red and yellow bars) are shown in Appendix C.2.

[25] At the same time, US tariffs lead to an appreciating US ToT, which pulls the EU's ToT downwards (this is also reflected in the weakening EUR/USD nominal exchange rate). The *effective* ToT, however, is a trade-weighted average of the bilateral rates vis-a-vis all trading partners: since the rest of the world outweighs the US market within the total trade of the EU, the appreciation against third countries dominates the depreciation against the US.

Figure 3.7: Trade diversion: tariff shock decomposition for the EU – unilateral tariffs

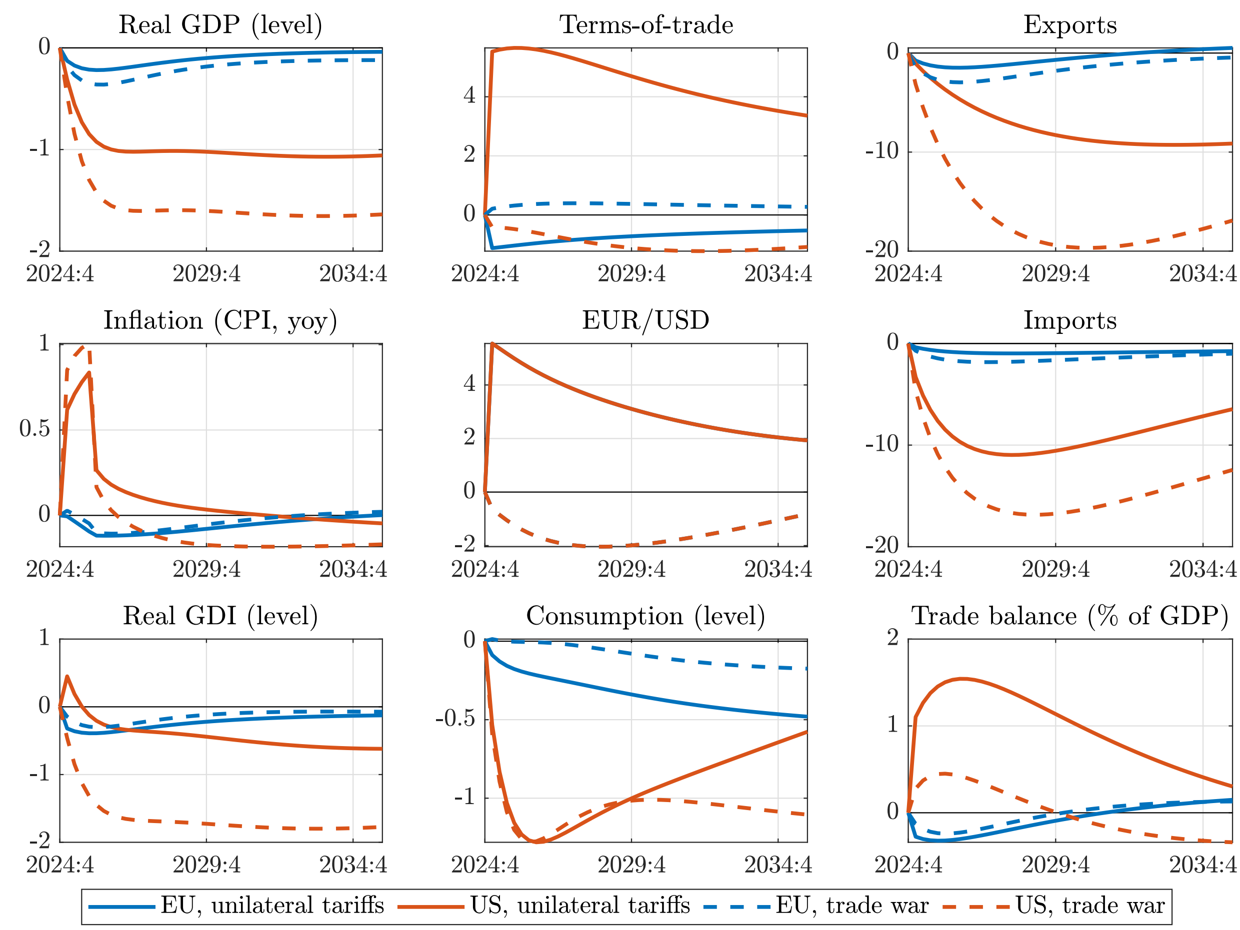


Note: In response to unilateral US tariff hikes, lines depict %-deviation of levels from a no-tariff baseline. For CPI inflation and the trade balance, lines depict percentage point deviations. Bars reflect *contributions* from tariffs levied directly on EU imports (blue bars), the same tariffs levied on imports from other US trading partners (red), and additional Apr 2 tariffs on others (yellow). *Source:* European Commission services.

4 ALTERNATIVE TARIFF SCENARIOS

4.1. TIT-FOR-TAT RETALIATIONS AND TRADE WAR

Figure 4.1: The effect of tit-for-tat retaliations by trading partners



Note: EU (blue) vs US (red) comparison. Lines depict %-deviation of levels from a no-tariff baseline in response to unilateral US tariffs (solid lines), or in a trade war scenario involving tit-for-tat retaliations by trading partners (dashed lines). For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

In an alternative scenario, illustrating a hypothetical "trade war", we simulate tit-for-tat retaliations by all trading partners on their imports from the US (see dashed lines in Figure 4.1).[26] Similarly to how US tariffs affected other countries, retaliations lead to deepening US GDP losses relative to the unilateral scenario (solid lines), mostly via weighing on the export demand of American firms. This mitigates the increase in the US trade balance, which is also driven lower due to the original US terms-of-trade gain being essentially eliminated.[27] In turn, through the combination of lower real GDP and a vanishing ToT-gain, coordinated global retaliations would hurt American real GDI markedly. This also means that in such a scenario, trading partners would manage to shift the burden of US tariffs entirely back to US households (see Figure C.6 and C.8).

However, the general result that tariffs hurt the economy of the country that implements them, applies also to retaliatory tariffs: retaliation deepens European GDP losses further, to 0.3-0.4% – despite somewhat

[26] Importantly, this is a hypothetical scenario. In reality, most retaliations were targeted and more contained than a one-for-one matching of US levies, especially for trading partners making subsequent deals with the US, such as the EU.

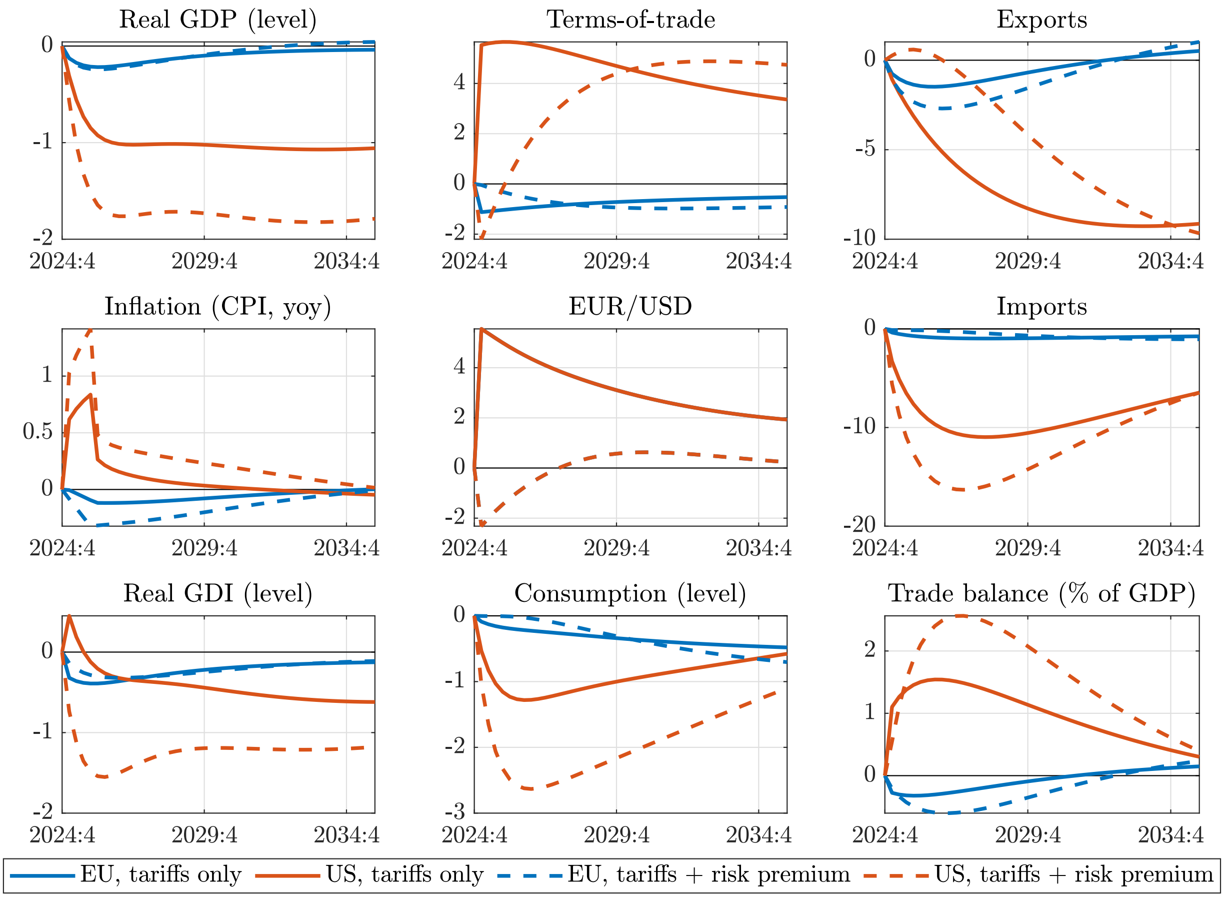
[27] Notice that in the retaliation scenario the US terms-of-trade actually *depreciates*, i.e. retaliations *more* than offset the original appreciation pressure stemming from unilateral US tariffs. The reason for this lies in the asymmetry between import and export elasticities, so that tariffs on imports and exports require different degrees of exchange rate offset to bring the goods market back in equilibrium. For more details see Jeanne and Son (2024).

mitigating the fall in the EU's trade balance, as tariffs shift demand from American imports towards EU-produced goods. At the same time, Europe's ToT-loss gets eliminated, so the EU's real gross disposable income (real GDI) declines somewhat less in this scenario, reflecting how the EU managed to shift the burden of US tariffs back to Americans. However, the burden from the EU's own new tariffs *is* born mostly by European households (see Figure C.7 and C.9).

As Figure 4.1 illustrates, the adverse effects on the US economy (both from the unilateral US tariffs as well as from the additional retaliations) are larger than on the EU. The reason for this is that tariffs affect the entirety of American trade flows, while for the EU (and other countries) they are levied only on their bilateral trade with the US. That said, retaliation by the EU and all other trading partners would be harmful for economic activity in the US, the EU and the rest of the world.

4.2. UNCERTAINTY AND US RISK PREMIUM SHOCKS

Figure 4.2: The effect of risk premium shocks on the US



Note: EU (blue) vs US (red) comparison. Lines depict %-deviation of levels from a no-tariff baseline in response to unilateral US tariff hikes (solid lines), with potential additional illustrative risk premium shocks on US financial assets and capital (dashed lines). For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

The above model simulations, which only consider the direct effect of tariffs, do not account for additional negative confidence effects, the extreme rise in global trade uncertainty, or expectations about adverse productivity effects in the US – all of which may hold back investment decisions and tighten financing conditions, further aggravating the adverse economic consequences. The reaction of financial markets following the April 2 tariff announcements (e.g. the unusual combination of a weakening dollar and rising US Treasury yields) seems to indicate such forces are highly relevant in the current context.

To qualitatively illustrate the implications of these additional channels, we simulate a stylised increase in the risk premium on USD denominated assets and US capital investments, *on top* of the unilateral US tariff hikes discussed above. In the absence of a comparable precedent, precise quantification is not feasible at this stage, so the exercise is purely illustrative.

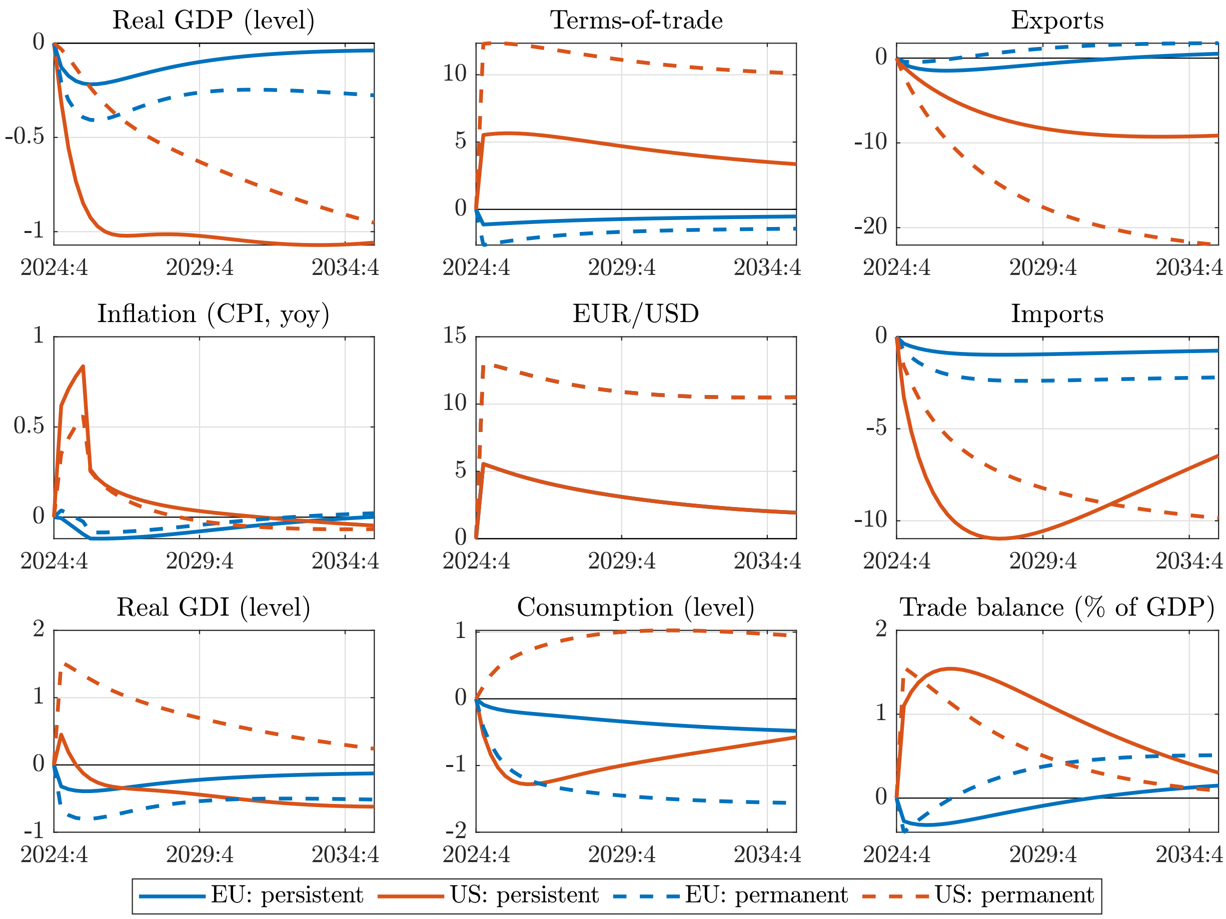
Higher risk premiums on US assets raise interest rates further and induce a dollar depreciation. Following the trade policy announcements of the US administration, such USD depreciation has more than offset the appreciation pressure that would be implied by unilateral US tariff hikes alone. However, as shown in Figure 4.2, despite a weaker *nominal* dollar exchange rate, the US *real* exchange rate (and terms-of-trade) is still expected to strengthen over time, only more gradually. While the more gradual terms-of-trade appreciation supports US net export volumes relative to a pure tariff shock, the higher interest rates induced by rising risk premiums dampen domestic capital accumulation and consumption demand – on balance deepening the decline in US GDP. Moreover, the smaller terms-of-trade gain lowers real gross disposable income even further. Therefore, although the US trade balance rises more in this scenario (due to both a more competitive terms-of-trade as well as lower domestic spending), it happens amid a more recessionary economic landscape.

For the EU, the consequences of the additional risk premium shocks are more contained, as long as the loss of confidence concerns primarily the US (as assumed here). The stronger euro weighs on the trade balance, but at the same time, investments also become relatively more attractive in Europe, leaving the GDP response similar.

5 SENSITIVITY ANALYSIS

5.1. TARIFF PERSISTENCE

Figure 5.1: The effect of tariff persistence – unilateral tariffs



Note: EU (blue) vs US (red) comparison. In response to unilateral US tariff hikes, lines depict %-deviation of levels from a no-tariff baseline with persistent tariff hikes (quarterly AR1 coefficient of 0.975, solid lines), or fully permanent tariff hikes (dashed lines). For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

In our main scenario we simulated highly persistent, but not fully permanent tariff hikes, in light of the uncertainties regarding US trade policies beyond the term of the current administration. In an alternative scenario we examine fully permanent tariffs (see Figure 5.1, dashed lines). The most consequential difference is that if tariffs are perceived to remain in place permanently, i.e. not expected to decline over time, they no longer act as an intertemporal tax on US consumption which encouraged saving in the persistent scenario via the intertemporal substitution channel. This would lead to a higher consumption path in the US. In addition, with permanent tariffs the US terms-of-trade would appreciate more than twice as much as under our persistent scenario, raising the purchasing power of American households and supporting US consumption demand further via the real income channel.

At the same time, a stronger ToT would erode the external competitiveness of the US even more, hurting exports and mitigating the fall in imports via the expenditure *switching* channel. Moreover, due to higher overall consumption, import leakage would also increase (expenditure *changing*). As a result, and unlike with persistent tariffs, exports would decline *more* than imports, creating a negative (!) volume contribution to the US trade balance and real GDP (see Figure C.10). Essentially, the ToT-gain-driven higher consumption would crowd out net export volumes. However, the larger American ToT gain would imply that the trade balance still increases, by a similar amount. On balance, the beneficial effects on consumption dominate the adverse effects on net exports, so the short-term economic outcomes for the US would be better, with the fall in real GDP being only half as negative as with non-permanent tariffs (and real GDI would even increase). Moreover, the larger US ToT gain also means that foreigners would now bear more than half of the burden of US tariffs, as their global purchasing power takes a larger hit (see Figure C.12).

Europe would experience the reverse of these effects, with the economic consequences being more negative under permanent US tariff hikes (see blue lines in Figure 5.1). This would entail a larger terms-of-trade loss, weighing on real incomes and consumption. [28] Although the weaker real exchange rate would *raise* net export volumes and the trade balance of the EU (see Figure C.11 and C.14), real GDP would still fall roughly twice as much (0.3-0.4%) as with non-permanent US tariffs.

5.2. INTERNATIONAL PRICING ASSUMPTIONS (DCP)

As shown by the exporter Phillips curve (2.12) and by (2.13), in our baseline setup of producer currency pricing (PCP) prices are sticky in the currency of the exporter (in $P_t^{X,f}$), and on top of this there is full exchange rate and tariff pass-through to import prices $P_t^{M,f} = \left(1 + \tau_t^{tariff,f}\right) \frac{\mathcal{E}_t}{\mathcal{E}_t^f} P_t^{X,f}$. This section explores the alternative of ***dominant currency pricing (DCP)***, where it is the pre-tariff US dollar price $\frac{P_t^{X,f}}{\mathcal{E}_t^f}$ that is rigid (see detailed derivation in Appendix A.7).

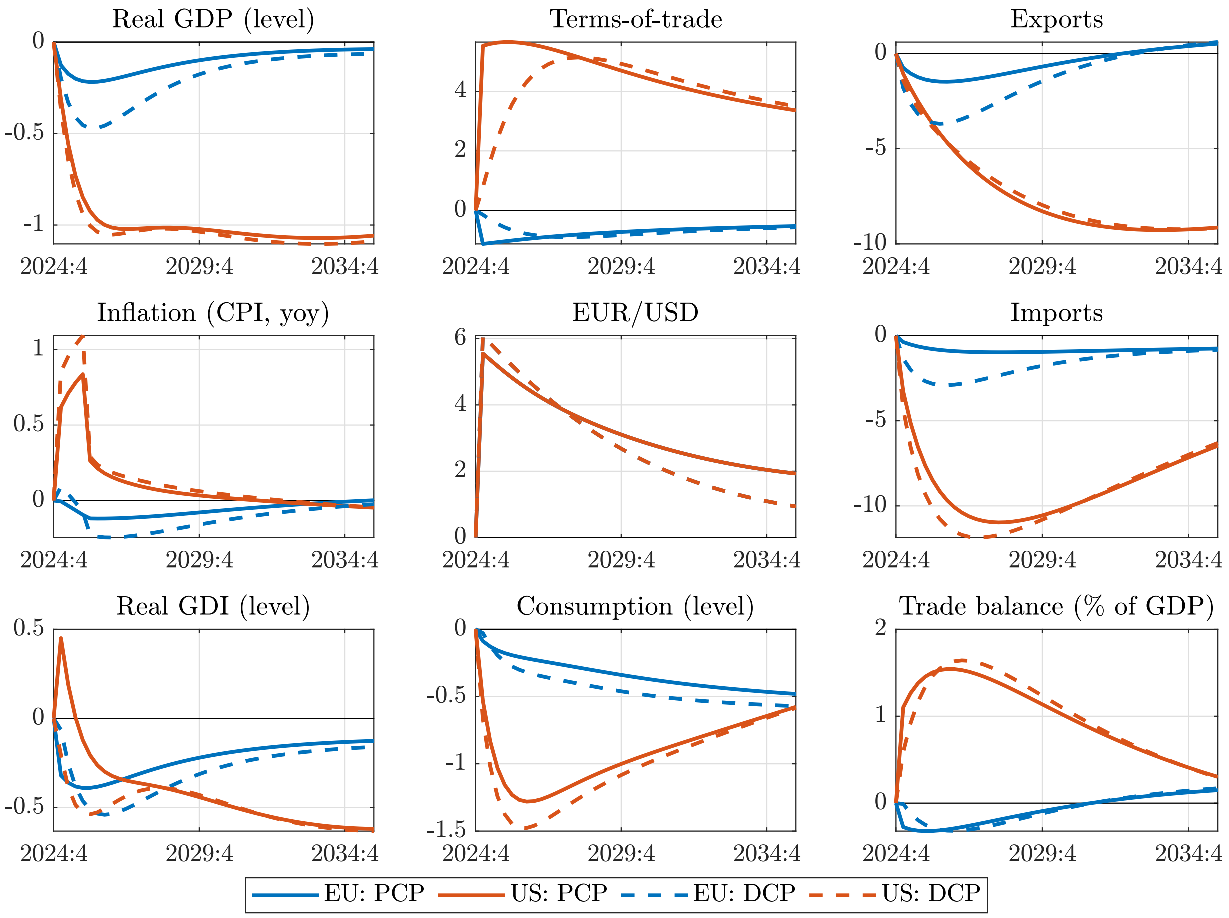
The most important implication of DCP is that it makes the terms-of-trade more rigid, as in Gopinath et al. (2020). From the perspective of America, a given nominal dollar appreciation no longer lowers pre-tariff USD import prices as quickly, given that those prices are sticky in USD terms. From the perspective of EU exporters, a dollar appreciation (i.e. a euro depreciation) allows them to raise their export prices in euro terms, mostly offsetting the rise in European import prices caused by a weaker euro. This means that, rather than jumping quickly through flexible nominal exchange rate changes, terms-of-trade movements occur more gradually through second-round domestic price pressures, as the relative demand shifts induced by US tariffs overheat the American economy and cool down foreign economies. Put differently, since import price rigidity is in the same currency across the world, flexible nominal exchange rate fluctuations across currencies can influence real exchange rates only to a smaller extent, and thereby DCP hinders global relative price adjustments.[29]

In turn, more rigid ToT adjustments have a harder time to offset the trade-shrinking effects of tariffs, so global trade declines more overall. With a smaller ToT depreciation, European exporters are less competitive relative to PCP, which weighs on EU exports and real GDP (see Figure 5.2). The income losses stemming from a more severe recession hurt consumption, which pull down imports as well (despite the relatively stronger ToT). A similar negative effect on imports is present also in third countries, which *amplifies* the hit to EU exports: as shown by Figure C.19, Europe would export less not just to the US, but initially also to other trading partners. This is in contrast to PCP, where the fall in EU exports was *cushioned* by selling more products to third countries, as European exporters gained market share in those places at the expense of more expensive US exporters. Conversely, with DCP the decline in overall trade (i.e. smaller market size of third countries) now dominates the market share-increasing effects of the (more rigid) ToT-depreciation

[28] EU consumption is also hurt relative to non-permanent tariffs by the absence of *beneficial* intertemporal substitution effects. As the expected decline of tariffs (an expected real depreciation) weighed on US consumption, so did the corresponding expected real appreciation support EU consumption. This channel is no longer there with permanent tariffs.

[29] More generally, as illustrated in Appendix A.7, for trade between non-US countries DCP means that it is only the exchange rate of the importer vis-a-vis the USD $\mathcal{E}_t$ (which is constant at 1 for the US) that fully passes through to local currency bilateral import prices $P_t^{M1,f}$, while the "remainder" of their bilateral exchange rate with the exporter $\mathcal{E}_t^f$ has no direct effect as that is swallowed by the exporter.

Figure 5.2: The effect of international pricing assumptions (PCP vs DCP) – unilateral tariffs



Note: EU (blue) vs US (red) comparison. In response to unilateral US tariff hikes, lines depict %-deviation of levels from a no-tariff baseline under producer currency pricing (PCP, solid lines), or dominant currency pricing (DCP, dashed lines). Details in Appendix A.7. For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

vis-a-vis the US.

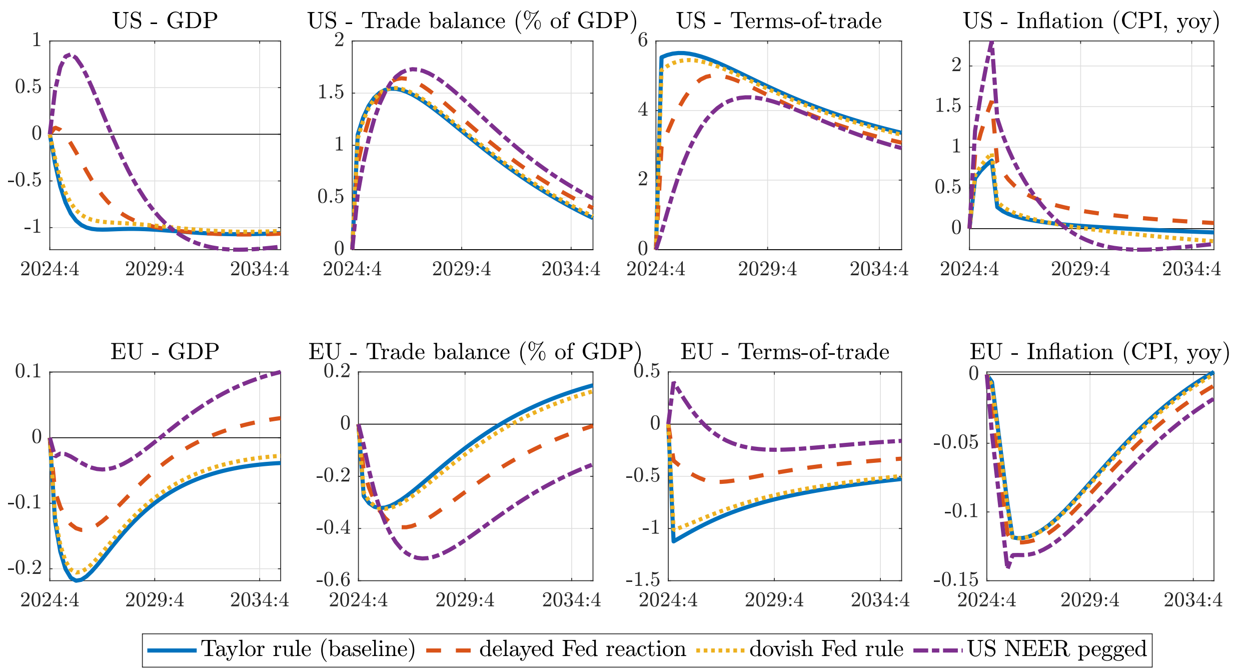
Beyond muting the competitiveness effect of ToT-adjustments through the expenditure switching channel, DCP also limits the effect of international relative prices on real incomes by reducing ToT-gains and losses. An initially markedly lower US ToT-gain now leads to a falling real GDI even in the short run. As shown in Figure C.17, this is mainly reflected in lower profits of US firms as a stronger dollar is no longer helpful in reducing the cost of imported intermediate inputs. The flipside of this is that under DCP foreigners bear a smaller share of the burden of US tariffs. EU exporters can now *raise* their prices in euro terms (to offset a stronger dollar), which pushes up their markups and leads to higher EU profits in the short run (see Figure C.18).[30]

5.3. MONETARY POLICY AND EXCHANGE RATE REGIMES

As discussed in Section 3.1, under nominal rigidities monetary policy is an important determinant of the transmission of tariff shocks via the intertemporal substitution channel. Relative to the baseline Taylor-type interest rate rule assumed in the main scenario, this section explores alternative US monetary policy strategies (see Figure 5.3). In the main scenario the Federal Reserve tightens monetary policy in response to

[30] Figure C.18 also shows how the EU's GVA-deflator is briefly positive (mainly due to the rising unit profits of European exporters). However, the contribution of GVA prices to EU CPI-inflation is still negative (Figure C.16), since exporter's prices have zero weight in the consumption basket.

Figure 5.3: The effect of alternative US monetary policies – unilateral tariffs



Note: In response to unilateral US tariff hikes, charts show the macroeconomic effects on the US (top row) and the EU (bottom row). Lines depict %-deviation of levels from a no-tariff baseline under different US monetary policy strategies. For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

inflationary pressures generated by the tariffs. In addition to hurting domestic demand, this policy response also channels the US terms-of-trade appreciation mostly into a stronger nominal dollar exchange rate, rather than into higher domestic prices, thereby containing the rise in CPI inflation (blue solid lines).

Alternatively, the Federal Reserve could adopt a relatively looser monetary stance, e.g. if it delays policy reaction by 6 quarters (red dashed lines), or by focusing on preventing nominal exchange rate appreciation amid fears that it would hurt the external competitiveness of US firms (NEER-peg, purple dash-dotted lines). Relative to the main scenario, these imply lower interest rates, amounting to an expansionary monetary policy shock that supports US domestic demand and mitigates the adverse consequences on GDP in the medium term – in the case of the exchange rate peg, US GDP even expands in the first few years. While these policies contain the nominal appreciation of the dollar, they also lead to higher domestic inflation, so the terms-of-trade would still appreciate, just more gradually. Therefore, the relative improvement in US competitiveness is temporary, but for a while it can support net export volumes. At the same time, more robust domestic spending implies a larger import leakage, resulting in a roughly similar outcome for the trade balance (which is also pulled down by a smaller terms-of-trade gain).

The spillovers from looser US monetary policy to Europe are positive (bottom panels). Relatively lower US interest rates induce a drop in European real interest rates as well,[31] which support consumption and investment demand in the EU, limiting the fall in GDP (intertemporal substitution channel). At the same time, lower US interest rates also lead to a relatively stronger terms-of-trade for the EU, which hurts the competitiveness of EU firms (expenditure *switching*), weighing on exports – although the import leakage from an overall more robust US demand somewhat mitigates the hit to European exporters (expenditure *changing*).

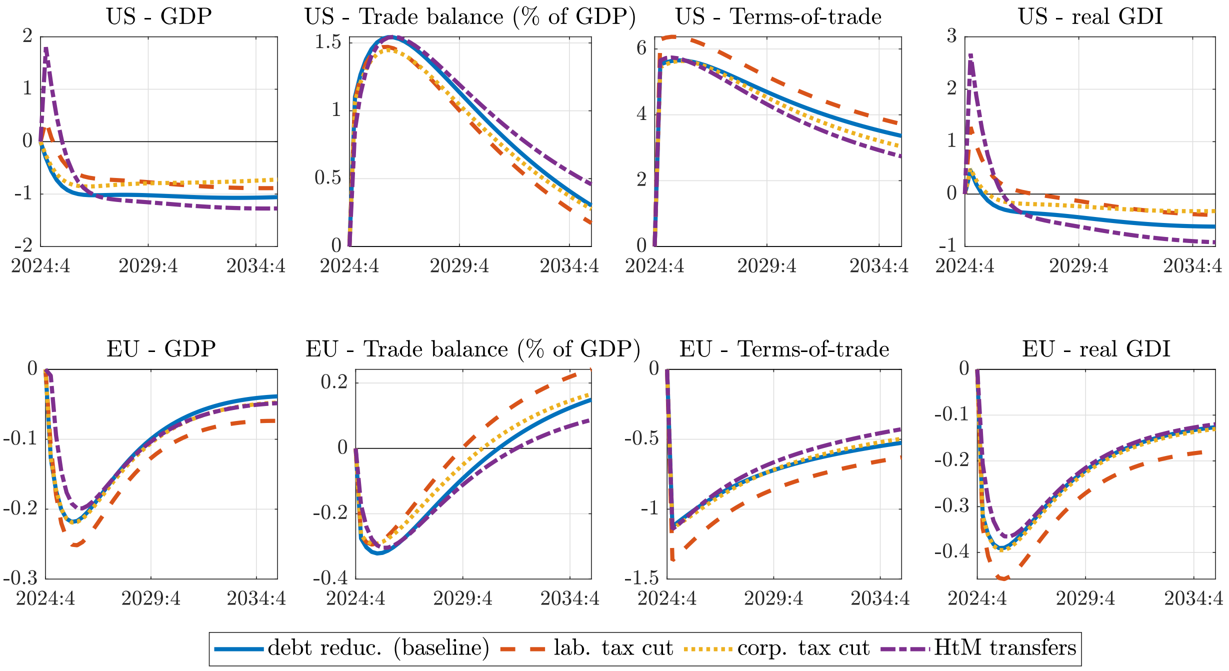
[31] One can think of this in terms of the real UIP condition, whereby a fall in foreign real interest rates is absorbed only partially by a stronger (CPI-based) real exchange rate, with the remainder of the effect lowering domestic real interest rates. This happens even without domestic monetary policy changing nominal interest rates at all, through higher inflation expectations. In turn, higher expected CPI inflation is the result of an expected exchange rate depreciation (following the initial jump to a stronger level) which would keep raising import prices back to their original level.

In addition, the higher ToT-gain provides a further boost to EU consumption by improving purchasing power (real income channel), and also implies that Europeans pay for US tariffs to a smaller extent.

In sum, looser monetary policy in the US would support domestic demand, propping up economic activity in the medium run, but at the expense of higher inflation, and shifting more of the burden of tariffs to US households. The spillovers to the GDP of trading partners are likely to be positive.

5.4. FISCAL POLICY – USE OF TARIFF REVENUES

Figure 5.4: The effect of alternative US fiscal policies – unilateral tariffs



Note: In response to unilateral US tariff hikes, charts show the macroeconomic effects on the US (top row) and the EU (bottom row). Lines depict %-deviation of levels from a no-tariff baseline under different US fiscal policies, which spend the extra tariff revenues on transfers or tax cuts in a balanced budget way. For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

The increase in import tariffs raises additional revenue for the US government that amounts to more than 2% of GDP (as already discussed in Section 3.2, see also Figure C.2). This additional fiscal room from tariff revenues can be used in various ways. This section explores how the timing and nature of these fiscal reactions matter for the macroeconomic impact.

Debt reduction. In the main scenario dicussed so far (blue solid lines in Figure 5.4), we assumed no further US fiscal policy reaction beyond the tariff hikes, which therefore lead to an almost 2 percentage point increase in the primary budget balance, contributing towards reducing public debt (see top row of Figure C.2). The rising primary balance implies a fiscal consolidation, the burden of which falls mostly on US firms and households as tariffs erode their purchasing power and lower their real incomes – while foreigners only "pay for" a quarter of tariffs in the form of a terms-of-trade gain to the US (see Section 3.2). While this fiscal consolidation creates some negative demand effects, it does so only to the extent that tariffs reduce the purchasing power of liquidity-constrained households – others tend to smooth out the fall in their real income, already anticipating lower taxes or higher transfers in the future. Instead, most of the negative GDP effects we have seen are caused through other channels, like the adverse supply side distortions generated by tariffs or general equilibrium price adjustments (see Section 3.1), that are *not* related to the size of the budget balance.

Tax cuts. Alternatively, tariff revenues can be used to finance cuts in labour income taxes, making it a budgetarily neutral reshuffling between different types of taxes (see middle row of Figure C.2). Relative to the main scenario, this would lower supply side distortions in labour markets (and raise the demand of liquidity constrained workers), that would somewhat mitigate the fall in GDP (red dashed lines in Figure 5.4). However, the difference is rather small since, due to much larger tax bases, labour income tax rates could only be lowered by a few percentage points while also respecting budget neutrality.

Transfers. Using tariff revenues (and the ToT-gain embodied in them) to finance transfers to US households would raise their disposable income. However, relative to the main scenario, this supports aggregate demand only to the extent that such transfers reach liquidity-constrained households (as the rest of the households were already anticipating these payouts). Therefore, transfers explicitly targeted at constrained households can achieve a large GDP boost (purple dash-dotted lines in Figure 5.4). That said, the improvement is relatively short-lived, as this amounts to a persistent redistribution across households, that eventually reduces the consumption of non-targeted groups.

EU spillovers. The spillovers to the EU from these US fiscal policy changes compared to the main scenario are small (bottom row of Figure 5.4). On the one hand, higher global interest rates induced by more robust economic activity in the US would constrain domestic demand in Europe (intertemporal substitution channel), which would also be weighed down by lower purchasing power due to a correspondingly weaker EU terms-of-trade (real income channel). On the other hand, a slightly more competitive ToT could boost net export volumes (expenditure *switching*), while EU exporters would also enjoy import leakage from stronger spending in the US (expenditure *changing*).

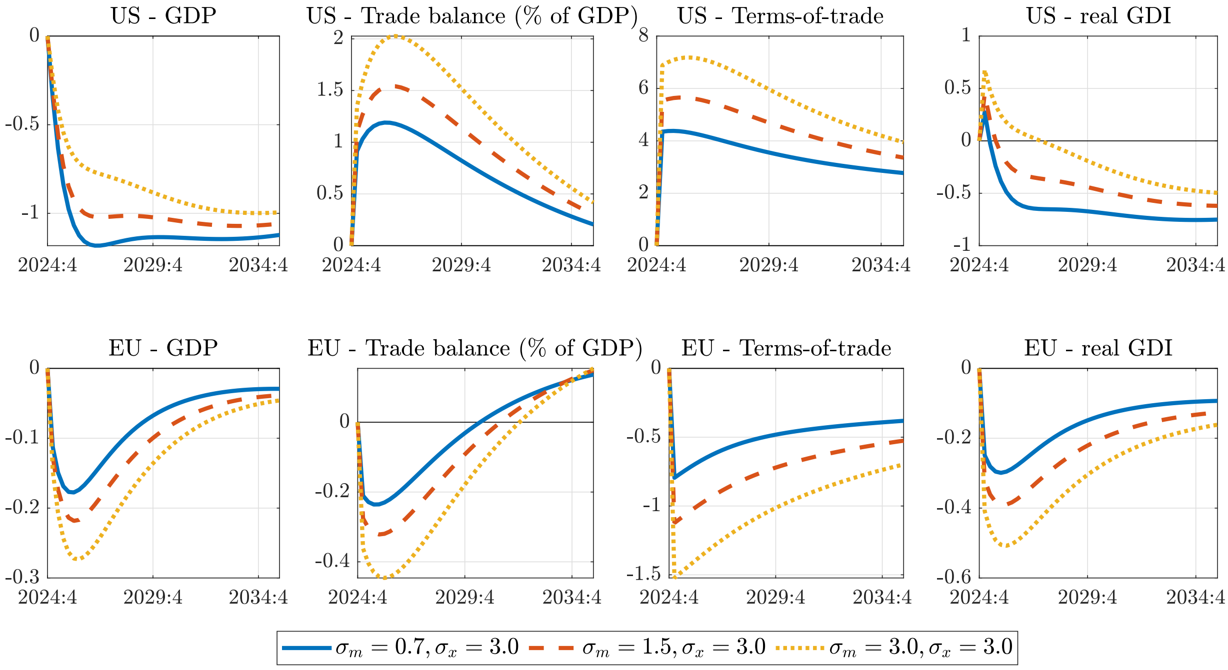
5.5. TRADE ELASTICITIES

The economic effects of tariffs depend fundamentally on how sensitive demand is to international relative prices – after all, tariffs work precisely by altering these prices. However, aggregate trade elasticities are subjuect to considerable uncertainty, which is why this section provides a stylised sensitivity analyis, aiming to illustrate the mechanisms through which alternative trade elasticities would affect the results. This analysis highlights that distinguishing between import and export elasticities is crucial, as they tend to affect GDP outcomes in opposite directions. Moreover, the impact of each elasticity on GDP can switch sign, depending on how trade elasticities interact with other features of the economy.

Import elasticities. A higher import elasticity of substitution, i.e. higher σ_m in (2.4), improves the ability of US tariffs hikes to shift demand from imports towards domestically produced goods (expenditure switching channel). While this leads to a sharper drop in US import volumes, tariffs also constitute a more forceful demand shock for American products, which entails a larger offsetting terms-of-trade appreciation in general equilibrium. At the same time, a relatively stronger ToT hurts the competitiveness of US exporters, so exports also decline more. On balance, with a higher σ_m tariff hikes lead to more positive response in the US trade balance (but with gross flows declining more), and the adverse effects on real GDP are mitigated (see Figure 5.5).

The EU experiences the reverse of these effects (bottom row in Figure 5.5). The sharper drop in overall US imports associated with a higher σ_m entails a larger hit to the EU's external demand, which makes European exports decline even more, and leads to a larger offsetting ToT depreciation. A weaker ToT, in turn, constrains EU imports as well, shifting demand more towards relatively cheaper European goods and cushioning the negative effects on aggregate demand. On balance, the marginal effect of a higher import elasticity for

Figure 5.5: The effect of import elasticities σ_m – unilateral tariffs



Note: In response to unilateral US tariff hikes, charts show the macroeconomic effects on the US (top row) and the EU (bottom row). Lines depict %-deviation of levels from a no-tariff baseline under different elasticities of substitution for imports σ_m, with the export elasticity kept constant at $\sigma_x = 3$. For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

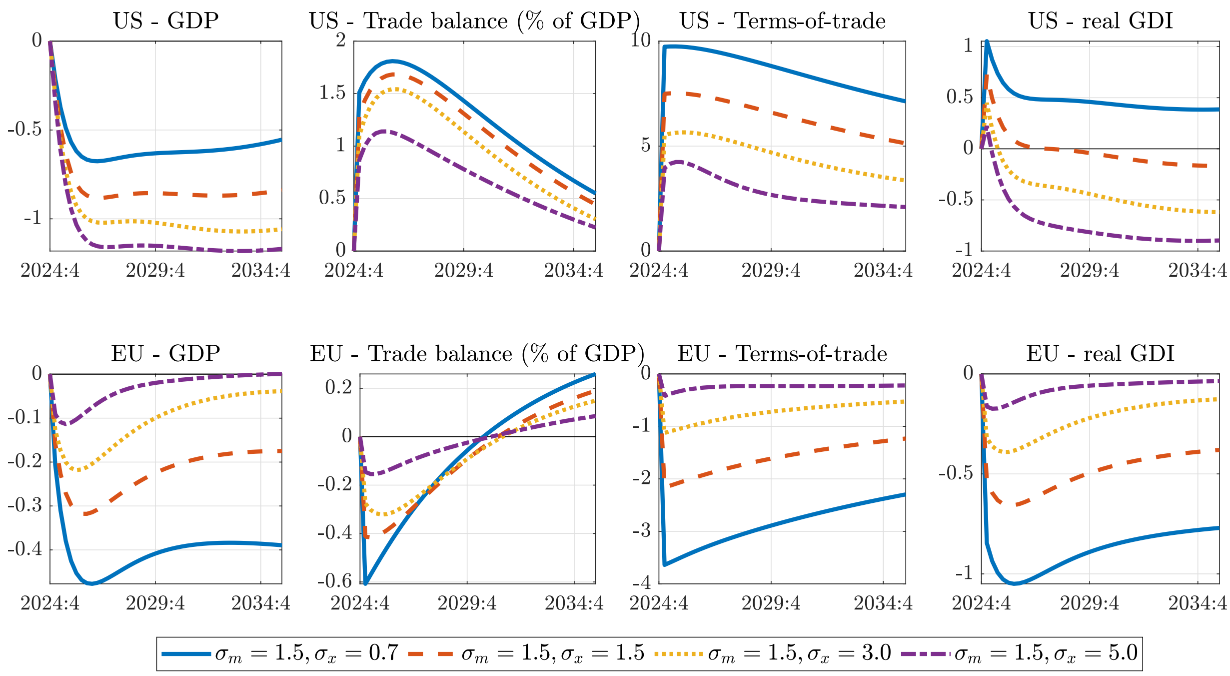
Europe is to deepen real GDP losses and to amplify the decline in the trade balance. On a global level, higher σ_m reduces gross trade flows.

Export elasticities. A higher elasticity of substitution *within* the import basket across different trading partners of a given country, i.e. a higher σ_x in (2.7), makes the exports of these trading partners more sensitive to terms-of-trade fluctuations (expenditure switching channel). Therefore, a *given* US ToT appreciation stemming from tariff-induced demand shifts, hurts American exporters in foreign markets more under higher σ_x, cooling down aggregate demand for US goods more efficiently. In turn, restoring goods market equilibrium then requires a *smaller* ToT appreciation to begin with, which leads to a smaller offset of tariffs on import prices, amplifying the fall in US import volumes as well. On balance, and in contrast to the case of σ_m, a higher σ_x deepens US real GDP losses, and mitigates the increase in the trade balance (see Figure 5.6).

For Europe a higher export elasticity operates via two opposing channels. On the one hand, the flipside of the smaller US ToT-appreciation is a *smaller* ToT-depreciation for the EU, which makes European exporters relatively less competitive in the US market (this is reflected in the above-discussed larger drop in US imports). A less weak EU ToT also leads to smaller drop in EU imports. On the other hand, the higher sensitivity to a *given* ToT-depreciation means that it is easier for EU exporters to gain market share against more expensive US exporters in third countries (this is reflected in the above-discussed larger drop in US exports). On balance, the second channel dominates, so with higher σ_x European exports decline less, and GDP losses are mitigated (see bottom row of Figure 5.6). In other words, easier substitution within traded goods baskets implies that US-tariff hikes reduce US gross trade flows with the rest of the world to a larger extent, while all other countries would trade relatively more with each other. [32]

[32] Figure C.3 provides impulse responses to unilateral US tariffs with more combinations of import and export elasticities. This yields a wider range of economic effects, but even with the more extreme and less realistic elasticity values, the qualitative messages outlined in this paper would still hold.

Figure 5.6: The effect of export elasticities σ_x – unilateral tariffs



Note: In response to unilateral US tariff hikes, charts show the macroeconomic effects on the US (top row) and the EU (bottom row). Lines depict %-deviation of levels from a no-tariff baseline under different elasticities of substitution for exports σ_x, with the import elasticity kept constant at $\sigma_m = 1.5$. For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

Interactions. It is important to note that the above results are valid with other parameters of the model remaining at their baseline calibration. The marginal effect of raising trade elasticities depends on other features of the economy, and can even switch sign in some cases. For instance, if the relative size of the US and third countries were different, then the balance of the above-discussed two competing channels for the EU (when raising σ_x) could also be reversed. In addition, the supply-side characteristics of an economy, which determine how aggregate supply is affected by ToT fluctuations, could interact with the effect of trade elasticities on aggregate demand.

The latter point is illustrated in Appendix B.2 in the context of a simple flexible price, incomplete market model. Depending on the balance between income and substitution effects on labour supply, the slope of aggregate supply with respect to the terms-of-trade (Φ_S^* in (AS-gdp-IM)) can be either positive, negative, or zero. Then, although raising the import elasticity σ_m would indeed entail a larger shift of aggregate demand in response to tariffs (Λ_τ^* in (AD-gdp-IM)), the resulting larger ToT-appreciation could leave aggregate supply (and with it, the actual GDP response) higher, lower, or the same (see Figure B.7). In the latter case trade elasticities would not matter for the equilibrium GDP response at all, and would just affect how much the equilibrating ToT-appreciation has to be.[33]

[33] Figure B.7 also confirms that the margainal effects of raising σ_m and σ_x have opposite signs. The simple AS-AD framework, as in Figure B.1 illustrates the reasons behind this nicely. Raising σ_m increases the tariff-induced outward shift in the AD curve, while also making the slope of the AD curve flatter (as demand becomes more sensitive to ToT, depending on openness): the "shift effect" dominates the "flattening effect", which *boosts* the GDP response if the AS curve has a positive slope. In contrast, raising σ_x only makes the slope of AD flatter, but does not affect its shift in response to tariffs: with a positively sloped AS, the same rightward shift of a flatter AD curve leads to a *smaller* GDP outcome.

6 CONCLUDING REMARKS

The simulations presented in this paper focus on the dynamic propagation of tariff shocks through international *macroeconomic* channels and trade linkages. In addition to these, however, there are other aspects of the tariff hikes, not captured by our model, which must be born in mind when interpreting the results.

Productivity effects. One important caveat is that the simulations do not consider productivity losses stemming from the slowdown in global trade flows. Less efficient exploitation of comparative advantages amid a more fragmented world economy could significantly reduce the *microeconomic* "gains from trade" that accrue due to specialising in certain sectors – this could exacerbate the negative economic consequences further. In addition, by shielding domestic-oriented firms from foreign competition, and hurting usually more productive export-oriented companies, tariffs can have adverse composition effects on average productivity in the US economy. For the analysis of such channels models with heterogeneous firms, firm entry and innovation would be more suitable.

Sectoral heterogeneity. A related limitation is that the simulations have focused on universal tariffs across the board, applying the rise in effective tariff rates to the single generic traded good in the model. This approach cannot distinguish between specific goods and sectors *within* a given country (only between goods *across* different countries). Therefore, to the extent there are product-specific or differentiated tariffs, the aggregate macroeconomic effects from our simulations could hide heterogeneous outcomes across sectors.[34] That said, higher sectoral granularity is perhaps less relevant in the current context when the US administration levied similar tariffs on the majority of imported goods.

Distributional effects. Another relevant factor, not sufficiently captured by our two-agent model, is household heterogeneity. As pointed out by Clausing and Lovely (2024), US tariff hikes amount to a sharply regressive tax policy change, the burden of which falls disproportionately on lower-income American households, given that they spend a higher share of their disposable income on (imported) consumption goods. Additional distributional effects could arise from different sensitivity of income and wealth to aggregate price movements, which could be better analysed by heterogeneous agent New Keynesian (HANK) models.

Optimal tariffs, welfare. Furthermore, the analysis presented here is descriptive in nature, and does not aim to assess optimal tariff policy from a normative perspective. Despite the efficiency losses stemming from declining international trade, the welfare-maximising tariff rate for a large country might be positive due to the terms-of-trade gain it can extract from its trading partners that raises real incomes per working hour. Relative to the traditional optimal tariff literature, which focuses on efficiency losses stemming from adverse productivity effects in the long run, dynamic New Keynesian models like QUEST capture the cyclical GDP losses and inflation costs due to demand shifts and nominal rigidities, which could lower the optimal tariff rate compared to traditional estimates. In addition, as pointed out by Itskhoki and Mukhin (2025), another channel lowering the optimal US tariff rate could be negative valuation effects from dollar appreciation

[34] In a similar vein, while the results suggests moderate aggregate effects on the EU as a whole (which is modeled as single region), the impacts on specific Member States could differ substantially, depending on their varying trade linkages with the United States (e.g. Ireland, Germany, Netherlands), or their different exposures to asymmetrically targeted sectors (e.g. car manufacturing, pharmaceuticals vs tourism).

realised on non-zero external USD debt.[35]

Geoeconomic fragmentation. Finally, the rise in trade protectionism might reverse the degree of global integration and push the world towards increased geoeconomic fragmentation – the impact of which is not shown in the simulations, and could be highly negative. Higher trade policy uncertainty, technological decoupling, or deglobalisation via reduction of international labour and capital flows would likely result in further global GDP losses.

What can Europe do? Section 4.1 presented scenario with tit-for-tat tariff retaliation, which concluded that this policy is economically costly, even if it manages to increase pressure on the US. While this might be beneficial for strategic reasons, there are other policy options for the EU, not explored in this paper, that might potentially mitigate the economic pain itself. Just as *raising* tariffs is harmful, by *lowering* trade barriers and pursuing trade deals with third countries, the EU could alleviate the negative effects stemming from US protectionism. In a similar vein, deepening the integration of the Single Market (e.g. in services or in capital markets) would also boost intra-EU trade that could counter the effects of rising external trade barriers. In addition, stronger domestic demand in Europe can help in offsetting the negative economic effects of the tariffs, as it could compensate for weakened external demand in utilising our productive capacities. Fiscal policy can play a role in this, by raising spending on growth-enhancing public investments – while paying attention to fiscal sustainability. More robust domestic demand could also reduce the reliance of the EU economy on external surpluses, and contribute towards a more balanced current account.

US current account rebalancing. Three of the main policy goals often mentioned in connection with US tariff hikes are to bring back manufacturing jobs, raise budgetary revenues paid by foreigners, and to close the wide US current account deficit. As analysed in detail in Section 3, tariffs have a limited ability to achieve these goals. What was not explored, however, is whether US policy *should* aim for more manufacturing or current account rebalancing to begin with.

First, while trade openness could have contributed to a sharper decline in US manufacturing jobs than elsewhere in the world, most of this decline is a global phenomenon, likely to be driven by technological improvements and structural change. Modern economies are becoming service economies, as higher manufacturing productivity affords countries to devote less of their time to produce goods, while also raising aggregate living standards (similarly to what happened with agriculture after the industrial revolution). Consequently, erecting trade barriers is unlikely to reverse this trend and resurrect manufacturing. But even the part of manufacturing jobs that *were* lost to foreigners, is mostly due to patterns of comparative advantages, meaning it must be beneficial for the US as a whole. While the gains from international trade were most certainly not evenly distributed, compensating the losers and addressing the serious inequalities among American households is better handled with redistributive policies than with regressive trade restrictions that also lower employment.

Second, and more fundamentally, to the extent international trade *is* behind US manufacturing decline, it is due to trade openness in general and not to macro-level *deficits* in particular. A current account (CA) deficit means that a country earns less income from its production than the value of its consumption (and investment). While in the short run larger CA-deficits can indeed be a reflection of a cyclical fall in production,

[35] This adverse NFA valuation channel is reversed in our model, since the steady state calibration focuses on matching trade balances to the data. With positive steady state interest rates, matching the observed US trade deficit necessarily requires a *positive* net foreign asset position in steady state – standing in contrast with observed net US external liabilities. This creates an inevitable trade-off between getting trade balances or NFA positions right, with the choice depending on the focus of the analysis.

over longer horizons the economy tends to converge to full employment, utilising its productive capacities (unless it is stuck in a liquidity trap). A persistent CA-deficit then is just a result of higher US consumption, and not of lower production or lost jobs. Consequently, closing these deficits is unlikely to increase the number of jobs.

Reducing CA-deficits might be desirable for other reasons. That the US can afford to consume persistently beyond their income, in exchange for cheap credit from abroad, is supported by foreigners' willingness to hold large amounts of US financial liabilities.[36] However, if eventually there is a limit to the amount of USD assets the world is willing to accumulate, then the pressure to run smaller CA-deficits would increase. But even in that case, this would require consuming less: rather than creating trade barriers, reducing the large US fiscal deficit is likely to be a more successful policy in raising national saving and balancing the current account (Obstfeld, 2025).

[36] A related debate is whether the US CA-deficit (i.e. overconsumption) is caused by Americans *willing* to consume more ("pulling in" credit, enabled by their exorbitant privilege), or by foreigners wanting to save more, demanding more dollar assets, which the US *must* accommodate by consuming and borrowing more ("pushed" by global savings glut, "exorbitant burden"). Obstfeld (2025) and Kumhof, Rungcharoenkitkul and Sokol (2020) argues the pull factors were more relevant than the push factors.

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Appendix

A FURTHER MODEL DETAILS

A.1. I/O TABLE

	INT_t^{NT}	INT_t^{TD}	INT_t^{X}	$C_t + G_t + IG_t$	I_t^{NT}	I_t^{TD}	X_t	adj_t
Y_t^{NT}	$INT_t^{NT,NT}$	$INT_t^{NT,TD}$		$C_t^{NT} + G_t^{NT} + IG_t^{NT}$	$I_t^{NT,NT}$	$I_t^{NT,TD}$		adj_t^{NT}
Y_t^{TD}	$INT_t^{TD,NT}$	$INT_t^{TD,TD}$	X_t	$C_t^{TD} + G_t^{TD} + IG_t^{TD}$	$I_t^{TD,NT}$	$I_t^{TD,TD}$		adj_t^{TD}
Y_t^{X}							X_t	
M_t	$INT_t^{M,NT}$	$INT_t^{M,TD}$		$C_t^{M} + G_t^{M} + IG_t^{M}$	$I_t^{M,NT}$	$I_t^{M,TD}$		
GVA_t^{labour}	$w_t N_t^{NT}$	$w_t N_t^{TD}$						
$GVA_t^{capital}$	$r_t^{NT} K_t^{NT}$	$r_t^{TD} K_t^{TD}$						
GVA_t^{profit}	$profit_t^{NT}$	$profit_t^{TD}$	$profit_t^{X}$					
	Y_t^{NT}	Y_t^{TD}	Y_t^{X}					

Table A.1: I/O table for QUEST (prices $P_t^{NT}, P_t^{TD}, P_t^{M}, P_t^{X}, P_t^{H}$ are supressed for brevity, and $w_t \equiv (1 + \tau_t^{ssc})\, W_t,\ r_t^j \equiv i_t^{K,j} P_t^{I,j}$)

- resources in row headers, uses in column headers
- rows represent the ***resource constraints*** for each type of good:
  - NT, TD : (2.22), M : (2.23), and for the further differentiated export good: $Y_t^X = X_t$
  - in each row, separate items represent the different *uses* of that good...
  - ...which must horizontally sum up to its total supply (*resources*): domestically produced gross output Y_t^j or total imports M_t
  - adding up the first 4 rows (with the relevant prices of each type of good, $P_t^{NT}, P_t^{TD}, P_t^{M}, P_t^{X}$) yields the aggregate nominal resource constraint of the economy (A.5)
- columns represent the ***CES baskets*** for different expenditure uses C_t, G_t, I_t^j... etc: (2.14)+(2.15)
  - in each column, separate items represent the different *types* of goods (NT, TD, M) that are needed for assembling the bundle for that particular use...
  - ...which must vertically sum up to the total bundle (shown in the header of the column) – in nominal terms (i.e. by applying the relevant prices which are now omitted for brevity)
  - in the first three columns, we can also add domestic value added (including labour and capital income as well as retailer profits): then, vertically summing these extended columns, we get gross output, representing the ***production technology***.

Steady state shares. We can rewrite Table A.1 in terms of steady state shares, such that the columns add up to 1:

Using parameters from:

- CES bundles (for each expenditure type $H \in \{C, G, IG, I^{NT}, I^{TD}, INT^{NT}, INT^{TD}\}$): (2.1) + (2.4)
- manufacturing technology (for each sector $j \in \{NT, TD\}$): (2.8)+(2.9)
- retailer markups in NKPC (for $j \in \{NT, TD, X\}$): (2.11), (2.12)

	INT^{NT}	INT^{TD}	INT^{X}	$C+G+IG$	I^{NT}	I^{TD}	X
NT	$\eta^{NT}\, s_{in}^{NT} \left(1 - s_{INT,NT}^{T}\right)$	$\eta^{TD}\, s_{in}^{TD} \left(1 - s_{INT,TD}^{T}\right)$	0	$\left(1 - s^{T}\right)$	$\left(1 - s_{I,NT}^{T}\right)$	$\left(1 - s_{I,TD}^{T}\right)$	0
TD	$\eta^{NT}\, s_{in}^{NT}\, s_{INT,NT}^{T}\, \left(1 - s^{M}\right)$	$\eta^{TD}\, s_{in}^{TD}\, s_{INT,TD}^{T}\, \left(1 - s^{M}\right)$	η^{X}	$s^{T}\, \left(1 - s^{M}\right)$	$s_{I,NT}^{T}\, \left(1 - s^{M}\right)$	$s_{I,TD}^{T}\, \left(1 - s^{M}\right)$	0
X	0	0	0	0	0	0	1
M	$\eta^{NT}\, s_{in}^{NT}\, s_{INT,NT}^{T}\, s^{M}$	$\eta^{TD}\, s_{in}^{TD}\, s_{INT,TD}^{T}\, s^{M}$	0	$s^{T}\, s^{M}$	$s_{I,NT}^{T}\, s^{M}$	$s_{I,TD}^{T}\, s^{M}$	0
GVA^{labour}	$\eta^{NT}\, \left(1 - s_{in}^{NT}\right) \alpha$	$\eta^{TD}\, \left(1 - s_{in}^{TD}\right) \alpha$	0				
$GVA^{capital}$	$\eta^{NT}\, \left(1 - s_{in}^{NT}\right) (1-\alpha)$	$\eta^{TD}\, \left(1 - s_{in}^{TD}\right) (1-\alpha)$	0				
GVA^{profit}	$1 - \eta^{NT}$	$1 - \eta^{TD}$	$1 - \eta^{X}$				
	NT	TD	X				

Table A.2: I/O steady state share parameters in QUEST (columns summing to 1)

Now partition the above table into 5 separate matrices $\mathbf{A^d}, \mathbf{A^m}, \mathbf{A^{gva}}, \mathbf{F^d}, \mathbf{F^m}$, while also compressing GVA terms by sector:

$$
\left[\begin{array}{cc}
\underbrace{\begin{bmatrix}
\eta^{NT}\, s_{in}^{NT} \left(1 - s_{INT,NT}^{T}\right) & \eta^{TD}\, s_{in}^{TD} \left(1 - s_{INT,TD}^{T}\right) & 0 \\
\eta^{NT}\, s_{in}^{NT}\, s_{INT,NT}^{T}\, \left(1 - s^{M}\right) & \eta^{TD}\, s_{in}^{TD}\, s_{INT,TD}^{T}\, \left(1 - s^{M}\right) & \eta^{X} \\
0 & 0 & 0
\end{bmatrix}}_{\mathbf{A^d}} &
\underbrace{\begin{bmatrix}
\left(1 - s^{T}\right) & \left(1 - s_{I,NT}^{T}\right) & \left(1 - s_{I,TD}^{T}\right) & 0 \\
s^{T}\, \left(1 - s^{M}\right) & s_{I,NT}^{T}\, \left(1 - s^{M}\right) & s_{I,TD}^{T}\, \left(1 - s^{M}\right) & 0 \\
0 & 0 & 0 & 1
\end{bmatrix}}_{\mathbf{F^d}} \\
\underbrace{\begin{bmatrix}
\eta^{NT}\, s_{in}^{NT}\, s_{INT,NT}^{T}\, s^{M} & \eta^{TD}\, s_{in}^{TD}\, s_{INT,TD}^{T}\, s^{M} & 0
\end{bmatrix}}_{\mathbf{A^m}} &
\underbrace{\begin{bmatrix}
s^{T}\, s^{M} & s_{I,NT}^{T}\, s^{M} & s_{I,TD}^{T}\, s^{M} & 0
\end{bmatrix}}_{\mathbf{F^m}} \\
\underbrace{\begin{bmatrix}
\left(1 - \eta^{NT}\, s_{in}^{NT}\right) & 0 & 0 \\
0 & \left(1 - \eta^{TD}\, s_{in}^{TD}\right) & 0 \\
0 & 0 & \left(1 - \eta^{X}\right)
\end{bmatrix}}_{\mathbf{A^{gva}}} &
\end{array}\right] \tag{I/O}
$$

Table A.3: Steady state share matrices in QUEST

Domestic value added vs import content. The ***import content of domestic demand*** components can be calculated as follows:

- $\mathbf{M}^{\mathbf{direct}}_{1\times4} = \mathbf{F}^{\mathbf{m}}_{1\times4}$
- $\mathbf{M}^{\mathbf{indirect}}_{1\times4} = \mathbf{A}^{\mathbf{m}}_{1\times3}\left(\mathbf{I}_{3\times3} - \mathbf{A}^{\mathbf{d}}_{3\times3}\right)^{-1}\mathbf{F}^{\mathbf{d}}_{3\times4}$
- $\mathbf{M}^{\mathbf{total}}_{1\times4} = \mathbf{M}^{\mathbf{direct}}_{1\times4} + \mathbf{M}^{\mathbf{indirect}}_{1\times4}$

where the indirect import content is calculated as in Baccaro and Hadziabdic (2024), based on the following relations, with $i, j \in \{NT, TD, X\}$ and $k \in \{(C+G+IG), I^{NT}, I^{TD}, X\}$:

$$\begin{aligned} y_{i,k} &= \sum_{j\in\{NT,TD,X\}} a^d_{i,j}\, y_{j,k} + f^d_{i,k} \\ \mathbf{Y} &= \mathbf{A^d}\,\mathbf{Y} + \mathbf{F^d} \\ \mathbf{Y} &= \left(\mathbf{I} - \mathbf{A^d}\right)^{-1}\mathbf{F^d} \end{aligned} \tag{A.1}$$

where $y_{i,k}$ denotes the output of domestic sector i that was used to satisfy final demand component k (both *directly* via $f^d_{i,k}$ and *indirectly* via other sectors' output $y_{j,k}$ which use inputs from sector i based on share $a^d_{i,j}$).

Then, the imported inputs necessary for the production of these domestic sectoral outputs $y_{j,k}$ are:[37]

$$\begin{aligned} m^{indirect}_{i,k} &= \sum_{j\in\{NT,TD,X\}} a^m_{i,j}\, y_{j,k} \\ \mathbf{M}^{\mathbf{indirect}} &= \mathbf{A^m}\,\mathbf{Y} = \\ &= \mathbf{A^m}\left(\mathbf{I} - \mathbf{A^d}\right)^{-1}\mathbf{F^d} \end{aligned} \tag{A.2}$$

Using an analogous formula, we can calculate the ***domestic value added content of domestic demand*** components:[38]

$$\begin{aligned} gva^{indirect}_{i,k} &= \sum_{j\in\{NT,TD,X\}} a^{gva}_{i,j}\, y_{j,k} \\ \mathbf{GVA} &= \mathbf{A^{gva}}\,\mathbf{Y} = \\ &= \mathbf{A^{gva}}\left(\mathbf{I} - \mathbf{A^d}\right)^{-1}\mathbf{F^d} \end{aligned} \tag{A.3}$$

By construction, the following identity must hold, showing steady state shares of expenditures:

$$\underbrace{\overbrace{1_{1\times3}\,\mathbf{GVA}_{3\times4}}^{\sum_{rows}\mathbf{GVA}} + \mathbf{M}^{\mathbf{indirect}}_{1\times4}}_{\text{dom. gross output}} + \mathbf{M}^{\mathbf{direct}}_{1\times4} = 1_{1\times4} \tag{A.4}$$

where $\sum_{rows}\mathbf{GVA} \equiv \underbrace{\mathbf{GVA}(1,:)_{1\times4}}_{GVA^{NT}} + \underbrace{\mathbf{GVA}(2,:)_{1\times4}}_{GVA^{TD}} + \underbrace{\mathbf{GVA}(3,:)_{1\times4}}_{GVA^{X}}$ is the column-wise sum, yielding for each expenditure type k the domestic value added content *combined* from all 3 domestic sectors.

[37] Here, i just denotes a single foreign "sector" from where imports come, and which are used in three domestic sectors $j \in \{NT, TD, X\}$.

[38] Here, i denotes the different sectoral types $i \in \{NT, TD, X\}$ of domestic gross value added (K,L,profit combination).

A.2. GDP DEFINITIONS EQUIVALENCE

Aggregate nominal resource constraint. Adding up the real resource constraints for sectors TD and NT (2.22) in nominal terms, then adding and subtracting total import M spending (inclusive of tariffs) (2.23):

$$
\begin{aligned}
P_t^{NT} Y_t^{NT} + P_t^{TD} Y_t^{TD} = {} & P_t^{NT} \underbrace{\left[\left[C_t^{NT} + G_t^{NT} + IG_t^{NT} \right] + \sum_{j \in \{NT,TD\}} I_t^{NT,j} + \sum_{j \in \{NT,TD\}} INT_t^{NT,j} + adj_t^{NT} \right]}_{Y_t^{NT}\ (2.22)} \\
& + P_t^{TD} \underbrace{\left[\left[C_t^{TD} + G_t^{TD} + IG_t^{TD} \right] + \sum_{j \in \{NT,TD\}} I_t^{TD,j} + \sum_{j \in \{NT,TD\}} INT_t^{TD,j} + X_t + adj_t^{TD} \right]}_{Y_t^{TD}\ (2.22)} + \\
& + P_t^{M} \underbrace{\left[\left[C_t^{M} + G_t^{M} + IG_t^{M} \right] + \sum_{j \in \{NT,TD\}} I_t^{M,j} + \sum_{j \in \{NT,TD\}} INT_t^{M,j} \right]}_{M_t\ (2.23)} - \\
& - P_t^{M} M_t
\end{aligned}
$$

Then reorganising the terms by expenditure categories, and combining them into composite goods:

$$
\begin{aligned}
P_t^{NT} Y_t^{NT} + P_t^{TD} Y_t^{TD} = {} & \underbrace{P_t^{NT} \left[C_t^{NT} + G_t^{NT} + IG_t^{NT} \right] + \underbrace{P_t^{TD} \left[C_t^{TD} + G_t^{TD} + IG_t^{TD} \right] + P_t^{M} \left[C_t^{M} + G_t^{M} + IG_t^{M} \right]}_{P_t^T [C_t^T + G_t^T + IG_t^T]\ (2.15)}}_{P_t^C [C_t + G_t + IG_t]\ (2.14)} \\
& + \sum_{j \in \{NT,TD\}} \underbrace{P_t^{NT} I_t^{NT,j} + \underbrace{P_t^{TD} I_t^{TD,j} + P_t^{M} I_t^{M,j}}_{P_t^T I_t^{T,j}\ (2.15)}}_{P_t^{I,j} I_t^j\ (2.14)} + \\
& + \sum_{j \in \{NT,TD\}} \underbrace{P_t^{NT} INT_t^{NT,j} + \underbrace{P_t^{TD} INT_t^{TD,j} + P_t^{M} INT_t^{M,j}}_{P_t^T INT_t^{T,j}\ (2.15)}}_{P_t^{INT,j} INT_t^j\ (2.14)} + \\
& + \left(P_t^{TD} X_t - P_t^{M} M_t \right) + P_t^{NT} adj_t^{NT} + P_t^{TD} adj_t^{TD}
\end{aligned}
$$

we arrive to the **nominal resource constraint of the economy:**

$$
\begin{aligned}
\sum_{J \in \{NT,TD\}} P_t^J Y_t^J = P_t^C \left[C_t + G_t + IG_t \right] + \sum_{J \in \{NT,TD\}} P_t^{I,J} I_t^J + \sum_{J \in \{NT,TD\}} P_t^{INT,J} INT_t^J + \\
+ \left(P_t^{TD} X_t - P_t^M M_t \right) + \sum_{J \in \{NT,TD\}} P_t^J adj_t^J \qquad \text{(A.5)}
\end{aligned}
$$

(A.5) is a fundamental identity in national accounts stating that ”for goods and services as a whole, total supply (Output + Imports) should equal total use (Intermediates + Final Consumption + Investments + Exports)”.

GDP equivalence. The equivalence of production, expenditure and income side GDP definitions in (2.26), (2.27), (2.28) can be verified by rearranging the nominal resource constraint (A.5):

$$\underbrace{\sum_j \left[P_t^j Y_t^j - P_t^{INT,j} INT_t^j\right] + (P_t^X - P_t^{TD})\, X_t + \left(\tau_t^{tariff}\, P_t^{M1}\right) M_t + \tau_t^{vat} P_t^C C_t}_{(2.26):\ nGDP_t^{prod}} =$$

$$= \underbrace{(1+\tau_t^{vat}) P_t^C C_t + P_t^C\left[G_t + IG_t\right] + \sum_j P_t^{I,j} I_t^j + \left(P_t^X X_t - P_t^{M1} M_t\right) + \sum_j P_t^j adj_t^j}_{(2.27):\ nGDP_t^{exp}}$$

where we have added $P_t^X X_t$ and $\tau_t^{vat} P_t^C C_t$ to both sides, and used (2.21): $P_t^M = \left(1 + \tau_t^{tariff}\right) P_t^{M1}$. The income side GDP definition follows from substituting the firm budget constraints into the production side GDP definition:

$$nGDP_t^{prod} \overset{(2.26)}{=} \sum_{j\in\{NT,TD\}} \underbrace{\left[P_t^j Y_t^j - P_t^{INT,j} INT_t^j\right]}_{profit_t^j + (1+\tau_t^{ssc}) W_t L_t^j + i_t^{K,j} P_t^{I,j} K_t^j} +$$

$$+ \underbrace{(P_t^X - P_t^{TD})\, X_t}_{profit_t^X} +$$

$$+ \tau_t^{vat} P_t^C C_t + \tau_t^{tariff} P_t^{M1} M_t =$$

$$= \sum_{j\in\{NT,TD\}} \left[(1+\tau_t^{ssc}) W_t L_t^j + i_t^{K,j} P_t^{I,j} K_t^j + profit_t^j\right] +$$

$$+ profit_t^X + \tau_t^{vat} P_t^C C_t + \tau_t^{tariff} P_t^{M1} M_t =$$

$$\overset{(2.28)}{=} nGDP_t^{inc}$$

A.3. TERMS-OF-TRADE GAIN DERIVATION

From trade balance. Using the definition for TB_t in (2.25), and steady state prices $P^X = \frac{1}{\eta^x}$ and $P^{M1} = \frac{1}{1+\tau^{tariff}}$, deviations in the nominal trade balance can be decomposed into contributions by trade volume changes, *average* price level growth and *relative* price changes (i.e. ToT-gain):

$$\begin{aligned}
\underbrace{TB_t - TB}_{\widehat{TB}_t} &= \left(P_t^X X_t - P_t^{M1} M_t\right) - \left(\frac{X}{\eta^x} - \frac{M}{1+\tau^{tariff}}\right) = \\
&= \left[\left(\frac{X_t}{\eta^x} - \frac{M_t}{1+\tau^{tariff}}\right) - \left(\frac{X}{\eta^x} - \frac{M}{1+\tau^{tariff}}\right)\right] + \left[\underbrace{\left(P_t^X X_t - P_t^{M1} M_t\right)}_{TB_t} - \frac{P_t^X X_t - P_t^{M1} M_t}{P_t^D}\right] + \\
&\quad + \left[\frac{P_t^X X_t - P_t^{M1} M_t}{P_t^D} - \left(\frac{X_t}{\eta^x} - \frac{M_t}{1+\tau^{tariff}}\right)\right] = \\
&= \underbrace{\left(\widehat{X}_t - \widehat{M}_t\right)}_{\widehat{TB}_t^{volume}} + \underbrace{\left(1 - \frac{1}{P_t^D}\right) TB_t}_{\widehat{TB}_t^{price}} + \underbrace{\left[\left(\frac{P_t^X}{P_t^D} - \frac{1}{\eta^x}\right) X_t - \left(\frac{P_t^{M1}}{P_t^D} - \frac{1}{1+\tau^{tariff}}\right) M_t\right]}_{\widehat{TB}_t^{ToT}}
\end{aligned} \tag{A.6}$$

where $\widehat{X}_t \equiv \frac{1}{\eta^x}(X_t - X)$ and $\widehat{M}_t \equiv \frac{1}{1+\tau^{tariff}}(M_t - M)$ and P_t^D is the implicit deflator for domestic expenditures, which is chosen to be the "average" price level.

$$P_t^D \equiv \frac{P_t^C\left[(1+\tau_t^{vat})C_t + G_t + IG_t\right] + \sum_J P_t^{I,j} I_t^j}{D_t} \tag{A.7}$$

$$D_t \equiv (1+\tau^{vat})C_t + G_t + IG_t + \sum_j I_t^j \tag{A.8}$$

From GDP. By combining the expenditure side nominal GDP definition (2.27) with the domestic expenditure definitions above, the trade balance can be also expressed as the difference between the two: $TB_t = P_t^{GDP}\, rGDP_t - P_t^D D_t$. From the real GDP definition (2.29), it is also clear that $rGDP_t - D_t = \frac{X_t}{\eta^x} - \frac{M_t}{1+\tau^{tariff}}$. Also, using that $P^{GDP} = P^D = 1$:

$$\begin{aligned}
\underbrace{TB_t - TB}_{\widehat{TB}_t} &= \left[\underbrace{P_t^{GDP}\, rGDP_t}_{nGDP_t} - P_t^D D_t\right] - \left(rGDP - D\right) = \\
&= \left[\underbrace{\left(rGDP_t - D_t\right)}_{\frac{X_t}{\eta^x} - \frac{M_t}{1+\tau^{tariff}}} - \underbrace{\left(rGDP - D\right)}_{\frac{X}{\eta^x} - \frac{M}{1+\tau^{tariff}}}\right] + \\
&\quad + \left[\underbrace{\left(P_t^{GDP}\, rGDP_t - P_t^D D_t\right)}_{TB_t} - \frac{P_t^{GDP}\, rGDP_t - P_t^D D_t}{P_t^D}\right] + \\
&\quad + \left[\frac{P_t^{GDP}\, rGDP_t - P_t^D D_t}{P_t^D} - \left(rGDP_t - D_t\right)\right] = \\
&= \underbrace{\left(\widehat{X}_t - \widehat{M}_t\right)}_{\widehat{TB}_t^{volume}} + \underbrace{\left(1 - \frac{1}{P_t^D}\right) TB_t}_{\widehat{TB}_t^{price}} + \underbrace{\left(\frac{P_t^{GDP}}{P_t^D} - 1\right) rGDP_t}_{\widehat{TB}_t^{ToT}}
\end{aligned} \tag{A.9}$$

Since every other term in (A.6) and (A.9) is the same, it follows that the two expressions for the terms-of-trade gain $\widehat{TB}_t^{ToT}$ must also be equivalent, and the latter can be used in (2.31).

A.4. PRICE DECOMPOSITIONS

This section provides different decompositions that split the change in consumer prices into various components, such as taxes, import prices and the price of domestic value added, taking into account both direct and indirect channels via intermediate inputs.

A.4.1. GVA deflator

Aggregate GVA deflator. For the "price" of domestic value added, we define the implicit GVA deflator based on (2.26) as

$$P_t^{GVA} \equiv \frac{\sum_{j\in\{NT,TD\}} \left[P_t^j Y_t^j - P_t^{INT,j} INT_t^j\right] \;+\; (P_t^X - P_t^{TD})\, X_t}{\sum_{j\in\{NT,TD\}} \left[Y_t^j - INT_t^j\right] \;+\; (P^X - 1)\, X_t} \tag{A.10}$$

which implicitly contains variation in wage costs, capital costs as well as markups and retailer profits – but not in taxes like tariffs (see (2.28)). Steady state export prices from (2.12) are $P^X = \frac{1}{\eta^x}$. This can be used to calculate the contribution to CPI changes stemming from domestic value added.

Sectoral GVA deflators. For an even finer decomposition sectoral GVA prices can also be calculated, which also provide further analytical insights.

$$P_t^{GVA,j} \equiv \frac{P_t^j Y_t^j - P_t^{INT,j} INT_t^j}{P^j\, Y_t^j - INT_t^j} =$$

$$= \frac{\frac{P_t^j}{P^j} - P_t^{INT,j} \frac{INT_t^j}{P^j Y_t^j}}{1 - \frac{INT_t^j}{P^j Y_t^j}} \qquad \text{for } j \in \{NT, TD, X\} \tag{A.11}$$

where for export retailers $j = X$ one substitute $Y_t^X = INT_t^X = X_t$, and $P_t^{INT,X} = P_t^{TD}$, while steady state prices are $P^{NT} = P^{TD} = 1$ and $P^X = \frac{1}{\eta^x}$, such that exporter GVA deflator simplifies to:

$$P_t^{GVA,X} = \frac{\eta^x}{1-\eta^x}\, \left(P_t^X - P_t^{TD}\right) \qquad \text{for } \eta^x \neq 1$$

which is not defined if steady state exporter markups are zero $(\eta^x = 1)$[39]

A log-linear approximation to (A.11) yields:

$$P^{GVA,j}\, \widehat{p}_t^{GVA,j} \approx \frac{1}{1-\eta^j s_{in}^j} \frac{P^j}{P^j} \widehat{p}_t^j \;-\; \frac{\eta^j s_{in}^j}{1-\eta^j s_{in}^j} P^{INT,j}\, \widehat{p}_t^{INT,j} \;+$$

$$+\; \underbrace{\frac{-\frac{P^{INT,j}}{P^j}\left(1-\eta^j s_{in}^j\right) - \left(1 - P^{INT,j}\eta^j s_{in}^j\right)\left(-\frac{1}{P^j}\right)}{\left(1-\eta^j s_{in}^j\right)^2}}_{=0} \;\frac{INT^j}{Y_t^j}\; d\log\left(\frac{INT_t^j}{Y_t^j}\right)$$

where steady state shares for $\frac{P^{INT,j}\, INT^j}{P^j\, Y^j} = \frac{INT^j}{P^j\, Y^j} = \eta^j s_{in}^j$ can be read off from the I/O table in Table A.2, also using $P^{INT,j} = 1$ for $\forall j$ which, when plugged into (A.11), also implies $P^{GVA,j} = 1$. Additionally applying that $s_{in}^x = 1$, the above expression simplifies to:

[39] In that case there is no exporter GVA in steady state relative (i.e. nominal as well as "real" GVA is zero). Since in retailer profit deviations, there is no "volume", only price (via the markup), therefore "real GVA volumes" always remain at zero. And the implicit price deflator measures the ratio of nominal values to real volumes.

$$\widehat{p}_t^{GVA,j} \approx \begin{cases} \frac{1}{1-\eta^j s^j_{in}} \left(\widehat{p}^j_t \; - \; \eta^j s^j_{in} \, \widehat{p}_t^{INT,j}\right) & \text{for } j \in \{NT, TD\} \\ \frac{1}{1-\eta^x} \left(\widehat{p}^x_t \; - \; \eta^x \, \widehat{p}_t^{TD}\right) & \text{for } j = X, \; \eta^x \neq 1 \end{cases} \tag{A.12}$$

Sectoral GVA deflator decomposition. Alternatively, instead of backing out GVA deflators from gross output and input prices, we can also build them up from their components as in (A.13), using time varying markups η^j_t, and VA denoting value added in the manufacturing stage from labour and capital only:

$$\begin{aligned}
P^j_t Y^j_t &= \overbrace{\underbrace{(1-\eta^j_t) P^j_t Y^j_t}_{profit^j_t} + P_t^{VA,j} VA^j_t}^{GVA^j_t} + P_t^{INT,j} INT^j_t \\
P^j Y^j_t &\equiv (1-\eta^j) P^j Y^j_t + VA^j_t + INT^j_t \\
&\Rightarrow \frac{INT^j_t}{P^j Y^j_t} = \eta^j - \frac{VA^j_t}{P^j Y^j_t} \\
\frac{P^j_t}{P^j} &= (1-\eta^j_t)\frac{P^j_t}{P^j} + P_t^{VA,j}\frac{VA^j_t}{P^j Y^j_t} + P_t^{INT,j}\frac{INT^j_t}{P^j Y^j_t} = \\
&= (1-\eta^j_t)\frac{P^j_t}{P^j} + P_t^{VA,j}\frac{VA^j_t}{P^j Y^j_t} + P_t^{INT,j}\left(\eta^j - \frac{VA^j_t}{P^j Y^j_t}\right) \\
1 &= (1-\eta^j_t) + \frac{P_t^{VA,j} VA^j_t}{P^j_t Y^j_t} + \frac{P_t^{INT,j} INT^j_t}{P^j_t Y^j_t} \\
1 &= (1-\eta^j) + \eta^j(1-s^j_{in}) + \eta^j s^j_{in}
\end{aligned} \tag{defl}$$

Then log-linearising (defl):

$$\begin{aligned}
\widehat{p}^j_t &\approx (1-\eta^j)\widehat{p}^j_t - \eta^j \widehat{\eta}^j_t + \eta^j (1-s^j_{in})\widehat{p}_t^{VA,j} + \eta^j s^j_{in} \widehat{p}_t^{INT,j} + 0 * d\log\left(\frac{VA^j_t}{Y^j_t}\right) \\
&\Rightarrow \widehat{p}^j_t = -\widehat{\eta}^j_t + (1-s^j_{in})\widehat{p}_t^{VA,j} + s^j_{in}\widehat{p}_t^{INT,j} \\
\widehat{p}^j_t &= \underbrace{-\widehat{\eta}^j_t + (1-\eta^j)\left[(1-s^j_{in})\widehat{p}_t^{VA,j} + s^j_{in}\widehat{p}_t^{INT,j}\right] + \eta^j(1-s^j_{in})\,\widehat{p}_t^{VA,j}}_{\text{(A.12): } \left(1-\eta^j s^j_{in}\right)\widehat{p}_t^{GVA,j}} + \eta^j s^j_{in}\,\widehat{p}_t^{INT,j}
\end{aligned}$$

$$\widehat{p}_t^{GVA,j} \approx \begin{cases} \frac{1}{1-\eta^j s^j_{in}}\left(-\widehat{\eta}^j_t + (1-\eta^j)\left[(1-s^j_{in})\,\widehat{p}_t^{VA,j} + s^j_{in}\,\widehat{p}_t^{INT,j}\right] + \eta^j(1-s^j_{in})\,\widehat{p}_t^{VA,j}\right) & \text{for } j \in \{NT, TD\} \\ \frac{1}{1-\eta^x}\left(-\widehat{\eta}^x_t + (1-\eta^x)\,\widehat{p}_t^{TD}\right) & \text{for } j = X, \eta^x \neq 1 \end{cases} \tag{A.13}$$

with a potential further break down based on (2.9): $\widehat{p}_t^{VA,j} \approx \alpha\,\widehat{w}_t + (1-\alpha)\widehat{r}_t^{K,j}$. According to (A.13), GVA prices can fluctuate due to:

- variations in markups $-\widehat{\eta}^j_t$ (changing profits)
- variations in production costs (labour, capital, inputs: $\widehat{p}_t^{VA,j}, \widehat{p}_t^{INT,j}$) to which unchaged markups are applied (changing profits)
- variations in wages and capital prices $\widehat{p}_t^{VA,j}$ (changing labour and capital incomes)

We can also define the "price of profits", only for $\eta^j \neq 1$ as:

$$\widehat{p}_t^j = \underbrace{-\widehat{\eta}_t^j + (1-\eta^j)\left[(1-s_{in}^j)\widehat{p}_t^{VA,j} + s_{in}^j \widehat{p}_t^{INT,j}\right]}_{(1-\eta^j)\,\widehat{p}_t^{profit,j}} + \eta^j (1-s_{in}^j)\,\widehat{p}_t^{VA,j} + \eta^j s_{in}^j\,\widehat{p}_t^{INT,j}$$

$$\widehat{p}_t^{profit,j} = \begin{cases} \frac{1}{1-\eta^j}\left(-\widehat{\eta}_t^j + (1-\eta^j)\left[(1-s_{in}^j)\widehat{p}_t^{VA,j} + s_{in}^j\widehat{p}_t^{INT,j}\right]\right) & \text{for } j \in \{NT, TD\} \\ \frac{1}{1-\eta^X}\left(-\widehat{\eta}_t^X + (1-\eta^X)\,\widehat{p}_t^{TD}\right) = \widehat{p}_t^{GVA,X} & \text{for } j = X \end{cases} \tag{A.14}$$

A.4.2. Import prices and tariffs.

Using (2.21) we can split the import price index into pre-tariff import price and tax components:

$$P_t^M = \left(1 + \tau_t^{tariff}\right) P_t^{M1} =$$

$$= P_t^{M1} + \tau_t^{tariff} P_t^{M1}$$

which can be approximated, denoting $\widetilde{\tau}_t^{tariff} = \tau_t^{tariff} - \tau^{tariff}$, and using $P^{M1} = \frac{1}{1+\tau^{tariff}}$:

$$\widehat{p}_t^M = P^{M1}\widehat{p}_t^{M1} + \tau^{tariff} P^{M1}\widehat{p}_t^{M1} + P^{M1}\underbrace{\tau^{tariff}\widehat{\tau}_t^{tariff}}_{\widetilde{\tau}_t^{tariff}} =$$

$$= \frac{1}{1+\tau^{tariff}}\widehat{p}_t^{M1} + \frac{\tau^{tariff}}{1+\tau^{tariff}}\widehat{p}_t^{M1} + \frac{1}{1+\tau^{tariff}}\widetilde{\tau}_t^{tariff} \tag{A.15}$$

Nominal exchange rates. The pre-tariff import price index P_t^{M1} from (2.18) can be broken down into the nominal effective exchange rate $NEER_t$ and an implicit foreign price index P_t^*, which captures the export prices of trading partners in their own currency.

$$NEER_t \equiv \sum_f \frac{\mathcal{E}_t}{\mathcal{E}_t^f}\frac{M_t^f}{M_t} \tag{A.16}$$

$$P_t^* \equiv \frac{P_t^{M1}}{NEER_t} \tag{A.17}$$

which can be approximated as

$$\widehat{p}_t^{M1} \approx \widehat{p}_t^* + \widehat{neer}_t \tag{A.18}$$

A.4.3. Composition effects in CPI with delayed substitution.

For the sake of illustration, let us break down tradable consumption expenditures into TD domestic products M imported goods, using (2.15), then plug in the demand functions (2.5), (2.6):

$$P_t^T = P_t^{TD}\frac{C_t^{TD}}{C^T} + P_t^M \frac{C_t^M}{C_t^T} =$$

$$= P_t^{TD} \underbrace{\left[(1-s^M)\left(\frac{P_t^{TD}}{\widetilde{\mathcal{P}}_t^T}\right)^{-\sigma_m}\right]^{1-\rho^{im}} \left[h_{t-1}^{TD}\right]^{\rho^{im}}}_{(2.5):\ h_t^{TD}} + P_t^M \underbrace{\left[s^M\left(\frac{P_t^{M}}{\widetilde{\mathcal{P}}_t^T}\right)^{-\sigma_m}\right]^{1-\rho^{im}} \left[h_{t-1}^{M}\right]^{\rho^{im}}}_{(2.6):\ h_t^{M}} =$$

$$= \begin{cases} \widetilde{\mathcal{P}}_t^T = \left[\left(1-s^M\right)\left(P_t^{TD}\right)^{1-\sigma_m} + \left(s^M\right)\left(P_t^M\right)^{1-\sigma_m}\right]^{\frac{1}{1-\sigma_m}} & \text{if } \rho^{im}=0 \\ \left(1-s^M\right)P_t^{TD} + s^M P_t^M & \text{if } \rho^{im}=1 \end{cases}$$

which shows no role for changing basket composition within the price index in these two extreme cases. In the intermediate case of $0 < \rho^{im} < 1$, composition effects should also be zero, up to first order, since $h_t^M \approx 1 - h_t^{TD}$. Using steady state $h^{TD} = 1 - s^M, h^M = s^M$, the log-linear approximation:

$$\widehat{p}_t^T \approx \left(1-s^M\right)\widehat{p}_t^{TD} + s^M \widehat{p}_t^M + \underbrace{\left(1-s^M\right)\widehat{h}_t^{TD} + s^M \widehat{h}_t^M}_{0} \tag{A.19}$$

where the cancelling out of composition effects can be verified by log-linearising h_t^{TD} and h_t^M:

$$\widehat{\mathcal{P}}_t^T \approx \left(1-s^M\right)\widehat{p}_t^{TD} + s^M \widehat{p}_t^M$$

$$\left[\left(1-s^M\right)\widehat{h}_t^{TD}\right] \approx \left(1-\rho^{im}\right)\sigma_m\left(1-s^M\right)\left(\widehat{\mathcal{P}}_t^T - \widehat{p}_t^{TD}\right) + \rho^{im}\left[\left(1-s^M\right)\widehat{h}_{t-1}^{TD}\right] =$$

$$= \underbrace{\left(1-\rho^{im}\right)\sigma_m\left(1-s^M\right)s^M\left(\widehat{p}_t^M - \widehat{p}_t^{TD}\right)}_{\equiv \widehat{x}_t} + \rho^{im}\left[\left(1-s^M\right)\widehat{h}_{t-1}^{TD}\right] =$$

$$\left[s^M\widehat{h}_t^M\right] \approx \left(1-\rho^{im}\right)\sigma_m s^M\left(\widehat{\mathcal{P}}_t^T - \widehat{p}_t^{M}\right) + \rho^{im}\left[s^M\widehat{h}_{t-1}^{M}\right]$$

$$= - \underbrace{\left(1-\rho^{im}\right)\sigma_m\left(1-s^M\right)s^M\left(\widehat{p}_t^M - \widehat{p}_t^{TD}\right)}_{\equiv \widehat{x}_t} + \rho^{im}\left[s^M\widehat{h}_{t-1}^{M}\right]$$

Then, solving backward, we arrive to:

$$\left(1-s^M\right)\widehat{h}_t^{TD} = \sum_{k=0}^{t-1}\left(\rho^{im}\right)^k \widehat{x}_{t-k}$$

$$s^M\widehat{h}_t^{M} = -\sum_{k=0}^{t-1}\left(\rho^{im}\right)^k \widehat{x}_{t-k}$$

verifying that $\left(1-s^M\right)\widehat{h}_t^{TD} + s^M\widehat{h}_t^M = 0$ for $\forall\, \rho^{im} \leq 1$ in (A.19).

Analogously to (A.19), ignoring the composition effects and focusing only on price effects, just by knowing the steady state shares of the components from Table A.2, and the dynamic path of their respective prices, an additive decomposition for the CPI trajectory can be provided.

A.4.4. CPI decompositions

Taking steady state shares from matrices derived in (A.4):

$$\overbrace{\underbrace{\mathbf{GVA}(1,:)_{1\times4}}_{GVA^{NT}}+\underbrace{\mathbf{GVA}(2,:)_{1\times4}}_{GVA^{TD}}+\underbrace{\mathbf{GVA}(3,:)_{1\times4}}_{GVA^{X}}+\mathbf{M}^{\mathbf{indirect}}_{1\times4}}^{\text{dom. gross output}}+\mathbf{M}^{\mathbf{direct}}_{1\times4}=1_{1\times4}$$

$$\widehat{\mathbf{p}}^{\mathbf{GVA}}_{\mathbf{t},1\times3}\ \mathbf{GVA}_{3\times4}\ +\ \widehat{p}^{M}_{t}\left(\mathbf{M}^{\mathbf{indirect}}_{1\times4}+\mathbf{M}^{\mathbf{direct}}_{1\times4}\right)=\widehat{\mathbf{p}}^{\mathbf{k}}_{\mathbf{t},1\times4} \tag{A.20}$$

with consumption being the first $k=1$ among expenditure uses $k\in\{(C+G+IG),I^{NT},I^{TD},X\}$, and with $\widehat{p}^{GVA,j}_t$ taken from (A.12). $\sum_j gva_{j,1}$ captures the domestic value added content embedded in consumption expenditures, while $m^{ind}_1+m^{dir}_1$ captures the import content in the consumption basket, both directly and indirectly via imported inputs used for domestically produced goods.

Domestic value added vs direct and indirect imports.

$$\widehat{p}^{C}_{t}\approx\underbrace{\sum_j gva_{j,1}\,\widehat{p}^{GVA,j}_{t}}_{\Theta^{GVA}_t}+\underbrace{m^{ind}_1\,\widehat{p}^{M}_{t}}_{\Theta^{M,ind}_t}+\underbrace{m^{dir}_1\,\widehat{p}^{M}_{t}}_{\Theta^{M,dir}_t} \tag{A.21}$$

Domestic value added vs net import prices vs tariffs.

$$\begin{aligned}\widehat{p}^{C}_{t}&\approx\sum_j gva_{j,1}\,\widehat{p}^{GVA,j}_{t}+\left(m^{ind}_1+m^{dir}_1\right)\underbrace{\left[\frac{1}{1+\tau^{tariff}}\widehat{p}^{M1}_{t}+\frac{\tau^{tariff}}{1+\tau^{tariff}}\widehat{p}^{M1}_{t}+\frac{1}{1+\tau^{tariff}}\widetilde{\tau}^{tariff}_{t}\right]}_{(A.15):\ \widehat{p}^{M}_{t}}\approx\\&\approx\underbrace{\sum_j gva_{j,1}\,\widehat{p}^{GVA,j}_{t}}_{\Theta^{GVA}_t}+\underbrace{\left(m^{ind}_1+m^{dir}_1\right)\frac{1}{1+\tau^{tariff}}\widehat{p}^{M1}_{t}}_{\Theta^{M1}_t}+\\&+\underbrace{\left(m^{ind}_1+m^{dir}_1\right)\left[\frac{\tau^{tariff}}{1+\tau^{tariff}}\widehat{p}^{M1}_{t}+\frac{1}{1+\tau^{tariff}}\widetilde{\tau}^{tariff}_{t}\right]}_{\Theta^{tariff}_t}\end{aligned} \tag{A.22}$$

where rising tariffs as captured by higher Θ^{tariff}_t can be paid for by:

- domestic households via falling nominal incomes, i.e. wages, capital rental rates or profits (decreasing Θ^{GVA}_t)
- domestic households via an erosion of their purchasing power due to higher consumer prices (higher $\widehat{p}^{C}_{t}$)
- foreigners via falling pre-tariff import prices (decreasing Θ^{M1}_t)

Nominal exchange rate effects. Using (A.18):

$$\begin{aligned}\widehat{p}^{C}_{t}&\approx\underbrace{\sum_j gva_{j,1}\,\widehat{p}^{GVA,j}_{t}}_{\Theta^{GVA}_t}+\underbrace{\left(m^{ind}_1+m^{dir}_1\right)\frac{1}{1+\tau^{tariff}}\widehat{p}^{*}_{t}}_{\Theta^{P*}_t}+\underbrace{\left(m^{ind}_1+m^{dir}_1\right)\frac{1}{1+\tau^{tariff}}\widehat{neer}_{t}}_{\Theta^{NEER}_t}+\\&+\underbrace{\left(m^{ind}_1+m^{dir}_1\right)\left[\frac{\tau^{tariff}}{1+\tau^{tariff}}\widehat{p}^{M1}_{t}+\frac{1}{1+\tau^{tariff}}\widetilde{\tau}^{tariff}_{t}\right]}_{\Theta^{tariff}_t}\end{aligned} \tag{A.23}$$

GVA components. If we want to break down GVA further, we would need to unstack the matrix $\mathbf{A}^{\mathbf{gva}}_{3\times 3}$ in (I/O), where everything was compressed into 3 types of sectoral GVA (with their sectoral prices $\widehat{p}^{GVA,j}_t$ calculated in (A.12)). Then we get a new matrix $\mathbf{A}^{\mathbf{gva}}_{7\times 3}$ which shows the 7 different types of domestic GVA sources in separate rows (each having its own separate price).

$$\begin{bmatrix} \widehat{w}_t \\ \widehat{r}^{K,NT}_t \\ \widehat{p}^{profit,NT}_t (\text{A.14}) \\ \widehat{w}_t \\ \widehat{r}^{K,TD}_t \\ \widehat{p}^{profit,TD}_t (\text{A.14}) \\ \widehat{p}^{profit,X}_t (\text{A.14}) \end{bmatrix} \Rightarrow \begin{bmatrix} \eta^{NT}\left(1-s^{NT}_{in}\right)\alpha & & \\ \eta^{NT}\left(1-s^{NT}_{in}\right)(1-\alpha) & & \\ 1-\eta^{NT} & & \\ & \eta^{TD}\left(1-s^{TD}_{in}\right)\alpha & \\ & \eta^{TD}\left(1-s^{TD}_{in}\right)(1-\alpha) & \\ & 1-\eta^{TD} & \\ & & 1-\eta^{X} \end{bmatrix}_{\mathbf{A}^{\mathbf{gva}}_{7\times 3}}$$

Then, through equation (A.3), we get a redefined $\mathbf{GVA}^*_{7\times 4}$ matrix, with each row showing the shares of different domestic GVA sources. This requires rewriting (A.4), with the rows added separately as:

$$\begin{aligned} &\underbrace{\mathbf{GVA}^*(1,:)_{1\times 4}}_{GVA^{NT,labour}} + \underbrace{\mathbf{GVA}^*(2,:)_{1\times 4}}_{GVA^{NT,capital}} + \underbrace{\mathbf{GVA}^*(3,:)_{1\times 4}}_{GVA^{NT,profit}} + \underbrace{\mathbf{GVA}^*(4,:)_{1\times 4}}_{GVA^{TD,labour}} + \\ &+ \underbrace{\mathbf{GVA}^*(5,:)_{1\times 4}}_{GVA^{TD,capital}} + \underbrace{\mathbf{GVA}^*(6,:)_{1\times 4}}_{GVA^{TD,profit}} + \underbrace{\mathbf{GVA}^*(7,:)_{1\times 4}}_{GVA^{X,profit}} + \mathbf{M}^{\mathbf{indirect}}_{1\times 4} + \mathbf{M}^{\mathbf{direct}}_{1\times 4} = 1_{1\times 4} \end{aligned}$$

Using these new weights and the prices of each component, we can rewrite the CPI decomposition (A.22) as:

$$\begin{aligned} \widehat{p}^C_t \approx\ & \underbrace{\left(gva^*_{1,1} + gva^*_{4,1}\right)\widehat{w}_t}_{\Theta^{labour}_t} + \underbrace{\left(gva^*_{2,1}\,\widehat{r}^{K,NT}_t + gva^*_{5,1}\,\widehat{r}^{K,TD}_t\right)}_{\Theta^{capital}_t} + \\ & + \underbrace{\left(gva^*_{3,1}\,\widehat{p}^{profit,NT}_t + gva^*_{6,1}\,\widehat{p}^{profit,TD}_t + gva^*_{7,1}\,\widehat{p}^{profit,X}_t\right)}_{\Theta^{profit}_t} + \\ & + \underbrace{\left(m^{ind}_1 + m^{dir}_1\right)\frac{1}{1+\tau^{tariff}}\,\widehat{p}^{M1}_t}_{\Theta^{M1}_t} + \\ & + \underbrace{\left(m^{ind}_1 + m^{dir}_1\right)\left[\frac{\tau^{tariff}}{1+\tau^{tariff}}\,\widehat{p}^{M1}_t + \frac{1}{1+\tau^{tariff}}\,\widetilde{\tau}^{tariff}_t\right]}_{\Theta^{tariff}_t} \end{aligned} \tag{A.24}$$

Note however, that this can only be done when all steady state markups are non-zero $\eta^j \neq 1$, otherwise the profit prices in (A.14) do not exist.

Investment and export prices. Based on (A.20), the same decompositions (A.21) through (A.24) can also be made for the other price indices $P^{I,j}_t$, P^X_t, by using the appropriate column $k \in \{2,3,4\}$.

A.5. REAL INCOME DECOMPOSITION

Note that for ease of notation, throughout this section $\Delta X_t \equiv X_t - X$ denotes *level deviation from the steady state* (instead of per period change).

A.5.1. Real gross domestic income

From (2.31):

$$rGDI_t = \frac{nGDP_t}{P_t^D} = \frac{P_t^{GDP}}{P_t^D}\, rGDP_t =$$

$$= rGDP_t + \underbrace{\left(\frac{P_t^{GDP}}{P_t^D} - 1\right) rGDP_t}_{\widehat{TB}_t^{ToT}}$$

Then, recalling the definition of D_t from (A.8), deviations from steady state are:

$$\Delta rGDI_t = \Delta rGDP_t + \widehat{TB}_t^{ToT} \tag{A.25}$$

$$= \underbrace{\Delta D_t + \widehat{TB}_t^{vol}}_{\text{(A.9): } \Delta rGDP_t} + \widehat{TB}_t^{ToT} \tag{A.26}$$

$$= \Delta D_t + \Delta\left(\frac{TB_t}{P_t^D}\right) \tag{A.27}$$

where the last equality follows from:

$$\frac{TB_t}{P_t^D} = \frac{P_t^{GDP} rGDP_t - P_t^D D_t}{P_t^D} =$$

$$= (rGDP_t - D_t) + \left(\frac{P_t^{GDP}}{P_t^D} - 1\right) rGDP_t$$

$$\Delta\left(\frac{TB_t}{P_t^D}\right) = \underbrace{\left(\Delta rGDP_t - \Delta D_t\right)}_{\widehat{TB}_t^{vol}} + \underbrace{\left(\frac{P_t^{GDP}}{P_t^D} - 1\right) rGDP_t}_{\widehat{TB}_t^{ToT}}$$

A.5.2. Real income allocation – Who pays for tariffs?

Starting from the income side nominal GDP definition (2.28):

$$nGDP_t^{inc} \equiv \sum_{j\in\{NT,TD\}} \left[(1+\tau_t^{ssc})W_t L_t^j + i_t^{K,j} P_t^{I,j} K_t^j + profit_t^j\right] +$$

$$+ profit_t^X + \tau_t^{vat} P_t^C C_t + \tau_t^{tariff} P_t^{M1} M_t$$

Income shares λ_t out of GDP sum to 1:

$$1 = \underbrace{\sum_{j\in\{NT,TD\}} \frac{(1+\tau_t^{ssc})W_t L_t^j}{nGDP_t}}_{\lambda_t^{labour}} + \underbrace{\sum_{j\in\{NT,TD\}} \frac{i_t^{K,j} P_t^{I,j} K_t^j}{nGDP_t}}_{\lambda_t^{capital}} + \underbrace{\sum_{j\in\{NT,TD,X\}} \frac{profit_t^j}{nGDP_t}}_{\lambda_t^{profit}} +$$

$$+ \underbrace{\frac{\tau_t^{vat} P_t^C C_t + \tau_t^{tariff} P_t^{M1} M_t}{nGDP_t}}_{\lambda_t^{tax}} \tag{A.28}$$

Real gross domestic income is split according to the same shares:

$$rGDI_t = \frac{P_t^{GDP}}{P_t^D} rGDP_t = \underbrace{\frac{P_t^{GDP}}{P_t^D} rGDP_t \, \lambda_t^{labour}}_{\Phi_t^{labour}} + \underbrace{\frac{P_t^{GDP}}{P_t^D} rGDP_t \, \lambda_t^{capital}}_{\Phi_t^{capital}} +$$

$$+ \underbrace{\frac{P_t^{GDP}}{P_t^D} rGDP_t \, \lambda_t^{profit}}_{\Phi_t^{profit}} + \underbrace{\frac{P_t^{GDP}}{P_t^D} rGDP_t \, \lambda_t^{tax}}_{\Phi_t^{tax}} \tag{A.29}$$

which shows how real income of certain economic actors Φ_t^i can change due to overall real production, the terms-of-trade, or their income shares.

Combining (A.29) with (2.31):

$$\underbrace{rGDP_t + \widehat{TB}_t^{ToT}}_{(2.31):\ rGDI_t} = \underbrace{\Phi_t^{labour} + \Phi_t^{capital} + \Phi_t^{profit} + \Phi_t^{tax}}_{(A.29):\ rGDI_t}$$

$$\Phi_t^{tax} = \underbrace{\widehat{TB}_t^{ToT}}_{\text{foreigners' loss}} + \underbrace{\left[-\left(\Phi_t^{labour} + \Phi_t^{capital} + \Phi_t^{profit} - rGDP_t \right) \right]}_{\text{domestic HH income } \textit{loss}\text{, } \underline{\text{not}} \text{ due to production}} \tag{A.30}$$

$$= \underbrace{rGDP_t + \widehat{TB}_t^{ToT}}_{rGDI_t} + \underbrace{\left[-\left(\Phi_t^{labour} + \Phi_t^{capital} + \Phi_t^{profit} \right) \right]}_{\text{domestic HH income } \textit{loss}} \tag{A.31}$$

which shows who pays for the tariffs:

- (A.30) shows that the increase in the government's real tax revenues Φ_t^{tax} stem from losses in domestic household real income (*beyond* what would be implied by changes in real production) on the one hand, and losses in foreign real incomes (the terms-of-trade gain) on the other hand.
- (A.31) captures that the government's real tax revenues Φ_t^{tax} can increase above changes in domestic real income $rGDI_t$ (that is either produced at home through real GDP, or paid by foreigners in the form of a ToT-gain) only to the extent that domestic households suffer a real income loss (since total $rGDI_t$ is the sum of household and government real incomes: (A.29)).

"ToT wedge". To control for changes in domestic production volumes as a source a "paying" for taxes,[40] focus on ***real unit prices***, dividing (A.29) through by real GDP:

$$\frac{rGDI_t}{rGDP_t} = \frac{nGDP_t / rGDP_t}{P_t^D} =$$

$$= \frac{P_t^{GDP}}{P_t^D} = \frac{P_t^{GDP}}{P_t^D} \lambda_t^{labour} + \frac{P_t^{GDP}}{P_t^D} \lambda_t^{capital} + \frac{P_t^{GDP}}{P_t^D} \lambda_t^{profit} + \frac{P_t^{GDP}}{P_t^D} \lambda_t^{tax} =$$

$$= \frac{ULC_t}{P_t^D} + \frac{UCC_t}{P_t^D} + \frac{UP_t}{P_t^D} + \frac{UT_t}{P_t^D} \tag{A.32}$$

[40] If the income of households falls because they also work less and enjoy more leisure, that is not "paying for" the tariffs in the strict meaning of the word. It is the income losses *beyond* what would be implied by changing production that matter here, as those can change the HH vs tax income shares.

which essentially splits the ToT-like wedge between the price of domestic production and expenditures $\frac{P_t^{GDP}}{P_t^D}$ into contributions by real unit prices of production factors and taxes. Rearrange this equation to real unit taxes:

$$\frac{UT_t}{P_t^D} = \underbrace{\frac{P_t^{GDP}}{P_t^D}}_{\text{foreigners' loss}} + \underbrace{\left[-\left(\frac{ULC_t + UCC_t + UP_t}{P_t^D}\right)\right]}_{\text{domestic HH loss}} \tag{A.33}$$

that provides a similar approach as in the (pre-VAT) CPI decompositions (A.22) and (A.24):

$$\Theta_t^{tariff} = \underbrace{-\Theta_t^{M1}}_{\text{foreigners loss}} + \underbrace{\left[-\left(\Theta_t^{GVA} - \widehat{p}_t^C\right)\right]}_{\text{domestic HH loss}} \tag{A.22}$$

$$\Theta_t^{tariff} = \underbrace{-\Theta_t^{M1}}_{\text{foreigners loss}} + \underbrace{\left[-\left(\Theta_t^{labour} + \Theta_t^{capital} + \Theta_t^{profit} - \widehat{p}_t^C\right)\right]}_{\text{domestic HH loss}} \tag{A.24}$$

A.6. TARIFFS IN GDP AND GVA DEFLATORS

A.6.1. GDP deflator decomposition

Analogously to (A.32), or by starting from $P_t^{GDP} = \frac{nGDP_t^{inc}}{rGDP_t}$, the GDP-deflator can be written as the sum of unit production costs and unit taxes:

$$\begin{aligned} P_t^{GDP} = &\underbrace{\sum_{j\in\{NT,TD\}} \frac{(1+\tau_t^{ssc})W_t L_t^j}{rGDP_t}}_{ULC_t} + \underbrace{\sum_{j\in\{NT,TD\}} \frac{i_t^{K,j} P_t^{I,j} K_t^j}{rGDP_t}}_{UCC_t} + \underbrace{\sum_{j\in\{NT,TD,X\}} \frac{profit_t^j}{rGDP_t}}_{UP_t} + \\ &+ \underbrace{\frac{\tau_t^{vat} P_t^C C_t \; + \; \tau_t^{tariff} P_t^{M1} M_t}{rGDP_t}}_{UT_t} = \\ &= \underbrace{\lambda_t^{labour} P_t^{GDP}}_{ULC_t} + \underbrace{\lambda_t^{capital} P_t^{GDP}}_{UCC_t} + \underbrace{\lambda_t^{profit} P_t^{GDP}}_{UP_t} + \underbrace{\lambda_t^{tax} P_t^{GDP}}_{UT_t} \end{aligned} \tag{A.34}$$

A.6.2. GVA deflator decomposition

The aggregate GVA deflator from (A.10) can also be written up using the income side definition (instead of production side). This allows us to perform an income-based decomposition, something akin to the sectoral GVA deflator (A.13), but averaged across all sectors j (i.e. aggregate).

$$\begin{aligned} P_t^{GVA} &= \frac{rGDP_t}{rGVA_t}\left[\underbrace{\sum_{j\in\{NT,TD\}} \frac{(1+\tau_t^{ssc})W_t L_t^j}{rGDP_t}}_{ULC_t} + \underbrace{\sum_{j\in\{NT,TD\}} \frac{i_t^{K,j} P_t^{I,j} K_t^j}{rGDP_t}}_{UCC_t} + \underbrace{\sum_{j\in\{NT,TD,X\}} \frac{profit_t^j}{rGDP_t}}_{UP_t}\right] = \\ &= \frac{rGDP_t}{rGVA_t}\left[\underbrace{ULC_t \; + \; UCC_t \; + \; UP_t}_{\text{(A.34): } (1-\lambda_t^{tax})\, P_t^{GDP}}\right] \end{aligned} \tag{A.35}$$

where based on the production side (A.10): $rGVA_t \equiv \sum_{j\in\{NT,TD\}} \left[Y_t^j - INT_t^j\right] \; + \; (P^X - 1)\, X_t$, and assuming expenditure and production side real GDP concepts to be similar, we also have that $rGVA_t \; \approx \; rGDP_t \; -$

$\left(\tau^{vat} C_t + \frac{\tau^{tariff}}{1+\tau^{tariff}} M_t\right) \equiv rGDP_t - rtaxes_t$. From this:

$$\frac{rGDP_t}{rGVA_t} = \frac{1}{1 - \frac{rtaxes_t}{rGDP_t}} \overset{(A.28)}{\approx} \frac{1}{1-\lambda^{tax}}$$

$$P_t^{GVA} \overset{(A.35)}{\approx} \frac{1-\lambda_t^{tax}}{1-\lambda^{tax}} P_t^{GDP}$$

where the approximation assumes roughly unchanged expenditure volume shares out of GDP. This shows how product tax rate fluctuations can drive a wedge between the GDP and GVA deflators.

A.6.3. Tariffs as a wedge between GDP and GVA deflators

- Rising tariffs rates seem to *mechanically* raise the GDP deflator (for given volumes), based on the production or income side nominal GDP definitions (2.26),(2.28).
- In the expenditure side nominal GDP definition (2.27) tariffs are implicitly part of the price indices for final expenditures $P_t^C, P_t^{I,j}$: they contain imported goods both directly and indirectly (through imported inputs used for domestic goods), whose price P_t^M includes the tariffs already, while in the trade balance the import bill is deducted at net-of-tariff prices P_t^{M1}.
- However, this mechanical direct effect (before any pass-through or second-round effects on domestic prices are considered) is not 1:1, but instead only works via tariffs on final imports (and not via tariffs on imported inputs, which cancel out). To see this, break down (2.26), using (2.14), (2.15) and (2.21) and (2.23):

$$\begin{aligned} nGDP_t\uparrow = & \underbrace{\sum_{j\in\{TD,NT\}} \left[P_t^j Y_t^j - \overbrace{\left(P_t^{NT} INT_t^{NT,j} + P_t^{TD} INT_t^{TD,j} + P_t^{M1} INT_t^{M,j} + \tau_t^{tariff} P_t^{M1} INT_t^{M,j}\uparrow\right)}^{P_t^{j,INT} INT_t^j}\right]}_{GVA_t\downarrow} + \\ & + \underbrace{\left[\left(\sum_{j\in\{TD,NT\}} \tau_t^{tariff} P_t^{M1} INT_t^{M,j}\uparrow\right) + \tau_t^{tariff} P_t^{M1}\left(C_t^M + G_t^M + IG_t^M + I_t^{M,NT} + I_t^{M,TD}\right)\uparrow\right]}_{(2.23):\ \tau_t^{tariff} P_t^{M1} M_t \uparrow\uparrow} \\ & + \tau_t^{vat} P_t^C C_t \end{aligned}$$

- The reason is that prices P_t^j are rigid at the gross output level (and not at the GVA level). In other words, when intermediate inputs become more costly due to tariffs, firms cannot fully pass on these higher marginal costs to their sticky prices, and this non-perfect pass-through eats into their markups, lowering profits and thereby nominal GVA and the implicit GVA deflator.
- So while the GDP-deflator is pushed *upwards* by tariffs on both final and intermediate imports, the GVA deflator is pushed *downwards* as tariffs on intermediate imports eat into profit margins in the presence of nominal rigidities.[41]
- **In summary, tariffs drive a wedge between the GVA and the GDP deflators.**

[41] To see how costlier imported inputs can hurt the GVA deflator under rigid output prices $\bar{P}_t$ (and before any composition effects via changing volumes), one could also consider the following stylised example for illustration, where nominal GVA is $P_t^{GVA} rGVA_t = \bar{P}_t Y_t - P_t^M Int_t^M$, with real volumes defined as $rGVA_t = Y_t - Int_t^M$, such that $\frac{Int_t^M}{rGVA_t} = \frac{Y_t}{rGVA_t} - 1 > 0$ so we get

$$\downarrow P_t^{GVA} = \bar{P}_t \frac{Y_t}{rGVA_t} - \uparrow P_t^M \left(\frac{Y_t}{rGVA_t} - 1\right)$$

A.7. DOMINANT CURRENCY PRICING (DCP)

Export agency. Exported final goods X_t are assembled by an export agency from a continuum $i \in [0,1]$ of varities $X_t(i)$ that are imperfect substitutes for each other, using a CES aggregator, with σ_x denoting the elasticity of substitution:

$$X_t \equiv \left[\int_0^1 X_t(i)^{\frac{\sigma_x - 1}{\sigma_x}} \, di \right]^{\frac{\sigma_x}{\sigma_x - 1}} \tag{A.36}$$

The standard expenditure minimisation problem of the export agency leads to the following demand function for each variety $X_t(i)$, given the overall demand for final exports X_t:

$$X_t(i) = \left[\frac{P_t^X(i)}{P_t^X} \right]^{-\sigma_x} X_t \tag{A.37}$$

Export retailers. A continuum $i \in [0,1]$ of monopolistically competitive export retailers (owned by domestic Ricardian households) differentiate domestically produced tradable (TD) goods for export purposes, using a linear transformation technology; sell the differentiated product $X_t(i)$, facing the export agency's demand curve (A.37); enjoy market power (monopolistic competition) due to imperfect substitutability of $X_t(i)$, giving them the ability to set prices.

Their pricing decision is subject to Rotemberg-style nominal rigidities,[42] with prices-adjustments being costly in terms of a dominant currency, the US dollar, i.e. it is costly to change $\frac{P_t^X(i)}{\mathcal{E}_t}$, which is also the *pre-tariff* import price in USD .[43] Quadratic adjustment costs (in terms of the export output) are:

$$adj_t^{PX}(i) = \frac{\gamma_{px}}{2} \left[\frac{P_t^X(i)}{P_{t-1}^X(i)} \frac{\mathcal{E}_{t-1}}{\mathcal{E}_t} - 1 \right]^2 X_t \tag{A.38}$$

Nominal profits of exporters (with a lump-sum rebate of adjustment costs from owners) evolve as:

$$profit_t^X = \left(P_t^X - P_t^{TD} \right) X_t \; - \; P_t^X \, adj_t^{PX} + LS_t^{adj,X} \tag{A.39}$$

Export retailers choose their price $P_t^X(i)$ to maximise the present value of their current and future expected profits, subject to the demand for their products (A.37) and nominal rigidities (A.38) (Rotemberg price adjustment costs). Expressed in terms of the exported good:

$$\max_{P_t^X(i)} \; E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^X \left\{ \left[\frac{P_t^X(i)}{P_t^X} - \eta_t^X \right] X_t(i) \; - \; \frac{\gamma_{px}}{2} \left[\frac{P_t^X(i)}{P_{t-1}^X(i)} \frac{\mathcal{E}_{t-1}}{\mathcal{E}_t} - 1 \right]^2 X_t \right\}$$

$$X_t(i) = \left[\frac{P_t^X(i)}{P_t^X} \right]^{-\sigma_x} X_t$$

where $\Lambda_{0,t}^X$ is the Ricardian household's stochastic discount factor, *compounded* between periods 0 and t, to discount profit flows expressed *in terms of exported goods*. $\eta_t^X = \frac{P_t^{TD}}{P_t^X}$ denotes the ***real marginal cost*** for export retailers, which depends on the price of domestically produced tradable goods P_t^{TD} (that are used as

[42] Note that these export sector nominal rigidities, as well as the differentiation "for export purposes", are *on top of* the price stickiness and differentiation already introduced with retailers in the domestic tradable sector, inherent in P_t^{TD}.

[43] See DCP with Calvo rigidities in Gopinath et al. (2020) and Gopinath and Itskhoki (2022), and applications in other models by Georgiadis and Mösle (2019), Casas et al. (2017), Egorov and Mukhin (2023).

an input), and which is the same for all exporters due to the linear production technology (so does not depend on i). We also define export price inflation as $1+\pi_t^X = \frac{P_t^X}{P_{t-1}^X}$.

Substituting out $X_t(i)$ with the demand function (A.37), the maximisation problem becomes:

$$\max_{P_t^X(i)} \; E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^X \left\{ \left[\frac{P_t^X(i)}{P_t^X} - \eta_t^X \right] \left[\frac{P_t^X(i)}{P_t^X} \right]^{-\sigma_X} X_t \; - \; \frac{\gamma_{px}}{2} \left[\frac{P_t^X(i)}{P_{t-1}^X(i)} \frac{\mathcal{E}_{t-1}}{\mathcal{E}_t} - 1 \right]^2 X_t \right\}$$

$$\max_{P_t^X(i)} \; E_0 \sum_{t=0}^{\infty} \Lambda_{0,t}^X \left\{ \left[\frac{P_t^X(i)}{P_t^X} \right]^{1-\sigma_X} X_t \; - \; \eta_t^X \left[\frac{P_t^X(i)}{P_t^X} \right]^{-\sigma_X} X_t \; - \; \frac{\gamma_{px}}{2} \left[\frac{P_t^X(i)}{P_{t-1}^X(i)} \frac{\mathcal{E}_{t-1}}{\mathcal{E}_t} - 1 \right]^2 X_t \right\}$$

Then, the first-order condition is:

$$\begin{aligned} 0 = (1-\sigma_X) \left[\frac{P_t^X(i)}{P_t^X} \right]^{-\sigma_X} \frac{X_t}{P_t^X} \; + \; \sigma_X \eta_t^X \left[\frac{P_t^X(i)}{P_t^X} \right]^{-\sigma_X - 1} \frac{X_t}{P_t^X} \; - \\ - \; \gamma_{px} \left[\frac{P_t^X(i)}{P_{t-1}^X(i)} \frac{\mathcal{E}_{t-1}}{\mathcal{E}_t} - 1 \right] \frac{X_t}{P_{t-1}^X(i)} \frac{\mathcal{E}_{t-1}}{\mathcal{E}_t} \; + \\ + \; \gamma_{px} E_t \left\{ \Lambda_{t,t+1}^X \left[\frac{P_{t+1}^X(i)}{P_t^X(i)} \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} - 1 \right] \frac{P_{t+1}^X(i)}{P_t^X(i)} \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \frac{X_{t+1}}{P_t^X(i)} \right\} \end{aligned} \tag{A.40}$$

In a symmetric equilibrium all firms face the same problem, so they will all choose the same price $P_t^X(i) = P_t^X$ for $\forall\, i$.[44] Exploiting this, using $(1+e_t) \equiv \frac{\mathcal{E}_t}{\mathcal{E}_{t-1}}$, then dividing through by $\gamma_{px} \frac{X_t}{P_t^X}$ and rearranging, we arrive to:

$$\begin{aligned} 0 = (1-\sigma_X) \frac{X_t}{P_t^X} \; + \; \sigma_X \eta_t^X \frac{X_t}{P_t^X} \; - \\ - \; \gamma_{px} \left[\frac{1+\pi_t^X}{1+e_t} - 1 \right] \frac{X_t}{P_{t-1}^X} \frac{1}{1+e_t} \; + \\ + \; \gamma_{px} E_t \left\{ \Lambda_{t,t+1}^X \left[\frac{1+\pi_{t+1}^X}{1+e_{t+1}} - 1 \right] \frac{X_{t+1}}{P_t^X} \frac{1+\pi_{t+1}^X}{1+e_{t+1}} \right\} \\ 0 = \frac{1-\sigma_X}{\gamma_{px}} \; + \; \frac{\sigma_X}{\gamma_{px}} \eta_t^X \; - \\ - \; \left[\frac{1+\pi_t^X}{1+e_t} - 1 \right] \frac{1+\pi_t^X}{1+e_t} \; + \; E_t \left\{ \Lambda_{t,t+1}^X \frac{X_{t+1}}{X_t} \left[\frac{1+\pi_{t+1}^X}{1+e_{t+1}} - 1 \right] \frac{1+\pi_{t+1}^X}{1+e_{t+1}} \right\} \end{aligned}$$

$$\underbrace{\left[\frac{1+\pi_t^X}{1+e_t} - 1 \right] \frac{1+\pi_t^X}{1+e_t}}_{\approx \pi_t^X - e_t} = E_t \left\{ \Lambda_{t,t+1}^X \frac{X_{t+1}}{X_t} \underbrace{\left[\frac{1+\pi_{t+1}^X}{1+e_{t+1}} - 1 \right] \frac{1+\pi_{t+1}^X}{1+e_{t+1}}}_{\approx \pi_{t+1}^X - e_{t+1}} \right\} \; + \; \frac{\sigma_X}{\gamma_{px}} \left[\eta_t^X - \frac{\sigma_X - 1}{\sigma_X} \right] \tag{A.41}$$

which is the ***New Keynesian Phillips Curve*** for export retailers, pinning down export prices.

- export price inflation depends on the deviation of real marginal costs $\eta_t^X = \frac{P_t^{TD}}{P_t^X}$ from *desired* real marginal costs $\frac{\sigma_X - 1}{\sigma_X}$ (which is the inverse of desired exporter markups $1+\mu^X = \frac{\sigma_X}{\sigma_X - 1}$)

[44] This is consistent with the definition of $P_t^X = \left[\int_0^1 P_t^X(i)^{1-\sigma_X} \, di \right]^{\frac{1}{1-\sigma_X}}$

- the slope of the NKPC is inversely proportional to the degree of price adjustment costs γ_{px}
- an approximation of $\left[\frac{1+\pi_t^X}{1+e_t} - 1\right] \frac{1+\pi_t^X}{1+e_t} \approx \pi_t^X - e_t$ is used

Rearranging to express the real marginal cost η_t^X, the NKPC looks as follows:

$$\eta_t^X = \frac{\sigma_x - 1}{\sigma_x} - \frac{\gamma_{px}}{\sigma_x}\left[E_t\left\{\Lambda_{t,t+1}^X \frac{X_{t+1}}{X_t}\left(\pi_{t+1}^X - e_{t+1}\right)\right\} - \left(\pi_t^X - e_t\right)\right] \tag{A.42}$$

- which shows that with perfectly substitutable varieties $\eta_t^X = \lim_{\sigma_x \to \infty} \frac{\sigma_x - 1}{\sigma_x} = 1$, meaning that export prices are equal to domestically produced tradable prices: $P_t^X = P_t^{TD}$;
- in the absence of price rigidities $\gamma_{px} = 0$, real marginal costs (thus, inverse exporter markups) are constant at $\eta_t^X = \frac{\sigma_x - 1}{\sigma_x}$;
- with a high degree of price stickiness $\gamma_{px} \to \infty$, an unexpected exchange rate depreciation $e_t \uparrow$ (relative to the dollar) would allow an increase of export prices in the producer's currency $\pi_t^X \uparrow$ (so that the USD price remains more sticky). This would also raise exporter markups and profits.

Implications of DCP for terms-of-trade. ToT will be more rigid. Given a nominal dollar appreciation $\downarrow \mathcal{E}_t^{\$/€} = \frac{1}{\mathcal{E}_t^{€/\$}\uparrow}$:

$$\uparrow ToT_t^{US,PCP} = \frac{\overline{P}_t^{X,\$}}{\downarrow \mathcal{E}_t^{\$/€}\, \overline{P}_t^{X,€}}$$

$$\downarrow ToT_t^{EU,PCP} = \frac{\overline{P}_t^{X,€}}{\uparrow \mathcal{E}_t^{€/\$}\, \overline{P}_t^{X,\$}}$$

$$\overline{ToT}_t^{US,DCP} = \frac{\overline{P}_t^{X,\$}}{\left(\downarrow \mathcal{E}_t^{\$/€}\, P_t^{X,€}\uparrow\right)}$$

$$\overline{ToT}_t^{EU,DCP} = \frac{\uparrow P_t^{X,€}}{\uparrow \mathcal{E}_t^{€/\$}\, \overline{P}_t^{X,\$}}$$

- under PCP, from the perspective of the importer, there is full pass-through of the bilateral nominal exchange rate $\frac{\mathcal{E}_t}{\mathcal{E}_t^f}$ to <u>*pre-tariff*</u> import prices

$$\uparrow\uparrow P_t^{M1,f} = \frac{\uparrow \mathcal{E}_t}{\downarrow \mathcal{E}_t^f}\overline{P}_t^{X,f}$$

- under DCP import prices are rigid in USD, so it is only the exchange rate of the importer vis-a-vis the USD $\mathcal{E}_t$ (for the US, this is constant at 1) that fully passes through to local currency bilateral import prices $P_t^{M1,f}$, while the "remainder" of their bilateral exchange rate with the exporter $\mathcal{E}_t^f$ has smaller effect, as that is swallowed by the exporter. Given a large change $\uparrow\uparrow$ in the bilateral exchange rate $\frac{\uparrow \mathcal{E}_t}{\downarrow \mathcal{E}_t^f}$:

$$\frac{\uparrow P_t^{M1,f}}{\uparrow \mathcal{E}_t} = \frac{\downarrow P_t^{X,f}}{\downarrow \mathcal{E}_t^f}$$

B TARIFF TRANSMISSION CHANNELS: STYLISED ILLUSTRATIONS

B.1. TARIFFS AND TERMS-OF-TRADE OFFSET IN A SIMPLE AS-AD FRAMEWORK

This Annex provides a static, stylised framework to illustrate the core transmission channels of tariffs. This is not the precise representation of QUEST, which is a much richer and dynamic model, but some key messages can be shown. Consider a permanent increase in tariffs. Then we can compare the steady state economy from before the tariff change with its new steady state featuring the higher tariffs, after the economy has already went through the transition dynamics ("comparative statics").[45]

- Aggregate demand depends negatively on the terms-of-trade (ToT) $\Rightarrow$ negative slope of AD
  - **(–)** ***Expenditure switching channel:*** Stronger ToT makes domestic goods less competitive at home (against imports) and abroad (exports) – depends on import and export elasticities + openness.
  - **(+)** ***Real income channel:*** A stronger ToT raises real domestic income (via relatively lower import prices) and consumption.
    - Expenditure switching dominates the real income channel for realistically high long-term trade elasticities, making the AD slope negative.
- Aggregate supply depends ambiguously on ToT $\Rightarrow$ roughly vertical slope of AS
  - **(+)** ***Substitution effect on labour supply:*** A stronger ToT raises the purchasing power of wages (in terms of imports), boosting labour supply by making leisure relatively more expensive.
  - **(+)** ***Intermediate inputs in production:*** A stronger ToT makes imported inputs cheaper, lowering production costs and boosting aggregate supply.[46]
  - **(–)** ***Income effect on labour supply:*** Higher real wages also make workers richer, while the savings on imported input costs raise profits. With higher real incomes households can afford to consume not just more goods, but also more leisure, constraining labour supply.
    - A positive AS slope requires an intertemp. elasticity of substitution above one. These channels exactly offset each other if the elasticity is unity (making the AS curve vertical).
- Higher tariffs raise aggregate demand $\Rightarrow$ AD shift to the right
  - **(+)** ***Expenditure switching:*** Higher tariffs induce substitution away from imports towards home-produced goods, as well as from imported intermediates towards domestic value added, supporting aggregate demand (depends on import elasticity and openness).
- Higher tariffs lower aggregate supply $\Rightarrow$ AS shift to the left
  - **(–)** ***Substitution effect on labour supply:*** Higher tariffs erode the purchasing power of wages, lowering the opportunity cost of leisure, which hurts labour supply.[47]

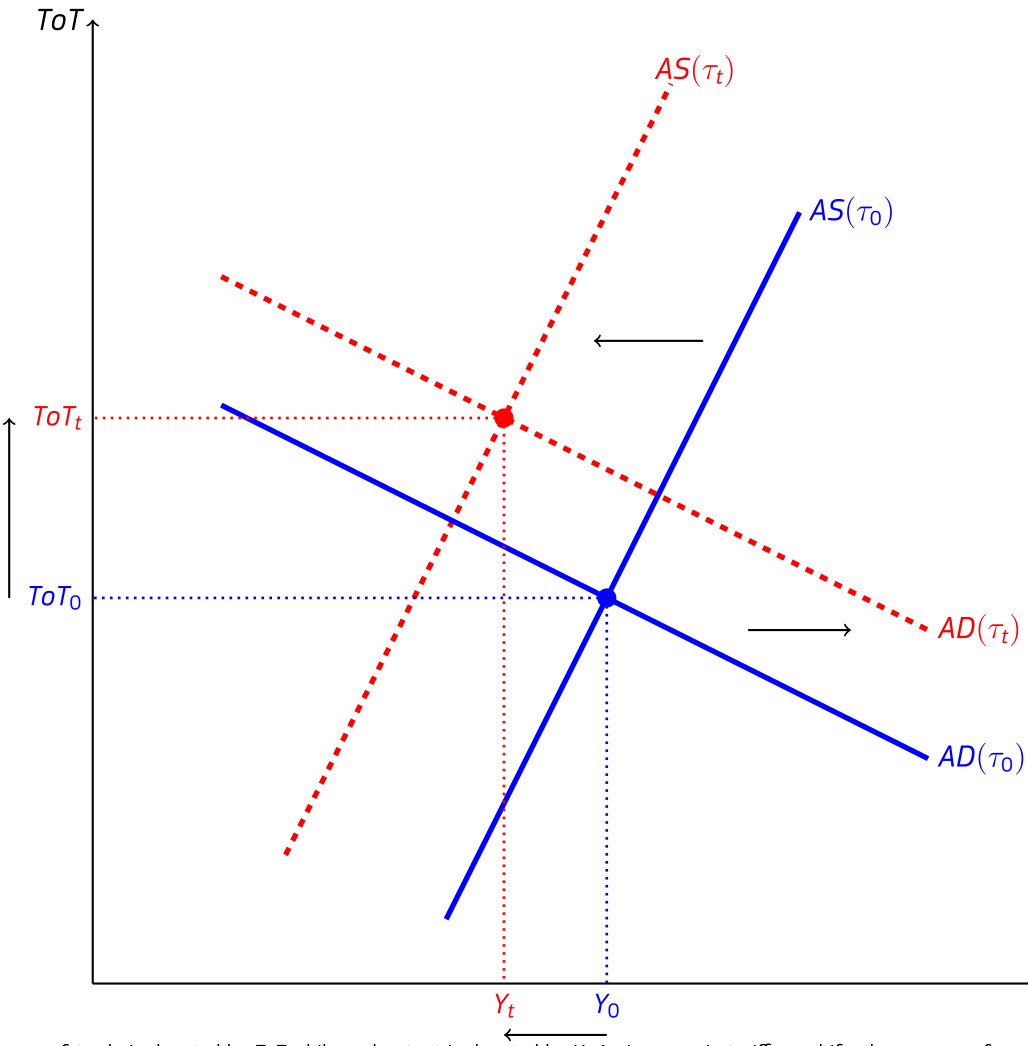
[45] This eliminates all intertemporal channels e.g. intertemporal substitution due to time-varying interest rates, or consumption-smoothing of temporary income changes. This also means that "Ricardian" households would behave identically to liquidity-constrained households, since all income changes are now permanent income changes which they would also fully consume.

[46] With intermediate inputs, a distinction between gross output and value added (GVA, GDP) is worthwhile: though less expensive inputs boost the supply of gross output, they also induce substitution away from less competitive domestic factors of production (labour and capital), lowering the share of domestic value added, something akin to "supply-side expenditure switching". With sufficiently complementary inputs, however, this channel does not dominate.

[47] Notice that while tariffs reduce the real wage, this exerts neither the usual beneficial income effect on labour supply, nor the demand-weakening effects via the real income channel (unlike a ToT depreciation would). The reason is that Ricardian households

(–) ***Intermediate inputs in production:*** Higher tariffs make imported inputs more expensive, raising production costs and hurting aggregate supply.

Figure B.1: Goods market equilibrium, tariffs and terms-of-trade



Note: The terms-of-trade is denoted by *ToT* while real output is denoted by *Y*. An increase in tariffs τ_t shifts the economy from the original (time 0, blue) equilibrium to the new (time t, red) equilibrium, by raising aggregate demand and adversely affecting aggregate supply. An appreciating terms-of-trade (rise in *ToT*) is the mechanism which restores goods market equilibrium. *Source:* European Commission services.

Higher tariffs raise aggregate demand and also act as an adverse supply shock. As shown in Figure 10, this corresponds to a rightward shift in the AD curve and a leftward shift in the AS curve, illustrating the demand-supply imbalance created by tariffs that would prevail under the original terms-of-trade.

- To restore equilibrium in the goods market, the terms-of-trade must appreciate, bringing aggregate demand back in line with supply. This terms-of-trade appreciation offsets some of the original expenditure switching induced by tariffs, lowering demand, while it also offsets some of the original adverse supply shock by supporting real wages and making imported inputs cheaper.
  - The magnitude of the ToT offset depends positively on the import elasticity. More elastic imports amplify the impact of the tariff in shifting AD to the right (although also make AD flatter), leading to more excess demand that requires a larger ToT offset.
  - The ToT offset depends negatively on the export elasticity. With more elastic exports less

internalise the government's budget constraint, expecting tariff revenues to be passed on to them in the form of other tax cuts or transfers in the future, so their lifetime permanent income does not change.

appreciation is enough to offset the excess demand from tariffs (i.e. AD is flatter).

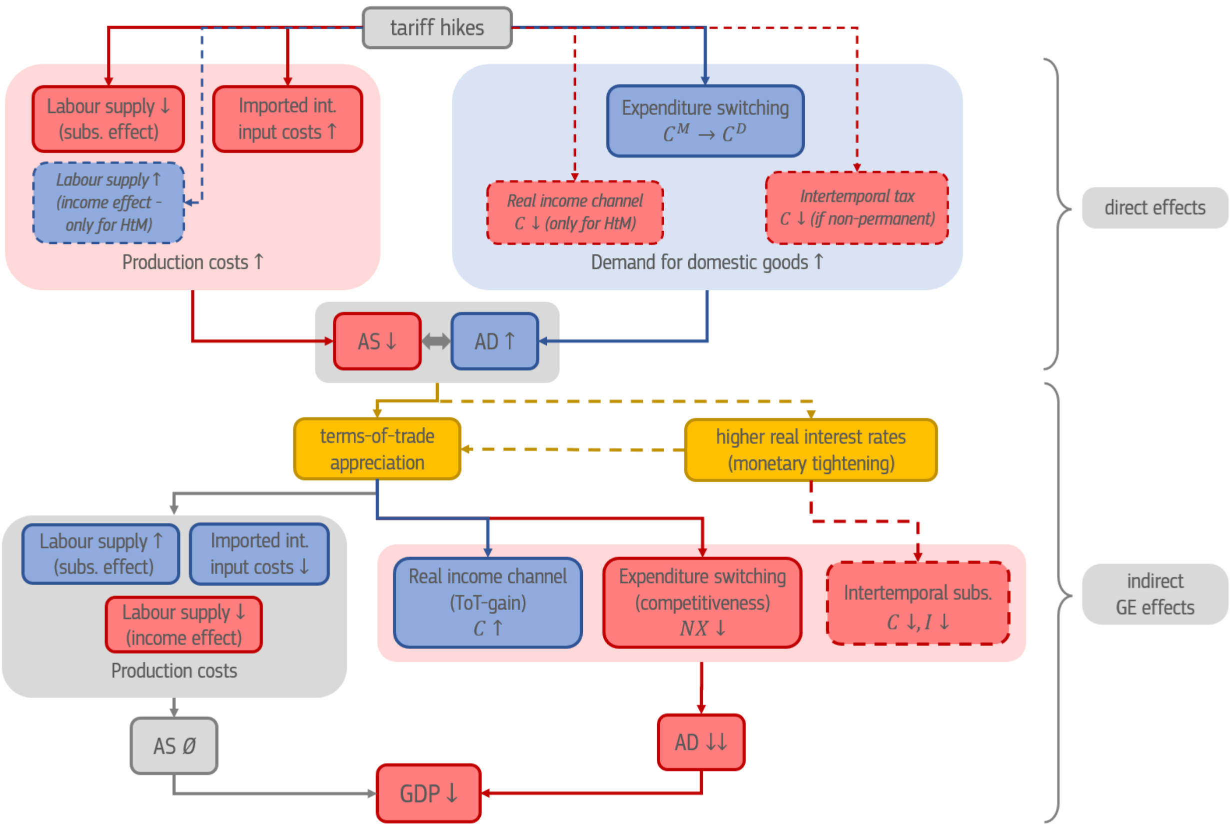
- In theory, the effect on output is ambiguous, but under realistic elasticities output would decline, as in our simulations.[48]

In a dynamic setting, there are additional channels:

(–) ***Intertemporal substitution:*** dynamic effects on domestic demand

- non-permanent tariffs: if tariffs are expected to decline in the future, they act as an intertemporal tax on consumption, prompting households to postpone purchases.
- monetary tightening: inflation-targeting central banks (under sticky prices) can raise the real interest rate in response to the inflationary effects of tariffs, constraining consumption and investment demand further, while also leading to a sharper terms-of-trade appreciation (along with the above-described effects associated with that)

Figure B.2: A stylised illustration of tariff transmission channels



Note: Red (blue) nodes and arrows indicate contractionary (expansionary) channels and effects, while yellow ones capture the general equilibrium price adjustments. Dashed lines are operational only in certain cases: a) if tariff revenues are not handed back *immiediately* to liquidity constrained (hand-to-mouth, HtM) households proportionally to their tariff-induced purchasing power losses; b) if tariffs are not expected to stay permanently; c) if monetary policy tightens in response to inflationary pressures. The influence of terms-of-trade movements on production costs is ambigous, and are roughly zero if income and substitution effects on labour supply cancel out. *Source:* European Commission services.

[48] A sufficiently high import elasticity (increasing the rightward AD shift from tariff hikes) combined with a sufficiently low export elasticity (making AD steeper) could achieve a large enough ToT-appreciation to raise output along a positively sloped AS curve. However, if AS is vertical (e.g. with exactly offsetting labour supply effects and without intermediate inputs), output unambiguously declines.

B.2. TARIFFS IN SIMPLE NEW KEYNESIAN MODELS

The above channels and simple AS-AD framework can also be illustrated within a tractable open economy New Keynesian model, like the one in Monacelli (2025) or Jeanne and Son (2024), which also lies at the core of the QUEST model. This section provides an overview of the *flexible price* version those models, which also highlight the **crucial difference between assuming complete markets (perfect international risk sharing) or incomplete financial markets,** especially in the case of permanent tariffs. Whether allowing for the terms-of-trade gain to be consumed, these assumptions affect the path of consumption materially, leading to qualitatively different outcomes.

Relative prices.

$$P_t^C = \left[(1-\alpha) \left(P_t^H\right)^{1-\sigma_m} + \alpha \left((1+\tau_t) \underbrace{P_t^* \mathcal{E}_t}_{P_t^M} \right)^{1-\sigma_m} \right]^{\frac{1}{1-\sigma_m}} \tag{B.1}$$

$$S_t \equiv \frac{P_t^M}{P_t^H} = \frac{\mathcal{E}_t P_t^*}{P_t^H} \tag{B.2}$$

$$\mathcal{G}(S_t, \tau_t) \equiv \frac{P_t^C}{P_t^H} = \left[(1-\alpha) + \alpha \, (1+\tau_t)^{1-\sigma_m} \, (S_t)^{1-\sigma_m} \right]^{\frac{1}{1-\sigma_m}} \tag{B.3}$$

$$Q_t \equiv \frac{\mathcal{E}_t P_t^*}{P_t^C} = \frac{P_t^M}{P_t^C} = \frac{S_t}{\mathcal{G}(S_t, \tau_t)} \tag{B.4}$$

where we have one domestic sector (Home, H), and P_t^M now denotes the <u>*pre-tariff*</u> import price, $S_t = \frac{1}{ToT_t}$ is the inverse terms-of-trade, while Q_t denotes the CPI-based real exchange rate. $\mathcal{G}(S_t, \tau_t)$ captures how the CPI-PPI ratio $\frac{P_t^C}{P_t^H}$ depends on the terms-of-trade and tariffs. These can be log-linearised as follows:

$$g(s_t, \tau_t) = \alpha \, (s_t + \tau_t) \tag{B.5}$$

$$\begin{aligned} q_t &= s_t - g(s_t, \tau_t) = \\ &= (1-\alpha) \, s_t - \alpha \, \tau_t \end{aligned} \tag{B.6}$$

Demand functions.

$$C_t^M = \alpha \left(\frac{(1+\tau_t) P_t^M}{P_t^C} \right)^{-\sigma_m} C_t \qquad = \alpha \, (1+\tau_t)^{-\sigma_m} \left(\frac{S_t}{\mathcal{G}(S_t, \tau_t)} \right)^{-\sigma_m} C_t \tag{B.7}$$

$$C_t^H = (1-\alpha) \left(\frac{P_t^H}{P_t^C} \right)^{-\sigma_m} C_t \qquad = (1-\alpha) \left[\mathcal{G}(S_t, \tau_t) \right]^{\sigma_m} C_t \tag{B.8}$$

$$X_t = \alpha^* \left(\frac{P_t^H}{\mathcal{E}_t P_t^*} \right)^{-\sigma_x} Y_t^* \qquad = \alpha^* \, S_t^{\sigma_x} \, Y_t^* \tag{B.9}$$

$$c_t^M \approx c_t - \sigma_m \, (1-\alpha) \, s_t - \sigma_m \, (1-\alpha) \, \tau_t \tag{B.10}$$

$$c_t^H \approx c_t + \sigma_m \, \alpha \, s_t + \sigma_m \, \alpha \, \tau_t \tag{B.11}$$

$$x_t \approx y_t^* + \sigma_x \, s_t \tag{B.12}$$

From now on, assume $y_t^* = 0$.

Goods market clearing (demand side).

$$Y_t = C_t^H + X_t = \\ = (1-\alpha)\left[\mathcal{G}(S_t,\tau_t)\right]^{\sigma_m} C_t \; + \; \alpha^* S_t^{\sigma_x} Y_t^* \tag{B.13}$$

$$y_t \approx \underbrace{(1-\alpha)\,c_t \; + \; \left[(1-\alpha)\alpha\,\sigma_m + \alpha^*\,\sigma_x\right] s_t}_{\text{GE effects}} \; + \; \underbrace{(1-\alpha)\alpha\,\sigma_m\,\tau_t}_{\text{direct effect (exp. switching)}} \tag{B.14}$$

- tariffs directly raise demand for domestic goods via expenditure switching (depending on import elasticity σ_m)
- indirect, general equilibrium effects depend on
  - the terms-of-trade offset, via expenditure switching (governed by trade elasticities σ_m, σ_x)
  - and the behaviour of consumption (crucially influenced by market completeness, see later)

Real marginal costs – flexible prices (supply side).

$$1 = MC_t^r = \frac{W_t/P_t^C}{A_t}\,\frac{P_t^C}{P_t^H} \; = \; \frac{1}{A_t}\,\underbrace{C_t^{\sigma_i}\left(\frac{Y_t}{A_t}\right)^{\varphi}}_{L^S:\ \frac{W_t}{P_t^C}}\,\mathcal{G}(S_t,\tau_t) \tag{B.15}$$

$$0 = mc_t^r \approx \sigma_i\,c_t \; + \; \varphi\,y_t \; - \; (1+\varphi)\,a_t \; + \; \alpha\,(s_t+\tau_t)$$

Setting $a_t = 0$ and using that under flexible prices $mc_t^r = 0$, the natural output is:

$$y_t^n = \underbrace{\underbrace{-\frac{\sigma_i}{\varphi}\,c_t}_{\text{inc. on } L^S} \; \underbrace{-\frac{\alpha}{\varphi}\,s_t}_{\text{subs. on } L^S}}_{\text{GE effects}} \; \underbrace{\underbrace{-\frac{\alpha}{\varphi}\,\tau_t}_{\text{subs. on } L^S}}_{\text{direct effect}} \tag{B.16}$$

- tariffs directly hurt supply, through adverse substitution effects on labour supply (by eroding the purchasing power of real wages)
- indirect general equilibrium effects depend on
  - the terms-of-trade offset, via substitution effect on labour supply (by altering the purchasing power of wages)
  - crucially, the behaviour of consumption, determining the income effect on labour supply

Market completeness. Up until this point Monacelli (2025) and Jeanne and Son (2024) are identical. From here on they diverge on the assumption of market completeness, which has major implications for consumption.

Consumption: complete markets. Monacelli (2025) assumes ***perfect international risk sharing***, such that the following condition holds *period-by-period*, providing full insurance:

$$C_t^{\sigma_i} = (C_t^*)^{\sigma_i} \frac{\mathcal{E}_t P_t^*}{P_t^C} = (C_t^*)^{\sigma_i} \frac{S_t}{\mathcal{G}(S_t, \tau_t)} \tag{B.17}$$

$$c_t \approx c_t^* + \sigma_i^{-1} \Big[(1-\alpha)\, s_t - \alpha\, \tau_t \Big] \tag{B.18}$$

- a ToT-appreciation $s_t \downarrow$ entails an *insurance payment to abroad*, to *lowering* consumption to keep it the same in terms of foreign goods (instead of enjoying the positive real income effect from the ToT-gain)
- higher tariffs also entail an insurance payment to foreigners, since they directly lead to a stronger REER by raising after-tax domestic consumer prices (even if ToT did not change, see (B.4))

Consumption: incomplete markets, permanent tariffs. Jeanne and Son (2024) consider incomplete markets. With permanent tariffs (and no other frictions), we jump immediately to the new steady state, where consumption is pinned down by the ***balance-of-payments constraint*** of the economy, mandating that the present value of future trade surpluses covers outstanding external debt. Assume that to be zero, so that trade must be balanced:

$$0 = TB_t = P_t^H X_t - P_t^M C_t^M =$$

$$\Big[\underbrace{\underbrace{P_t^H (C_t^H + X_t)}_{P_t^H Y_t \equiv GVA_t} + \tau_t P_t^M C_t^M}_{nGDP_t^{prod}} \Big] - \Big[\underbrace{P_t^H C_t^H + (1+\tau_t) P_t^M C_t^M}_{P_t^C C_t} \Big]$$

$$C_t = \underbrace{\frac{P_t^H}{P_t^C} Y_t + \tau_t \frac{P_t^M}{P_t^C} C_t^M}_{rGDI_t = \frac{nGDP_t}{P_t^C}} = \frac{Y_t}{\mathcal{G}(S_t, \tau_t)} + \tau_t \frac{S_t}{\mathcal{G}(S_t, \tau_t)} C_t^M \tag{B.19}$$

$$\begin{aligned} c_t + g(s_t, \tau_t) &\approx y_t + \alpha\, \tau_t \\ c_t &= y_t - \alpha\, s_t \end{aligned} \tag{B.20}$$

- a ToT-appreciation $s_t \downarrow$ now entails a ***terms-of-trade gain***, which can be consumed (real income effect)
- tariffs have no direct effect on consumption (since housholds internalise the government budget constraint, expecting tariff revenues to be handed back to them in the form of transfers)
- note the contrast with the full insurance case in (B.18), where ToT movements have *opposite*-signed effects on consumption (and where the BoP constraint does not matter ex post)
- this real income channel is important also with temporary tariff shocks as long as market incompleteness hampers consumption smoothing (e.g. via liquidity-constrained households)

AD curve. Substitute out c_t from the goods market clearing (B.14), using either (B.18) or (B.20), to arrive at an aggregate demand curve in the (y_t, s_t) space.

For complete markets:

$$y_t = \underbrace{\frac{(1-\alpha)}{\sigma_i}\left[(1-\alpha)\, s_t - \alpha\, \tau_t\right]}_{\text{(B.18): GE int'l risk-sharing}} + \underbrace{\left[(1-\alpha)\alpha\, \sigma_m + \alpha^*\, \sigma_x\right] s_t}_{\text{GE exp. switching}} + \underbrace{(1-\alpha)\alpha\, \sigma_m\, \tau_t}_{\text{direct exp. switching}} =$$

$$= \underbrace{\left[\frac{(1-\alpha)^2}{\sigma_i} + (1-\alpha)\,\alpha\,\sigma_m + \alpha^* \sigma_x\right]}_{\Omega_s\text{: AD slope}} s_t + \underbrace{\left[\left(\sigma_m - \frac{1}{\sigma_i}\right)(1-\alpha)\,\alpha\right]}_{\Omega_\tau\text{: AD shift}} \tau_t \qquad \text{(AD-CM)}$$

- which is exactly the condition from Monacelli (2025)
- the slope $\Omega_s > 0$, capturing expenditure switching effects from ToT-movements, as well as insurance from international risk sharing in response to ToT-movements
- whether AD is shifted right or left by tariffs, depends on $\sigma_m \lessgtr 1/\sigma_i$.
  - Note that the shift Ω_τ includes more than the direct effect of tariffs on aggrgate demand (via expenditure switching), containing also some GE insurance effects induced by tariffs on consumption.

For incomplete markets, permanent tariffs:

$$y_t = \underbrace{(1-\alpha)\left[y_t - \alpha\, s_t\right]}_{\text{(B.20): GE real income (ToT-gain)}} + \underbrace{\left[(1-\alpha)\alpha\, \sigma_m + \alpha^*\, \sigma_x\right] s_t}_{\text{GE exp. switching}} + \underbrace{(1-\alpha)\alpha\, \sigma_m\, \tau_t}_{\text{direct exp. switching}}$$

$$y_t = \underbrace{\left[(\sigma_m - 1)(1-\alpha) + \frac{\alpha^*}{\alpha}\sigma_x\right]}_{\Omega_s^*\text{: AD slope}} s_t + \underbrace{(1-\alpha)\,\sigma_m}_{\Omega_\tau^*\text{: AD shift}} \tau_t \qquad \text{(AD-IM)}$$

- which is exactly the condition from Jeanne and Son (2024)
- the slope Ω_s^* depends on the relative strength of the real income channel from ToT-movements (missing under complete markets) and the expenditure switching channel (governed by the trade elasticities σ_m, σ_x)
  - $\Omega_s^* > 0$ for reasonably large elasticities: $\sigma_m + \frac{\alpha^*}{\alpha\,(1-\alpha)}\,\sigma_x > 1$
- tariffs shift (AD-IM) unambigously to the right ($\Omega_\tau^* > 0$), due to expenditure switching from imports to domestically produced goods

AS curve. Substitute out c_t from real marginal costs (B.16), using either (B.18) or (B.20), to arrive at an aggregate supply curve in the (y_t, s_t) space.

For complete markets:

$$\begin{aligned} y_t^n &= \underbrace{-\frac{\sigma_i}{\varphi}\frac{1}{\sigma_i}\Big[(1-\alpha)\,s_t - \alpha\,\tau_t\Big]}_{\text{(B.18): GE inc. on } L^S \text{ (int'l risk-sh.)}} \quad \underbrace{-\frac{\alpha}{\varphi}\,s_t}_{\text{GE subs. on } L^S} \quad \underbrace{-\frac{\alpha}{\varphi}\,\tau_t}_{\text{direct subs. on } L^S} \\ &= \underbrace{-\frac{1}{\varphi}}_{\Theta_s\text{: AS slope}} \; s_t \end{aligned} \tag{AS-CM}$$

- which is consistent with Monacelli (2025)
- (AS-CM) is not shifted by tariffs either way:
  - the adverse substitution effect on labour supply from higher tariffs is offset by a positive income effect on labour supply due to lower consumption (that is induced by tariffs via the risk-sharing condition)
- the slope $\Theta_s < 0$
  - the beneficial substitution effect on labour supply from a stronger ToT is reinforced by a beneficial income effect due to lower consumption (that is induced by a stronger ToT via the risk-sharing condition)

For incomplete markets, permanent tariffs:

$$y_t^n = \underbrace{-\frac{\sigma_i}{\varphi}\Big(y_t - \alpha\,s_t\Big)}_{\text{(B.20): GE inc. on } L^S \text{ (ToT-gain)}} \quad \underbrace{-\frac{\alpha}{\varphi}\,s_t}_{\text{GE subs. on } L^S} \quad \underbrace{-\frac{\alpha}{\varphi}\,\tau_t}_{\text{direct subs. on } L^S}$$

$$y_t^n = \underbrace{\frac{\alpha}{\varphi+\sigma_i}\Big(\sigma_i - 1\Big)}_{\Theta_s^*\text{: AS slope}}\; s_t \underbrace{-\frac{\alpha}{\varphi+\sigma_i}}_{\Theta_\tau^*\text{: AS shift}}\;\tau_t \tag{AS-IM}$$

- which is exactly the condition from Jeanne and Son (2024)
- tariffs act as an adverse supply shock, shifting (AS-IM) to the left: $\Theta_\tau^* < 0$
  - this is the adverse substitution effect on labour supply, by eroding the purchasing power of wages
  - tariffs exert no income effect on labour supply, since they have no direct effect on consumption (as housholds internalise the government budget constraint, expecting tariff revenues to be handed back to them in the form of transfers)
- the slope of (AS-IM) with respect to ToT, Θ_s^* depends on the relative strength of income and substitution effects on labour supply, i.e. wehther $\sigma_i \lesseqgtr 1$
  - with $\sigma_i = 1$ (AS-IM) is vertical, and the terms-of-trade has no effect on aggregate supply
  - with $\sigma_i > 1$ (AS-IM) is negatively sloped in the (y_t, tot_t) plane, with $tot_t = -s_t$

Equilibrium – natural output. We can find the equilibrium natural output y_t^n in the two cases by substituting out s_t, setting (AD-CM)=(AS-CM), or (AD-IM)=(AS-IM).

For complete markets:

$$y_t^n = \underbrace{\left[\frac{(1-\alpha)^2}{\sigma_i} + (1-\alpha)\,\alpha\,\sigma_m + \alpha^*\sigma_x\right]}_{\Omega_s} \underbrace{\left(-\varphi\, y_t^n\right)}_{s_t \text{ (AS-CM)}} + \underbrace{\left[\left(\sigma_m - \frac{1}{\sigma_i}\right)(1-\alpha)\,\alpha\right]}_{\Omega_\tau} \tau_t$$

$$y_t^n = \frac{\Omega_\tau}{1+\varphi\,\Omega_s}\,\tau_t \qquad \text{(Y-CM)}$$

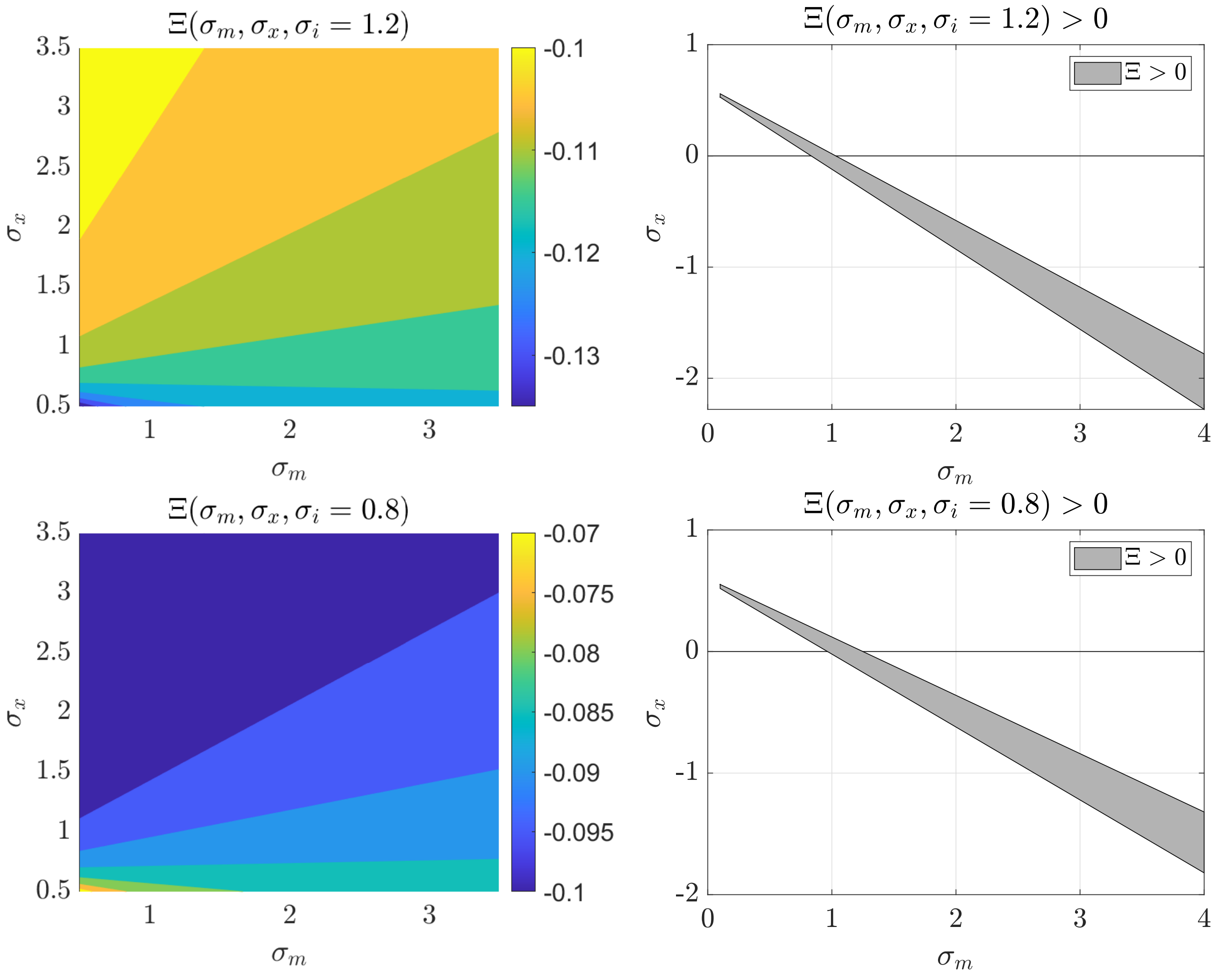
- Under complete markets, the effect of tariffs on natural output is ***positive*** when $\Omega_\tau > 0$, which happens when $\sigma_m > 1/\sigma_i$. This is the exact condition in Monacelli (2025).
  - Tariffs shift AD to the right, leading to a ToT appreciation along a fixed AS curve, which boosts labour supply (both via substitution and income effects!), and thereby output.
  - The crucial ingredient here is that the adverse L^S substitution effect from tariffs is offset by a beneficial income effect (due to the insurance payment), thereby preventing the leftward shift of the AS.
- So the result relies heavily on the assumption of perfect international risk sharing, which lowers consumption.

For incomplete markets:

$$y_t^n = \frac{\Theta_s^*\Omega_\tau^* - \Theta_\tau^*\Omega_s^*}{\Theta_s^* - \Omega_s^*}\,\tau_t = \underbrace{\left[\frac{\alpha\left[\sigma_x + (1-\alpha)\left(\sigma_i\sigma_m - 1\right)\right]}{\alpha(\sigma_i - 1) - (\varphi+\sigma_i)\left[\sigma_x + (1-\alpha)(\sigma_m - 1)\right]}\right]}_{\Xi}\tau_t \qquad \text{(Y-IM)}$$

- Under incomplete markets, the effect of tariffs on natural output is ***negative*** when $\Xi < 0$ which holds in most cases, except for a small region of the parameter space (see Figure B.3).
  - Tariffs shift AD unambigously to the right due to expenditure switching, and leading to offsetting ToT appreciation.
  - However, tariffs also act as an adverse supply shock (via adverse substitution effects on L^S), shifting AS to the left, and leading to even more ToT appreciation.
  - The balance of these forces on output (Ξ), is in principle ambiguous, but is negative for reasonable parameters (see Figure B.3).
- The real income channel is at play now: households can consume the terms-of-trade gain (instead of decreasing consumption due to having to make insurance payouts).
  - Therefore, and crucially, tariffs now have no direct L^S-boosting income effect, since they do not lead directly to lower consumption: households internalise the government budget constraint, and don't feel their lifetime disposable income dropping, neither do they have to make insurance payment to foreigners.

Figure B.3: Tariff semi-elasticity of output under incomplete markets and permanent tariffs (Ξ), depending on trade elasticities (σ_m, σ_x), with $\alpha = 0.4, \varphi = 3, \sigma_i = \{0.8, 1.2\}$



Source: European Commission services.

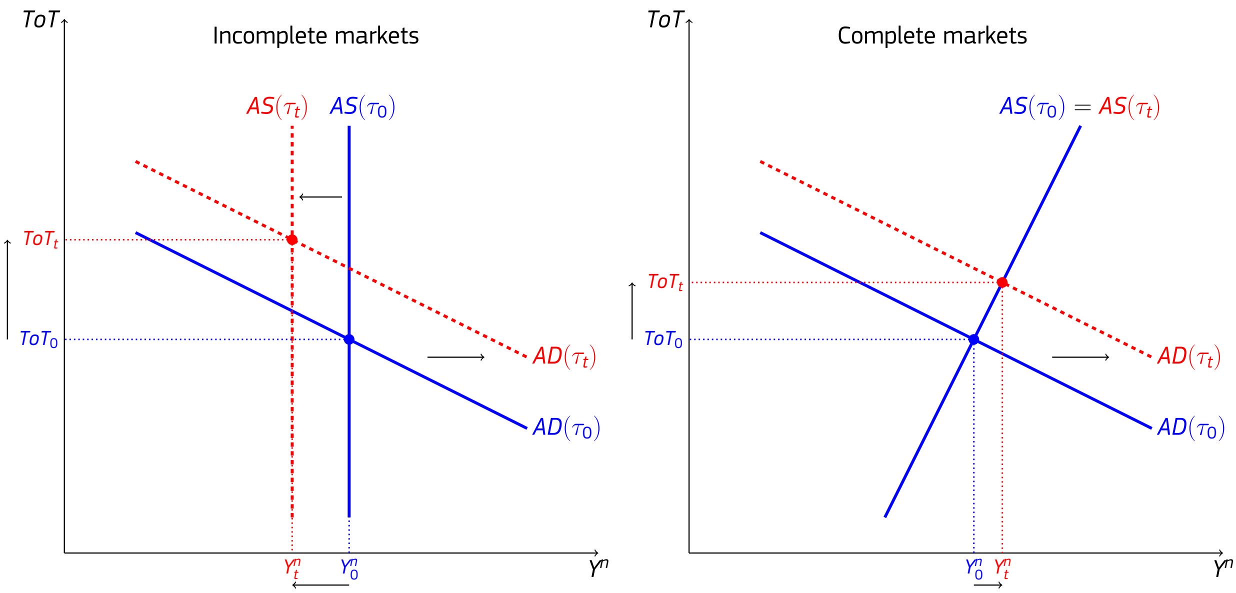
As Figure B.3 shows, the sign of the the output effect Ξ's derivative with respect to trade elasticities is ambiguous: it depends on the other parameters.

- more elastic imports $\sigma_m \uparrow$ increase the rightward shift in AD induced by tariff – but also make AD flatter, depending in openness.
- more elastic exports $\sigma_x \uparrow$ make AD flatter
- all these effects interact with the slope of the AS, too (which depends crucially on σ_i)

Further channels. On top of the above described *natural* (flexible price) equilibrium in representative agent New Keynesian economies, additional model ingredients entail further channels:

- **temporary/persistent tariffs:** If tariffs are expected to decline in the future, they act as an intertemporal tax on consumption, inducing ***intertemporal substitution***. This can approximate the effect of perfect risk-sharing even under incomplete markets.
- **sticky prices and monetary policy:** Monetary tightening can then raise the real interest rate above its natural level, inducing additional ***intertemporal substitution*** effects, and hurting aggregate demand. (This is the main focus in Monacelli (2025).)
- **imported intermediate inputs:** Tariffs influence production costs not just indirectly via labour supply,

Figure B.4: *Natural* goods market equilibrium, tariffs and terms-of-trade ($\sigma_i = 1 < \sigma_m$)



Note: Comparison between incomplete markets, permanent tariffs (Jeanne and Son, 2024) and complete markets (Monacelli, 2025), representing the *natural* goods market equilibrium under flexible prices. The terms-of-trade is denoted by *ToT* while real output is denoted by *Y*. An increase in tariffs τ_t shifts the economy from the original (time 0, blue) equilibrium to the new (time t, red) equilibrium. Curves are illustrations of (AD-IM),(AS-IM), (AD-CM), (AS-CM), with $ToT_t = -s_t$, and $\sigma_i = 1 < \sigma_m$. *Source:* European Commission services.

but also directly via imported inputs. However, GDP (value added) is no longer the same as output, and there might be ***"supply side expenditure switching"*** from more expensive imported inputs towards domestic value added, depending on elasticities inside the production technology (Auclert, Rognlie and Straub, 2025).

- **delayed import substitution:** Trade elasticities might be smaller in the short run than their long-run values, leading to an initially sluggish expenditure switching channel.
- **liquidity constraints:** High-MPC households, who consume their current income, cannot engage in consumption smoothing. This strengthens the real income channel from ToT-movements, even if they are temporary. In addition, these households cannot internalise the government budget constraint, so tariffs lower consumption unless they are immediately handed back by the government in the form of transfers. This modifies direct demand effects as well as income effects on labour supply from tariffs.

B.2.1. Intermediate inputs in production

Imported intermediate inputs introduce another channel through which tariffs can affect the economy. To see how this changes the above-described transmission mechanism, we append that simple model with imported intermediates. In addition to intermediates amplifying the adverse supply side effects induced by tariffs, expenditure switching now happens less through the consumption basket and more on the production side, as firms substitute from imported inputs towards domestic value added (GDP). In general, adding intermediates makes the GDP-response to tariffs more negative, so that even under complete markets and flexible prices it is harder to achieve a positive effect. Auclert, Rognlie and Straub (2025) present a similar model, but with no imports in consumption ($\alpha = 0$), and with sticky wages (while below we focus on the natural equilibrium without nominal rigidities).

Real marginal costs – flexible prices (supply side). CES production structure (formerly $\eta = 1, \psi = 0$), with demand functions, and real marginal costs:

$$Y_t = A_t \left[(1-\psi)^{\frac{1}{\eta}} N_t^{\frac{\eta-1}{\eta}} + \psi^{\frac{1}{\eta}} \left(Int_t^M\right)^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{1-\eta}} \tag{B.21}$$

$$N_t = (1-\psi)\, Y_t \left(\frac{\frac{W_t}{P_t^C}\, \mathcal{G}(S_t, \tau_t)}{MC_t^r} \right)^{-\eta} A_t^{\eta-1} \tag{B.22}$$

$$Int_t^M = \psi\, Y_t \left[\frac{(1+\tau_t)\, S_t}{MC_t^r} \right]^{-\eta} A_t^{\eta-1} \tag{B.23}$$

$$MC_t^r = \frac{1}{A_t} \left[(1-\psi) \left(\frac{W_t}{P_t^C}\, \mathcal{G}(S_t, \tau_t) \right)^{1-\eta} + \psi \left[(1+\tau_t)\, S_t \right]^{1-\eta} \right]^{\frac{1}{1-\eta}} \tag{B.24}$$

with log-linear forms as:

$$y_t \approx (1-\psi)\, n_t + \psi\, int_t^M \tag{B.25}$$

$$n_t \approx y_t - \eta \Big[(w_t - p_t^C) + g(s_t, \tau_t) \Big] + \eta\, mc_t^r + (\eta - 1) a_t \tag{B.26}$$

$$int_t^M \approx y_t - \eta \Big[s_t + \tau_t \Big] + \eta\, mc_t^r + (\eta - 1) a_t \tag{B.27}$$

$$\begin{aligned} mc_t^r &\approx (1-\psi) \Big[(w_t - p_t^C) + g(s_t \tau_t) \Big] + \psi \Big(s_t + \tau_t \Big) - a_t = \\ &= (1-\psi) \left(w_t - p_t^C \right) + \Big[(1-\psi)\alpha + \psi \Big] \left(s_t + \tau_t \right) - a_t \end{aligned} \tag{B.28}$$

Plug in $a_t = 0$, the labour supply, production function (B.25), and intermediate input demand (B.27), already setting $mc_t^r = 0$ (indicating flexible prices):

$$\begin{aligned} 0 &= (1-\psi)\, \underbrace{(\sigma_i\, c_t + \varphi\, n_t)}_{w_t - p_t^C} + \Big[(1-\psi)\alpha + \psi \Big] \left(s_t + \tau_t \right) = \\ &= (1-\psi)\, \sigma_i\, c_t + (1-\psi)\, \varphi\, \underbrace{\left[y_t + \eta \frac{\psi}{1-\psi} (s_t + \tau_t) \right]}_{\text{(B.25),(B.27): } n_t} + \Big[(1-\psi)\alpha + \psi \Big] \left(s_t + \tau_t \right) \end{aligned}$$

$$y_t^n = \underbrace{-\frac{\sigma_i}{\varphi} c_t}_{\text{inc. on } L^S} \underbrace{\underbrace{- \frac{\alpha}{\varphi} (s_t + \tau_t)}_{\text{subs. on } L^S} \underbrace{- \frac{\psi}{(1-\psi)\, \varphi} (s_t + \tau_t)}_{\text{dir. } Int_t^M \text{ costs}}}_{\text{adverse cost effect}} \underbrace{- \eta \frac{\psi}{1-\psi} (s_t + \tau_t)}_{\text{supply-side exp.switching}} \tag{B.29}$$

GDP (domestic value added) $rgdp_t$ is no longer the same as gross output y_t:

$$rgdp_t^n \equiv n_t^n = \left[y_t^n + \eta \frac{\psi}{1-\psi} (s_t + \tau_t) \right] = \tag{B.30}$$

$$= \underbrace{-\frac{\sigma_i}{\varphi} c_t}_{\text{inc. on } L^S} \underbrace{\underbrace{- \frac{\alpha}{\varphi} (s_t + \tau_t)}_{\text{subs. on } L^S} \underbrace{- \frac{\psi}{(1-\psi)\, \varphi} (s_t + \tau_t)}_{\text{dir. } Int_t^M \text{ costs}}}_{\text{adverse cost effect}} \tag{B.31}$$

- (B.29) features two additional terms relative to (B.16)

  - ***intermediate input costs*** also have an adverse effect on the aggregate supply of gross output (to the extent those inputs are used in production, $\psi > 0$)
  - there is ***supply-side expenditure switching***: as imported inputs Int_t^M become more expensive, production shifts towards using more domestic labour, pushing up wage costs, in relation with the elasticity of substitution, $\eta > 0$
- wage costs now depend on three factors:
  - income effect on L^S: higher consumption lowers labour supply
  - substitution effect on L^S: weaker ToT and higher tariffs erode the purchasing power of wages, hurting labour supply (to the extent households consume imported goods, $\alpha > 0$)
  - expenditure switching: more expensive imported inputs (due to weaker ToT or higher tariffs) raise firm demand for labour, pushing up wages as well ($\eta > 0$)
- Gross output is no longer equal to real GDP (domestic value added). Removing intermediate inputs from output, the only source of domestic value added is labour.
  - Real GDP can grow either because gross output y_t^n grows, or because for given output there is expenditure switching from imported inputs towards domestic value added (depending on $\eta > 0$). See (B.30).
  - Expenditure switching has beneficial effects for real GDP which exactly offset (!) its adverse effects (via raising wage costs) on gross output supply, thereby dropping out from the supply relation for $rgdp_t^n$, (B.31).
- ***intermediate input channel:*** $\psi > 0$ increases the sensitivity of aggregate supply to the price of production inputs, raised either by tariffs or ToT-depreciation.

Goods market clearing (demand side). (B.13) is unchanged with imported inputs. But the steady state is different: to achieve a zero trade balance, paying for imported inputs requires lower consumption out of gross output. [49] So the log-linear goods market clearing (B.14) around the zero trade balance steady state is modified as below:

$$y_t \approx (1-\alpha)(1-\psi)\, c_t + \underbrace{\Big[\alpha + \psi(1-\alpha)\Big]\, \sigma_x\, s_t}_{\text{demand-side exp. switching (via exports)}} + \underbrace{\alpha\,(1-\alpha)(1-\psi)\,\sigma_m\,\big(s_t+\tau_t\big)}_{\text{demand-side exp. switching (via imports)}} \tag{B.32}$$

$$rgdp^n_t \overset{(B.30),(B.32)}{=} (1-\alpha)(1-\psi)\, c_t + \underbrace{\Big[\alpha + \psi(1-\alpha)\Big]\, \sigma_x\, s_t}_{\text{demand-side exp. switching (via exports)}} +$$
$$+ \underbrace{\alpha\,(1-\alpha)(1-\psi)\,\sigma_m\,\big(s_t+\tau_t\big)}_{\text{demand-side exp. switching (via imports)}} + \underbrace{\eta\frac{\psi}{1-\psi}\big(s_t+\tau_t\big)}_{\text{supply-side exp. switching}} \tag{B.33}$$

- where the AD relation is very similar to (B.14)
  - But the coefficients on (imported) consumption are scaled down by $(1-\psi)$: consumption is now a lower share of output, since now not all output represents income for domestic agents. Consequently, any demand shift towards domestically produced consumption affects gross output to a lesser extent.
  - In contrast, exports now represent a larger weight (as they need to cover the additional bill for imported inputs)
- for real GDP (as before) there is an extra term capturing ***supply side expenditure switching*** from imported inputs towards domestic value added (depending on $\eta > 0, \psi > 0$)
  - expenditure switching happens less via imports in the consumption basket ($\alpha\,\sigma_m > 0$ scaled down by $(1-\psi)$)...
  - ...and more via imported intermediate inputs ($\psi\,\eta > 0$), and via exports ($\sigma_x > 0$)
  - but the overall strength of expenditure switching induced by tariffs can be higher or lower:

$$\alpha\,(1-\alpha)(1-\psi)\,\sigma_m + \eta\frac{\psi}{1-\psi} \lessgtr \alpha\,(1-\alpha)\,\sigma_m$$
$$\frac{\eta}{1-\psi} \lessgtr \alpha\,(1-\alpha)\,\sigma_m$$

[49] Using the steady state $X \equiv \alpha^* Y$.

$$Y = C^H + X \; = \; (1-\alpha)\, C + X$$
$$C = \frac{1-\alpha^*}{1-\alpha}\, Y$$

For zero trade balance in steady state the country would need to export $\alpha^* = 1-(1-\alpha)(1-\psi)$, such that

$$C = \frac{1-\alpha^*}{1-\alpha}\, Y = (1-\psi)\, Y$$
$$TB = Y - Int^M - C \; = \; (1-\psi)\, Y - (1-\psi)\, Y = 0$$

which requires $\alpha^* = \alpha$ when $\psi = 0$.

Balance of payments and national accounts.

$$nGDP_t \equiv \underbrace{P_t^H Y_t - (1+\tau_t) P_t^M\, Int_t^M}_{GVA_t} + \tau_t P_t^M \left(C_t^M + Int_t^M \right) = \tag{B.34}$$

$$= P_t^H Y_t - P_t^M\, Int_t^M + \tau_t P_t^M\, C_t^M =$$

$$= P_t^H \underbrace{\left(C_t^H + X_t \right)}_{Y_t} - P_t^M\, Int_t^M + \tau_t P_t^M\, C_t^M =$$

$$= \underbrace{\left[P_t^H C_t^H + (1+\tau_t) P_t^M C_t^M \right]}_{P_t^C C_t} + \underbrace{\left[P_t^H X_t - P_t^M \underbrace{\left(C_t^M + Int_t^M \right)}_{M_t} \right]}_{TB_t} \tag{B.35}$$

$$rGDP_t \equiv \underbrace{C_t^H + X_t}_{Y_t = N_t + Int_t^M} - Int_t^M = N_t \tag{B.36}$$

For long-term zero trade balance, consumption must comply with:

$$0 = TB_t = \underbrace{\left[P_t^H Y_t - P_t^M\, Int_t^M + \tau_t P_t^M\, C_t^M \right]}_{nGDP_t} - P_t^C\, C_t$$

$$C_t = \frac{P_t^H}{P_t^C}\, Y_t - \frac{P_t^M}{P_t^C} Int_t^M + \tau_t \frac{P_t^M}{P_t^C}\, C_t^M =$$

$$= \frac{1}{\mathcal{G}(S_t, \tau_t)}\, Y_t - \frac{S_t}{\mathcal{G}(S_t, \tau_t)}\, Int_t^M + \tau_t \frac{S_t}{\mathcal{G}(S_t, \tau_t)}\, C_t^M \tag{B.37}$$

$$c_t \approx \frac{1}{1-\psi} y_t - \frac{\psi}{1-\psi} (int_t^m + s_t) + \alpha \tau_t - \underbrace{\alpha (s_t + \tau_t)}_{g(s_t, \tau_t)} =$$

$$= y_t + \underbrace{\eta \frac{\psi}{1-\psi} (s_t + \tau_t)}_{\text{exp.switching from } Int^M \text{ to } GVA} - \underbrace{\left(\frac{\psi}{1-\psi} + \alpha \right) s_t}_{\text{ToT-gain on } Int^M \text{ and } C^M} =$$

$$= rgdp_t - \underbrace{\left(\frac{\psi}{1-\psi} + \alpha \right) s_t}_{\text{ToT-gain on } Int^M \text{ and } C^M} \tag{B.38}$$

- income from domestic value added production (GDP) as well as the ***terms-of-trade gain*** can be consumed (***real income channel***)
  - the ToT-gain is now realised both on imported consumption ($\alpha > 0$) and on imported inputs ($\psi > 0$)
- tariffs have no direct effect on consumption, since households internalise the government budget constraint, expecting tariff revenues to be eventually handed back to them in the form of transfers/tax cuts

AD curve. Substitute out c_t from the goods market clearing (B.33), using either (B.18) or (B.38), to arrive at an aggregate demand curve in the $(rgdp^n_t, s^n_t)$ space.

For complete markets:

$$rgdp^n_t = \underbrace{\frac{(1-\alpha)(1-\psi)}{\sigma_i}\Big[(1-\alpha)\,s_t - \alpha\,\tau_t\Big]}_{\text{(B.18): GE int'l risk sharing}} + \underbrace{\big[\alpha+\psi(1-\alpha)\big]\,\sigma_x\,s_t}_{\text{demand-side exp. switching (via exports)}} +$$

$$+ \underbrace{\alpha\,(1-\alpha)(1-\psi)\,\sigma_m\,\big(s_t+\tau_t\big)}_{\text{demand-side exp. switching (via imports)}} + \underbrace{\eta\,\frac{\psi}{1-\psi}(s_t+\tau_t)}_{\text{supply-side exp.switching}} =$$

$$= \underbrace{\left[(1-\psi)\left(\frac{(1-\alpha)^2}{\sigma_i} + \alpha\,(1-\alpha)\,\sigma_m\right) + \big[\alpha+\psi(1-\alpha)\big]\,\sigma_x + \eta\,\frac{\psi}{1-\psi}\right]}_{\Lambda_s\text{: AD slope}} s_t +$$

$$+ \underbrace{\left[\alpha\,(1-\alpha)\,(1-\psi)\,\left(\sigma_m - \frac{1}{\sigma_i}\right) + \eta\,\frac{\psi}{1-\psi}\right]}_{\Lambda_\tau\text{: AD shift}} \tau_t \qquad \text{(AD-gdp-CM)}$$

- the AD slope $\Lambda_s > 0$ capturing expenditure switching effects from ToT-movements (on imports and exports) as well as insurance from international risk sharing in response to ToT-movements
- whether AD is shifted right or left by tariff hikes, depends on $\Lambda_\tau \lesseqgtr 1$
  - demand-side expenditure switching (for domestic consumption goods) ($\alpha \neq 0, \sigma_m > 0$)
  - supply-side expenditure switching (for domestic value added, GDP) ($\psi \neq 0, \eta > 0$)
  - insurance effects induced by tariffs on consumption ($\alpha \neq 0, \sigma_i$)
- the effect of intermediate inputs $\psi > 0$ on both the slope and the shift Λ_s, Λ_τ is ambiguous
  - $\Rightarrow$ expenditure switching happens less via imports in the consumption basket and more via imported intermediate inputs, and via exports

For incomplete markets, permanent tariffs:

$$rgdp^n_t = \underbrace{(1-\alpha)(1-\psi)\left[rgdp^n_t - \left(\frac{\psi}{1-\psi} + \alpha\right) s_t\right]}_{\text{(B.38): GE real income (ToT-gain)}} + \underbrace{\big[\alpha+\psi(1-\alpha)\big]\,\sigma_x\,s_t}_{\text{demand-side exp. switching (via exports)}} +$$

$$+ \underbrace{\alpha\,(1-\alpha)(1-\psi)\,\sigma_m\,\big(s_t+\tau_t\big)}_{\text{demand-side exp. switching (via imports)}} + \underbrace{\eta\,\frac{\psi}{1-\psi}(s_t+\tau_t)}_{\text{supply-side exp.switching}} =$$

$$rgdp^n_t = \underbrace{\left\{\frac{1-\alpha}{\alpha+\psi\,(1-\alpha)}\Big[(1-\psi)\alpha\,(\sigma_m-1) + \psi(\eta-1)\Big] + \eta\,\frac{\psi}{1-\psi} + \sigma_x\right\}}_{\Lambda^*_s\text{: AD slope}} s_t +$$

$$+ \underbrace{\left\{\frac{1-\alpha}{\alpha+\psi\,(1-\alpha)}\Big[(1-\psi)\,\alpha\,\sigma_m + \eta\,\psi\Big] + \eta\,\frac{\psi}{1-\psi}\right\}}_{\Lambda^*_\tau\text{: AD shift}} \tau_t \qquad \text{(AD-gdp-IM)}$$

- AD slope Λ_s^* depends on the balance between ***expenditure switching*** (competitiveness) and ***real income*** (ToT-gain) channels
  - $\Lambda_s^* > 0$ for reasonably high enough σ_m, σ_x, η
  - intermediate inputs $\psi > 0$ matter by scaling *down* the exp.switching effects via imported consumption in line with σ_m, but raising them through imported inputs in line with η, and through exports in line with σ_x
- tariffs shift AD unambigously to the right $(\Lambda_\tau^* > 0)$ due to
  - expenditure switching from imported consumption towards domestically produced goods (demand side)
  - expenditure switching from imported inputs towards domestic value added (supply side) – which raises real income (for given output)
  - no real income effect from tariffs, as households internalise the government budget

AS curve. Substitute out c_t from real marginal costs (B.29), using either (B.18) or (B.38), to arrive at an aggregate supply curve in the (y_t, s_t) space, or in the $(rgdp_t, s_t)$ space.

For complete markets:

$$rgdp_t^n = \underbrace{-\frac{1}{\varphi}\left[(1-\alpha)\,s_t - \alpha\,\tau_t\right]}_{\text{(B.18): inc. on } L^S \text{ (int'l risk-sh.)}} \underbrace{\underbrace{-\frac{\alpha}{\varphi}(s_t+\tau_t)}_{\text{subs. on } L^S} \underbrace{-\frac{\psi}{(1-\psi)\,\varphi}(s_t+\tau_t)}_{\text{dir. } Int_t^M \text{ costs}}}_{\text{adverse cost effect}} =$$

$$= \underbrace{-\frac{1}{(1-\psi)\,\varphi}}_{\Phi_s\text{: AS slope}} s_t \underbrace{-\frac{\psi}{(1-\psi)\,\varphi}}_{\Phi_\tau\text{: AS shift}} \tau_t \qquad \text{(AS-gdp-CM)}$$

- tariffs shift (AS-gdp-CM) unambiguously to the left, $\Phi_\tau < 0$ (in contrast to (AS-CM)):
  - the adverse substitution effect on labour supply from higher tariffs is offset by a positive income effect on labour supply due to lower consumption (that is induced by tariffs via the risk-sharing condition)
  - but now there is an additional adverse supply side effect as ***intermediate input costs*** $\psi > 0$ are raised by tariffs
- the AS slope $\Phi_s < 0$
  - the beneficial substitution effect on labour supply from a stronger ToT is reinforced by a beneficial income effect due to lower consumption (that is induced by a stronger ToT via the risk-sharing condition)
  - a stronger ToT can now further lower production costs via intermediate inputs
- intermediate inputs $\psi > 0$ increase the sensitivity of aggregate supply to the price of production inputs, raised either by tariffs or by ToT depreciation

For incomplete markets, permanent tariffs:

$$rgdp_t^n = \underbrace{-\frac{\sigma_i}{\varphi}\left[rgdp_t^n - \left(\frac{\psi}{1-\psi} + \alpha\right) s_t\right]}_{\text{(B.38): inc. on } L^S \text{ (ToT-gain)}} \underbrace{- \underbrace{\frac{\alpha}{\varphi}(s_t + \tau_t)}_{\text{subs. on } L^S} - \underbrace{\frac{\psi}{(1-\psi)\,\varphi}(s_t + \tau_t)}_{\text{dir. } Int_t^M \text{ costs}}}_{\text{adverse cost effect}} =$$

$$rgdp_t^n = \underbrace{-\left[\frac{(1-\sigma_i)}{\sigma_i + \varphi}\,\frac{\alpha + \psi\,(1-\alpha)}{1-\psi}\right]}_{\Phi_s^*:\text{ AS slope}} s_t \; \underbrace{-\left[\frac{1}{\sigma_i + \varphi}\,\frac{\alpha + \psi\,(1-\alpha)}{1-\psi}\right]}_{\Phi_\tau^*:\text{ AS shift}} \tau_t \qquad \text{(AS-gdp-IM)}$$

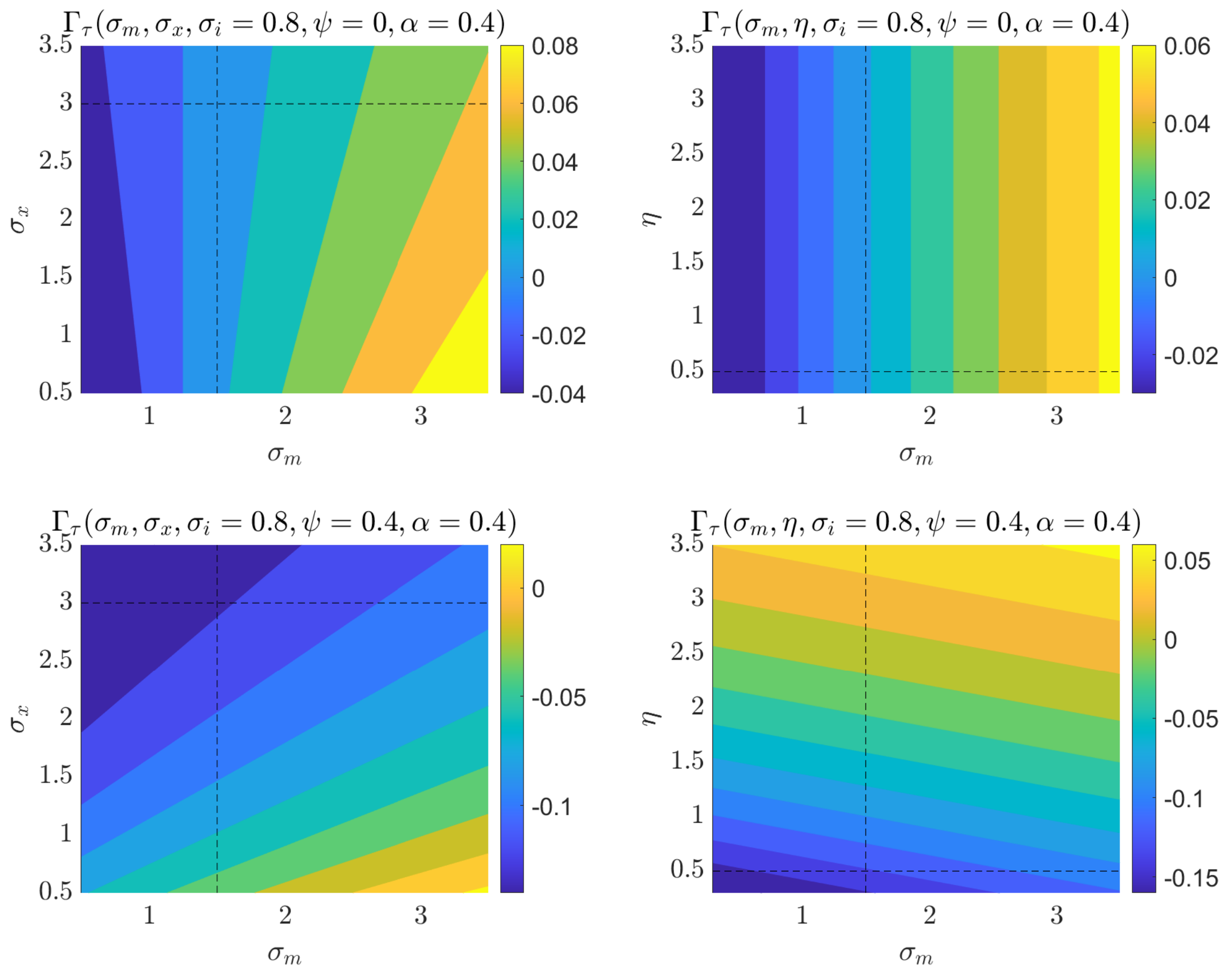
- ToT-gain can now be consumed
  - The ToT-gain is now realised not just on imported consumption $\alpha > 0$ but also on the cost of imported inputs $\psi > 0$ (leaving more income for consumption).
  - This *raises* consumption, creating an adverse income effect on labour supply...
  - ...that has to be set off against the beneficial substitution effect on labour supply, stemming from a stronger ToT raising the purchasing power of wages (via the consumption basket $\alpha > 0$)...
  - ...as well as against the beneficial cost effect on intermediate inputs ($\psi > 0$) as a stronger ToT makes imports cheaper.
- sign of AS slope Φ_s still depends on $\sigma_i \lessgtr 1$, but supply is generally more sensitive to ToT for higher $\psi > 0$
- AS is shifted by tariffs unambiguously to the left $\Phi_s < 0$, the more so with higher $\psi > 0$: tariffs act as an adverse supply shock.
  - higher tariffs generate an adverse substitution effect on labour supply as they erode the purchasing power of wages (via the consumption basket $\alpha > 0$)...
  - ...as well as an adverse cost effect on intermediate inputs ($\psi > 0$) as tariffs make import cheaper
  - tariffs exert no income effect on labour supply, since they have no direct effect on consumption (as housholds internalise the government budget constraint, expecting tariff revenues to be handed back to them in the form of transfers)

Equilibrium – "Natural GDP". Plugging (AD-gdp-CM) into (AS-gdp-CM) for complete markets, and (AD-gdp-IM) into (AS-gdp-IM) for incomplete markets:

$$rgdp_t^{n,CM} = \underbrace{\frac{\Phi_s\,\Lambda_\tau - \Phi_\tau\,\Lambda_s}{\Phi_s - \Lambda_s}}_{\Gamma_\tau} \tau_t \qquad \text{(GDP-CM)}$$

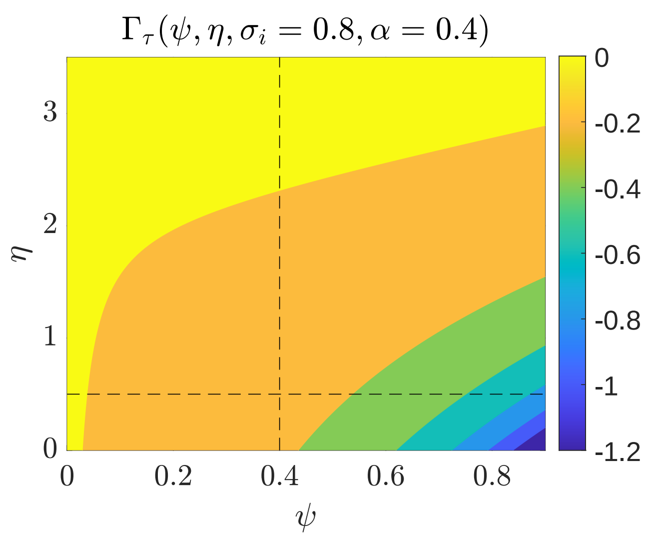
$$rgdp_t^{n,IM} = \underbrace{\frac{\Phi_s^*\,\Lambda_\tau^* - \Phi_\tau^*\,\Lambda_s^*}{\Phi_s^* - \Lambda_s^*}}_{\Gamma_\tau^*} \tau_t \qquad \text{(GDP-IM)}$$

Figure B.5: Tariff semi-elasticity of real GDP under **complete markets** (Γ_τ), depending on trade elasticities $(\sigma_m, \sigma_x, \eta)$, with $\psi = \{0, 0.4\}, \alpha = 0.4, \varphi = 3, \sigma_i = 0.8$. GDP response is higher if import elasticities $(\sigma_m, \eta) \uparrow$ are higher or if export elasticity $\sigma_x \downarrow$ is lower.



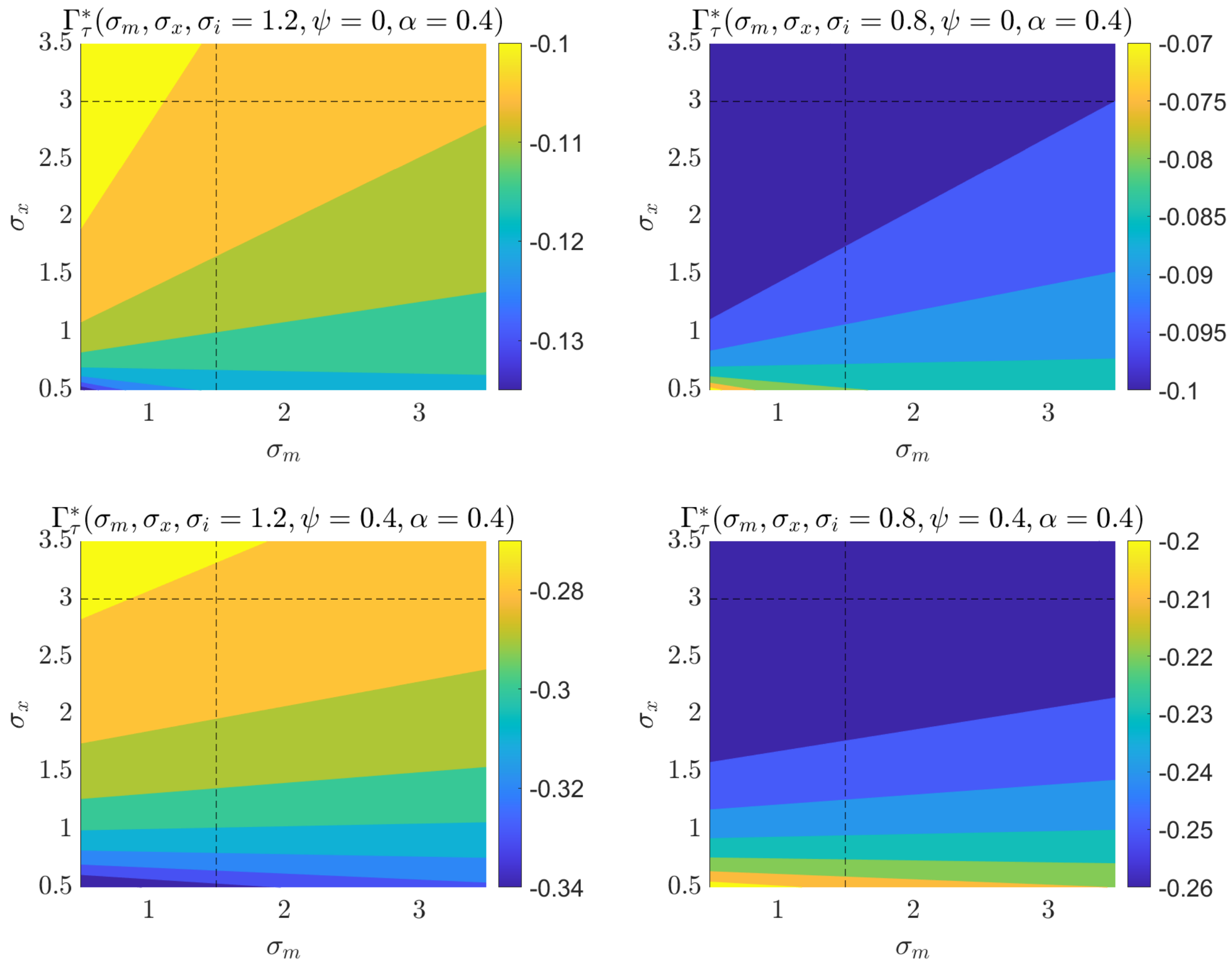
Source: European Commission services.

Figure B.6: Tariff semi-elasticity of real GDP under **complete markets** (Γ_τ), depending (ψ, η), with $\alpha = 0.4, \sigma_m = 1.5, \sigma_x = 3, \varphi = 3, \sigma_i = 0.8$. Imported inputs $\psi > 0$ lower the GDP response.



Source: European Commission services.

Figure B.7: Tariff semi-elasticity of real GDP under **incomplete markets** (Γ_τ^*), depending on trade elasticities (σ_m, σ_x), with $\eta = 0.5, \alpha = 0.4, \psi = \{0, 0.4\}, \varphi = 3, \sigma_i = \{0.8, 1.2\}$. GDP response is affected in opposite ways by import elasticity σ_m and export elasticity σ_x, but the sign of these effects depends on the slope of AS (Φ_s^*), which in turn depends on $\sigma_i \lessgtr 1$ (left vs right panels). For $\sigma_i = 1$ trade elasticities do not matter for Γ_τ^*.



Source: European Commission services.

Figure B.8: Tariff semi-elasticity of real GDP under **incomplete markets** (Γ_τ^*), depending on trade elasticities (σ_m, η), with $\sigma_x = 3, \alpha = 0.4, \psi = \{0, 0.4\}, \varphi = 3, \sigma_i = \{0.8, 1.2\}$. GDP response is affected in opposite ways by import elasticity η and export elasticity σ_x, but the sign of these effects depends on the slope of AS (Φ_s^*), which in turn depends on $\sigma_i \lessgtr 1$ (left vs right panels). For $\sigma_i = 1$ trade elasticities do not matter for Γ_τ^*.

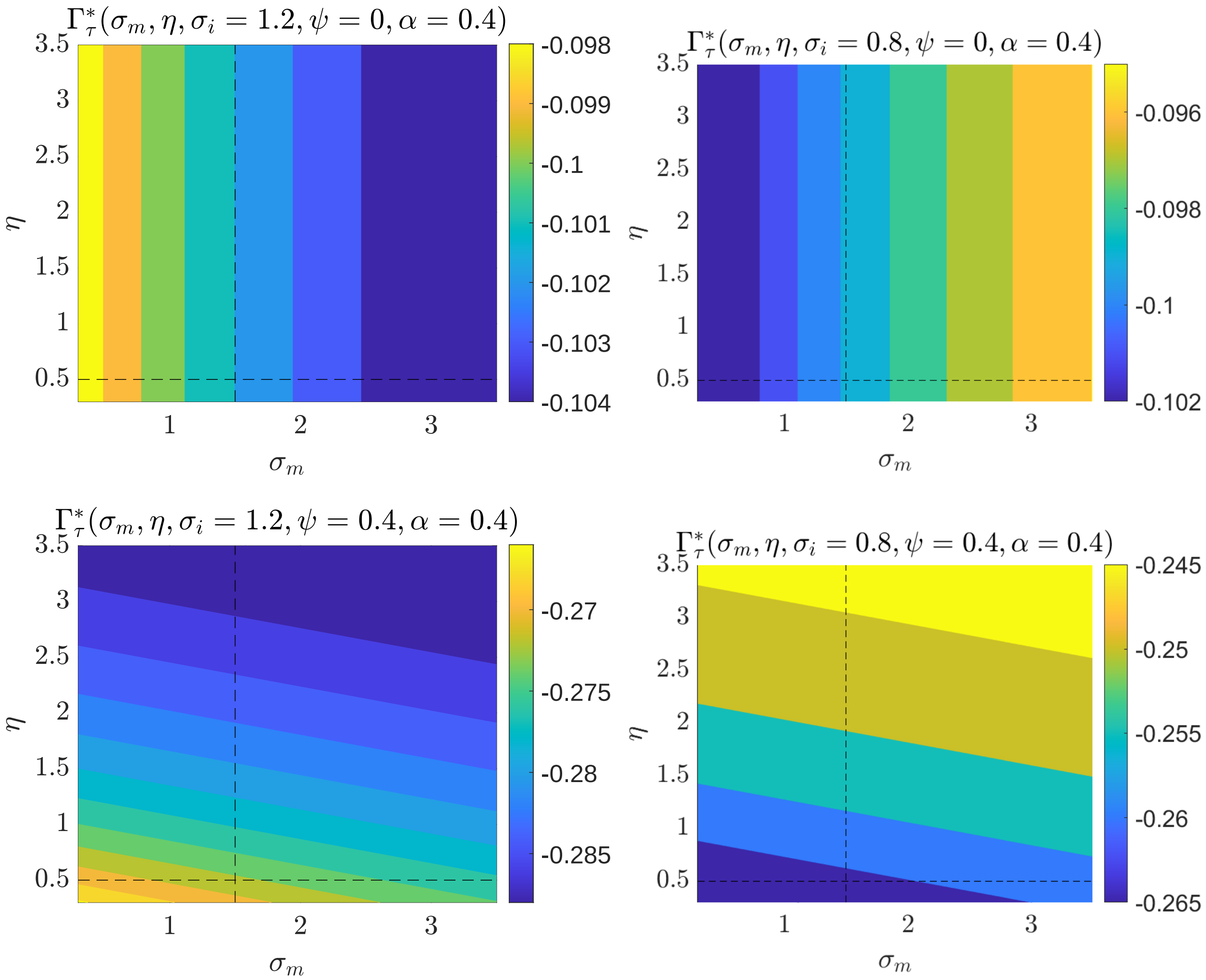
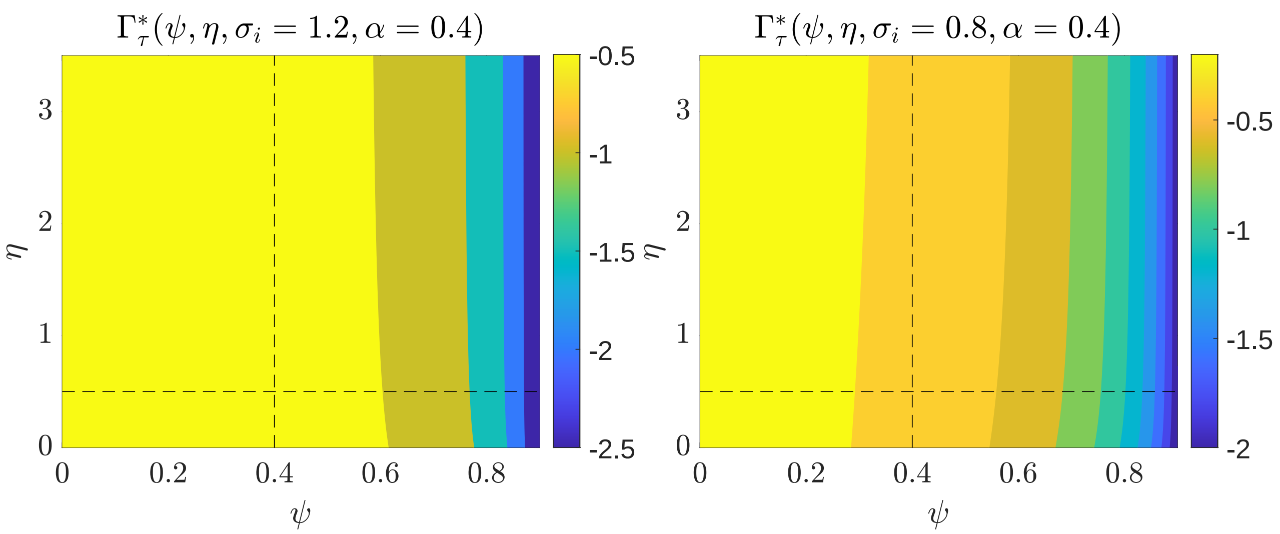


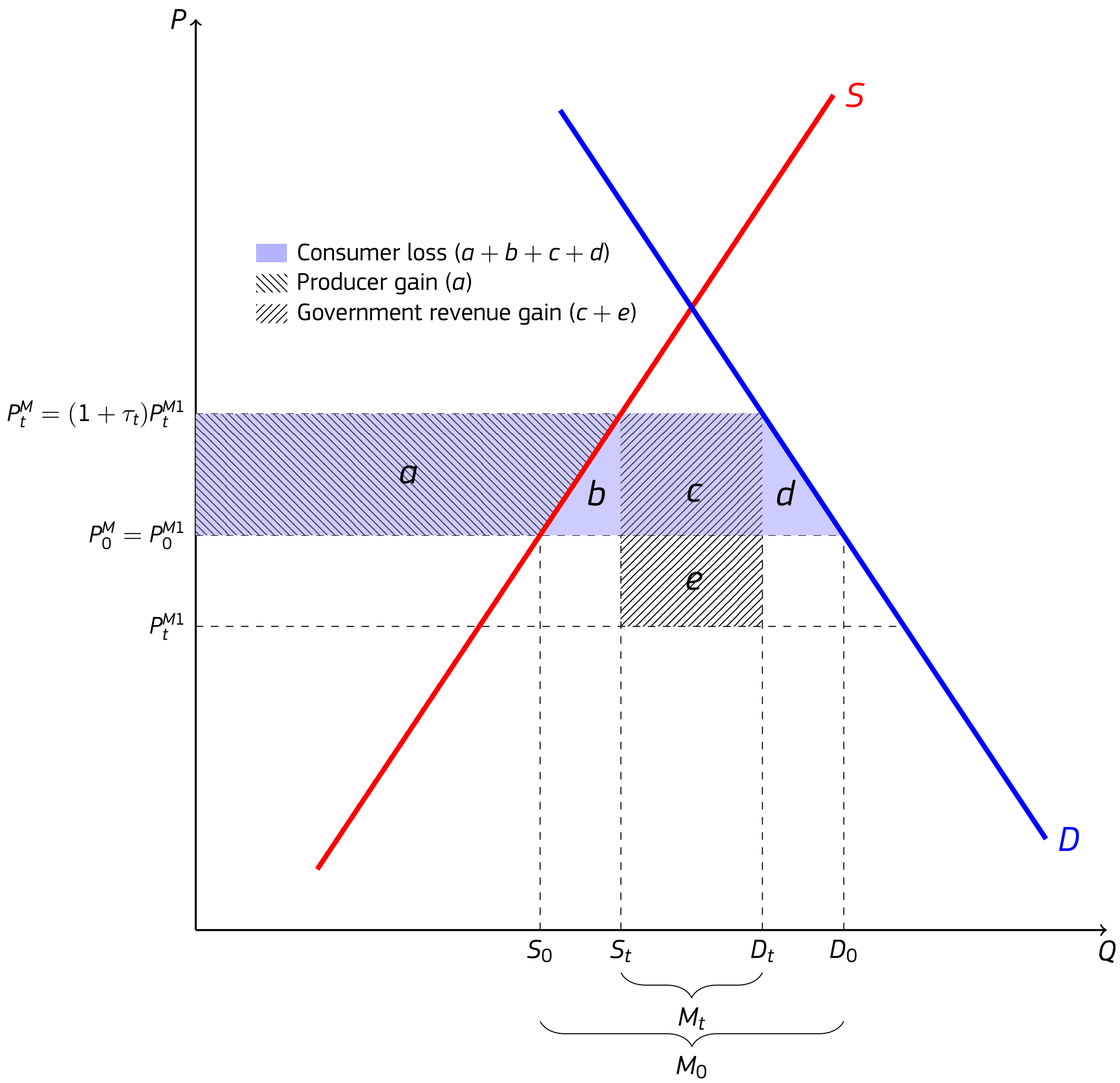
Figure B.9: Tariff semi-elasticity of real GDP under **incomplete markets** (Γ_τ^*), depending (ψ, η), with $\alpha = 0.4, \sigma_m = 1.5, \sigma_x = 3, \varphi = 3, \sigma_i = \{0.8, 1.2\}$. Imported inputs $\psi > 0$ lower the GDP response.



B.3. TERMS-OF-TRADE GAIN AND TARIFF REVENUES

Consider now the simple partial equilibrium framework familiar from international trade textbooks, on Figure B.10 .[50]

Figure B.10: Tariff welfare accounting in partial equilibrium



Note: Areas '*b*' and '*d*' capture the efficiency loss from tariffs, while area '*e*' is the **terms-of-trade gain**, which is embodied in higher government revenue. The terms-of-trade moves inversely with the net-of-tariff import price P_t^{M1} which is depressed by the tariff. Despite this, the after-tariff import price P_t^M still increases, so households feel the terms-of-trade gain only indirectly, via the government budget. *Source:* European Commission services.

- Tariffs create a wedge between import prices faced by domestic firms/households ("after-tariff price") and import prices paid to foreign trading partners ("net-of-tariff price").
- While tariffs raise the after-tariff price, they depress the net-of-tariff price, as they reduce the import

[50] This is a partial equilibrium framework for a single traded good, from which many general equilibrium macro-level channels are missing. First, relative prices of home and foreign goods faced by households do not change (they are identical), so there is no substitution via the expenditure switching channel. Households just face higher prices, and want to consume less of the good, with the reduction in their demand (from D_0 to D_t) going fully into lower imports (from M_0 to M_t), by assumption. Also, there are no exporters whose competitiveness is hurt by rising domestic prices. At the same time, domestic firms want to expand their supply in response to higher prices, raising domestic output (from S_0 to S_t), displacing imports 1:1, and without having to deal with the adverse supply side effects of tariffs (e.g. lower labour supply, or more expensive imported inputs).

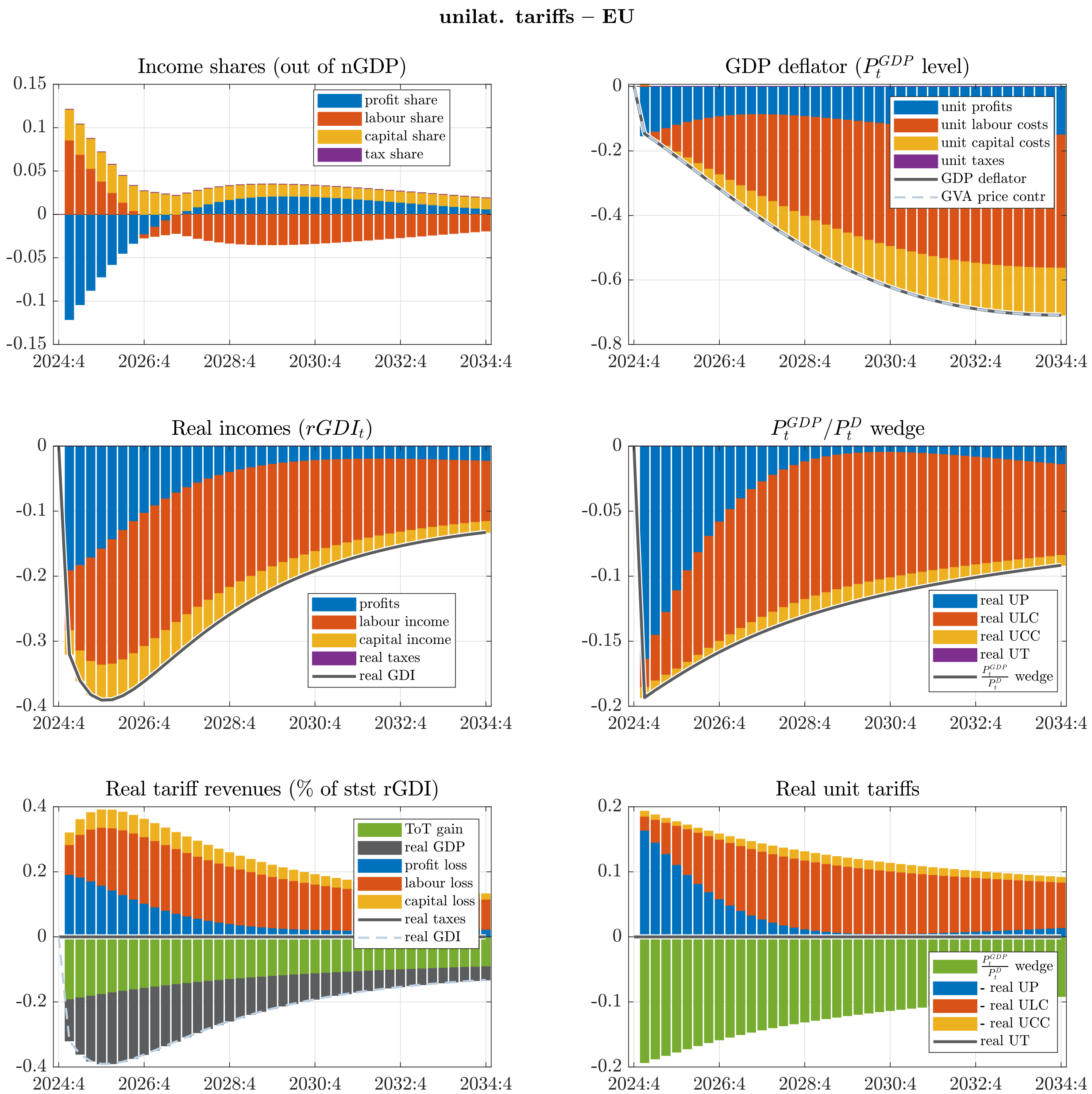
demand of a large country on global markets.

- Since the terms-of-trade is the ratio of export to import prices *net of any taxes*, this decline of the net-of-tariff import price implies a terms-of-trade gain (area 'e' in Figure B.10).
- The terms-of-trade gain shows up directly in the government budget as tariff revenue, but it can (eventually) be passed on to households in the form of tax cuts or transfers.
  - To put it another way, import costs for households grow by less (area 'c') than the tariff revenues (areas $c + e$) from which they can expect other tax cuts or transfers.
- Whether passed on to households or staying in the government budget, the terms-of-trade raises the real gross disposable income (GDI) of the country, i.e. the purchasing power of GDP in terms of the consumption basket (that includes now relatively cheaper imports beyond domestically produced goods).
- Households suffer real income losses as consumers (due to rising after-tariff import prices), but enjoy the gains accruing to firms (as shareholders), and to the government (as taxpayers and transfer recipients), so the net welfare gain is '$e - (b + d)$'.

C FURTHER FIGURES

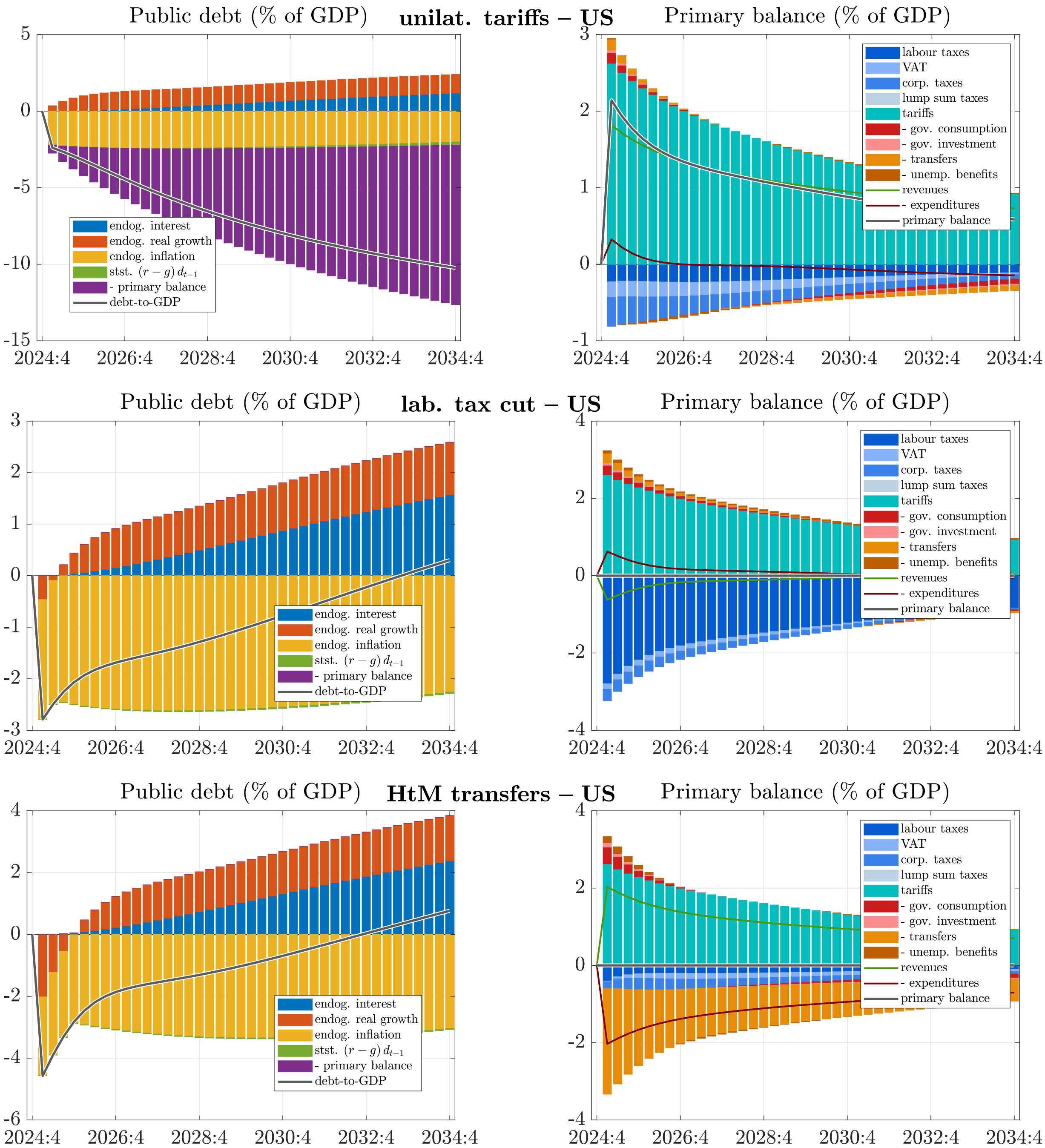
C.1. UNILATERAL US TARIFFS

Figure C.1: EU-27 real income decomposition – unilateral tariffs



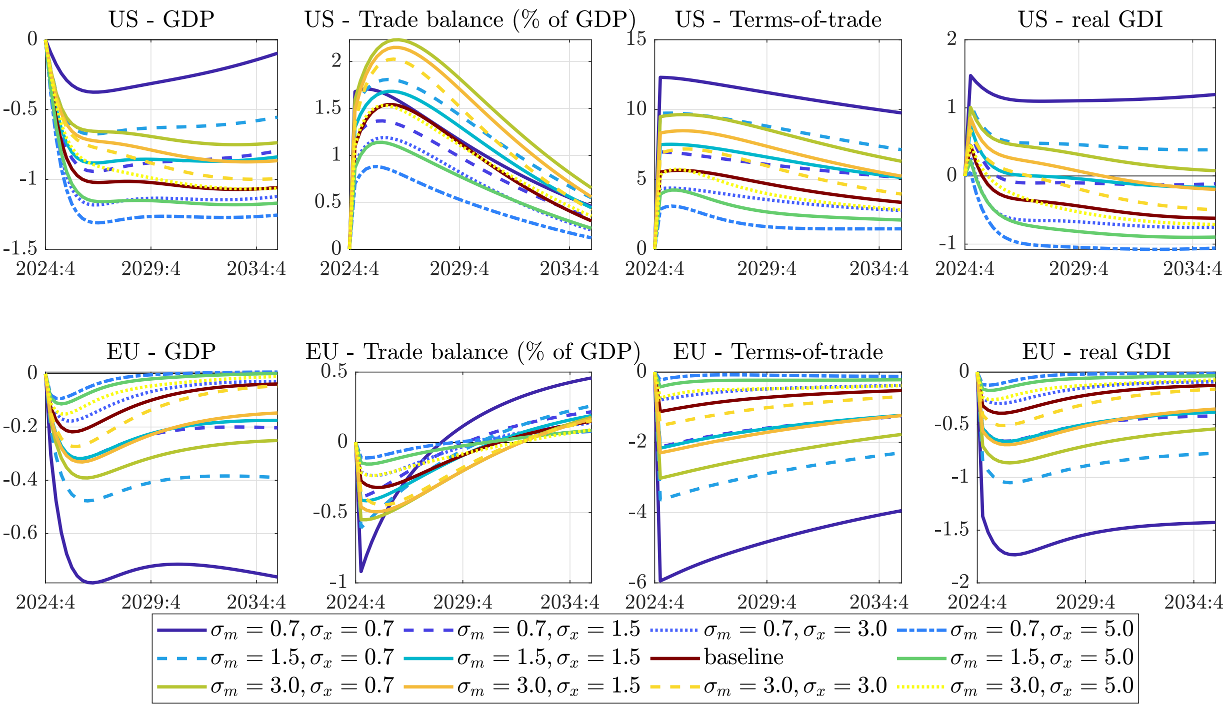
Note: Panels show the decomposition of income between production factors and product taxes. Left columns focus on income *levels* (except for income *shares*), while right columns adjust for production volumes (dividing through by $rGDP_t$) to display *unit prices*. The first row looks at GDP, while the second row looks at real GDI (deflated by P_t^D). The third row is just a rearranged version of the second row such that bars add up to tariffs, showing how those are "paid for". In the bottom left panel, real GDI is split to real GDP and the terms-of-trade gain. Results are in terms of percentage point deviation from baseline. Underlying equations are (A.28), (A.34), (A.29), (A.32), (A.31), (A.33). *Source:* European Commission services.

Figure C.2: US fiscal balances under different policies – unilateral tariffs



Note: Panels percentage point deviations in the US public debt-to-GDP and the primary budget balance-to-GDP ratios in response to unilateral US tariff hikes, with bars capturing contributions by various budgetary items. Debt-to-GDP deviations are driven by the primary balance as well as by the snowball term, where the latter is further split into endogenous changes in the interest-growth differential induced by tariff shocks and the steady state $r-g$. *Source:* European Commission services.

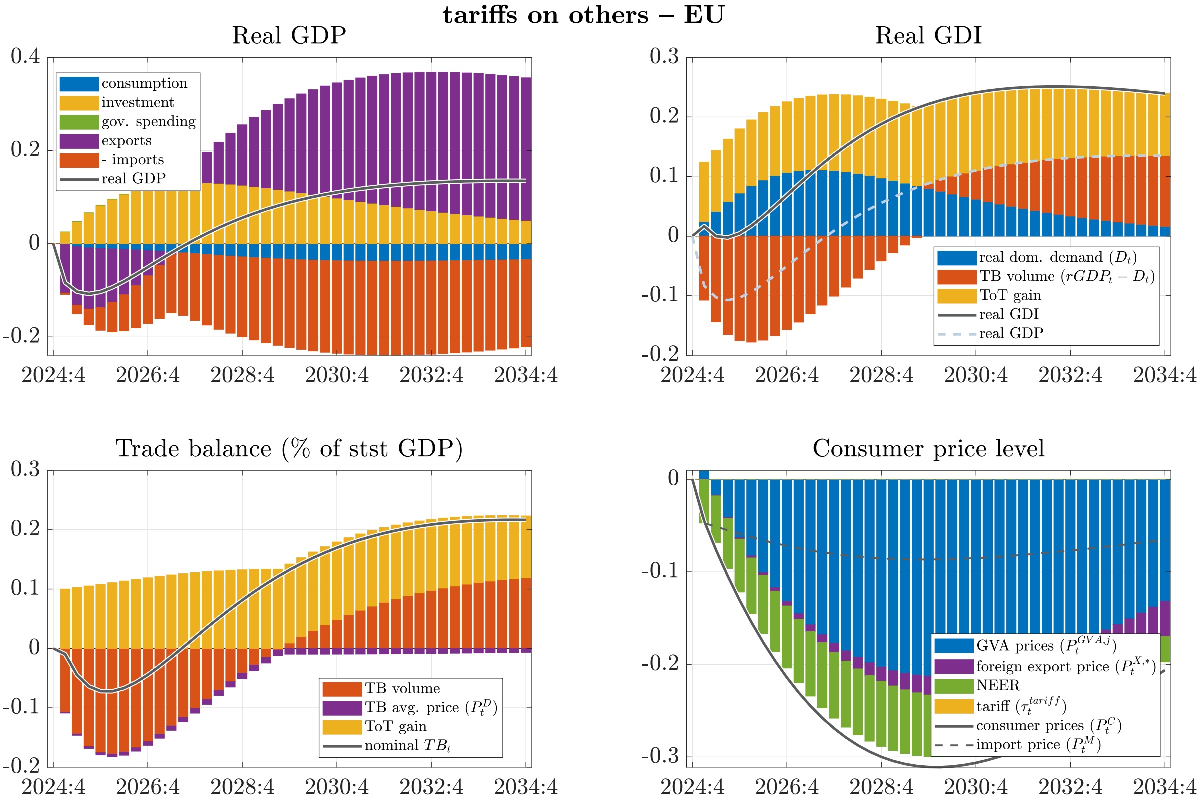
Figure C.3: The effect of trade elasticities – unilateral tariffs



Note: In response to unilateral US tariff hikes, charts show the macroeconomic effects on the US (top row) and the EU (bottom row). Lines depict %-deviation of levels from a no-tariff baseline under different elasticities of substitution for imports σ_m and exports σ_x. For CPI inflation and the trade balance, lines depict percentage point deviations. *Source:* European Commission services.

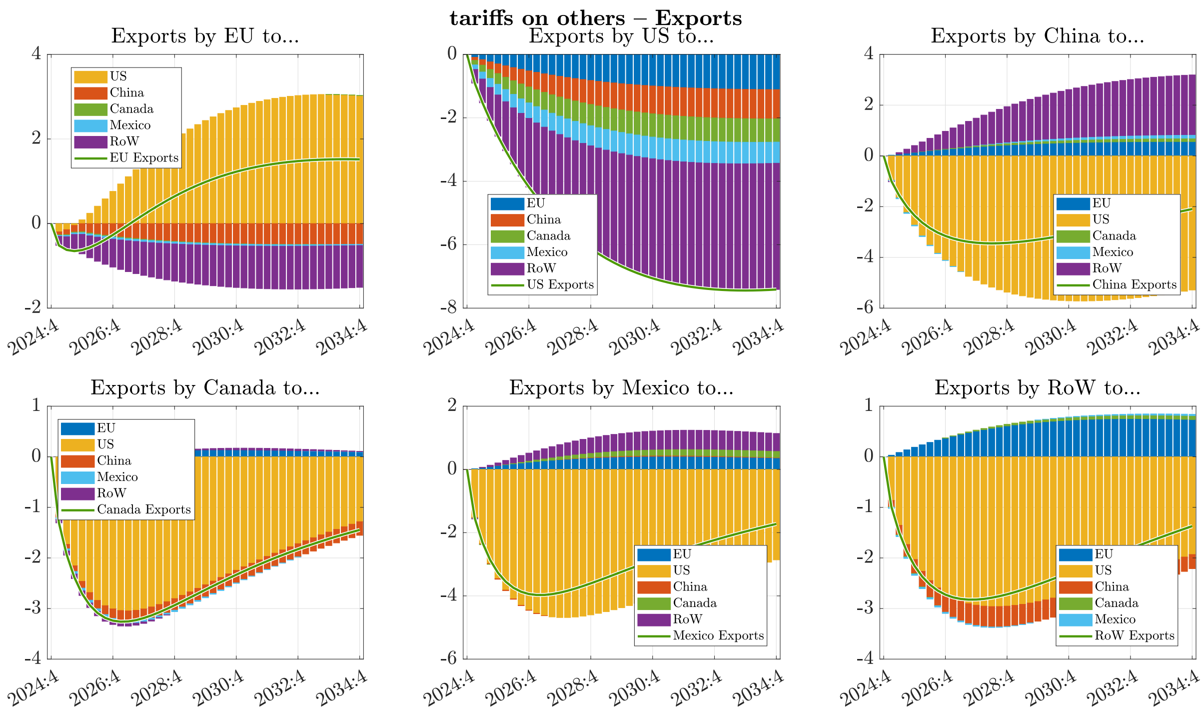
C.2. TARIFFS ON NON-EU COUNTRIES ONLY (TRADE DIVERSION EFFECTS)

Figure C.4: Macroeconomic effects in the EU-27 – unilateral US tariffs only on non-EU countries



Source: European Commission services.

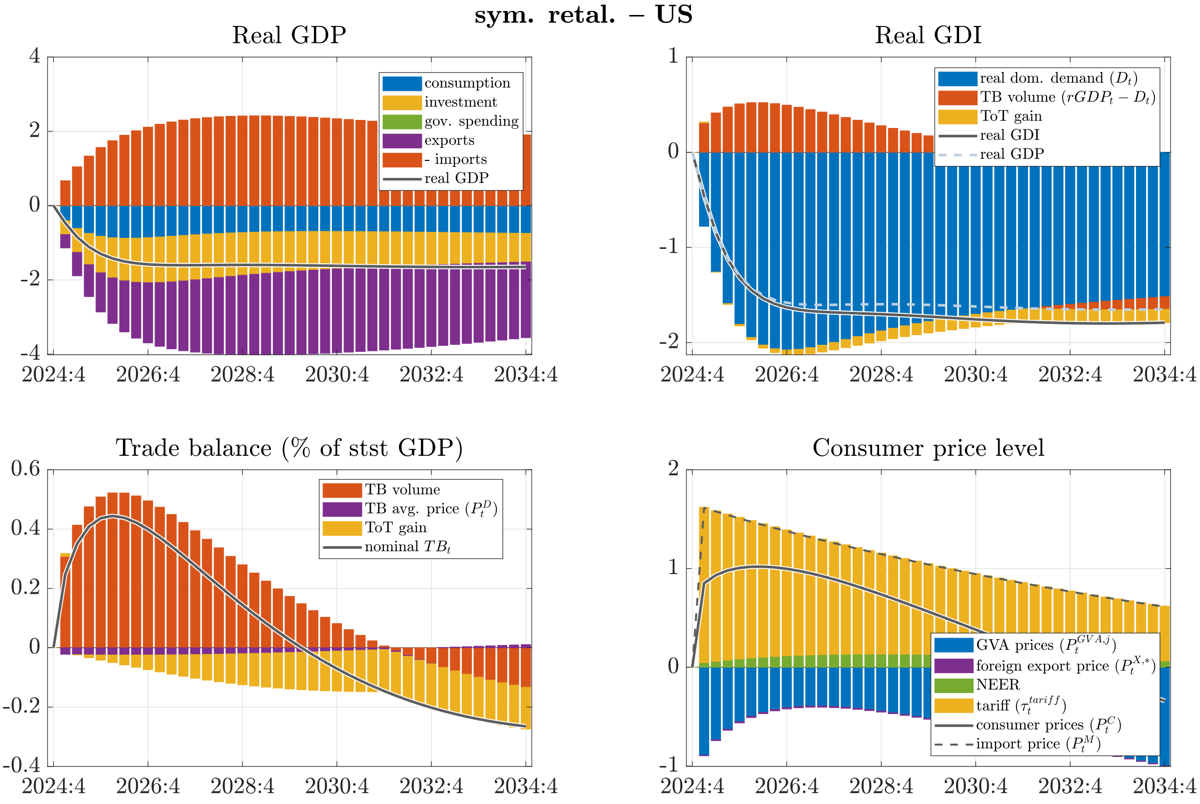
Figure C.5: Trade diversion: exports by destination markets – unilateral US tariffs only on non-EU countries



Source: European Commission services.

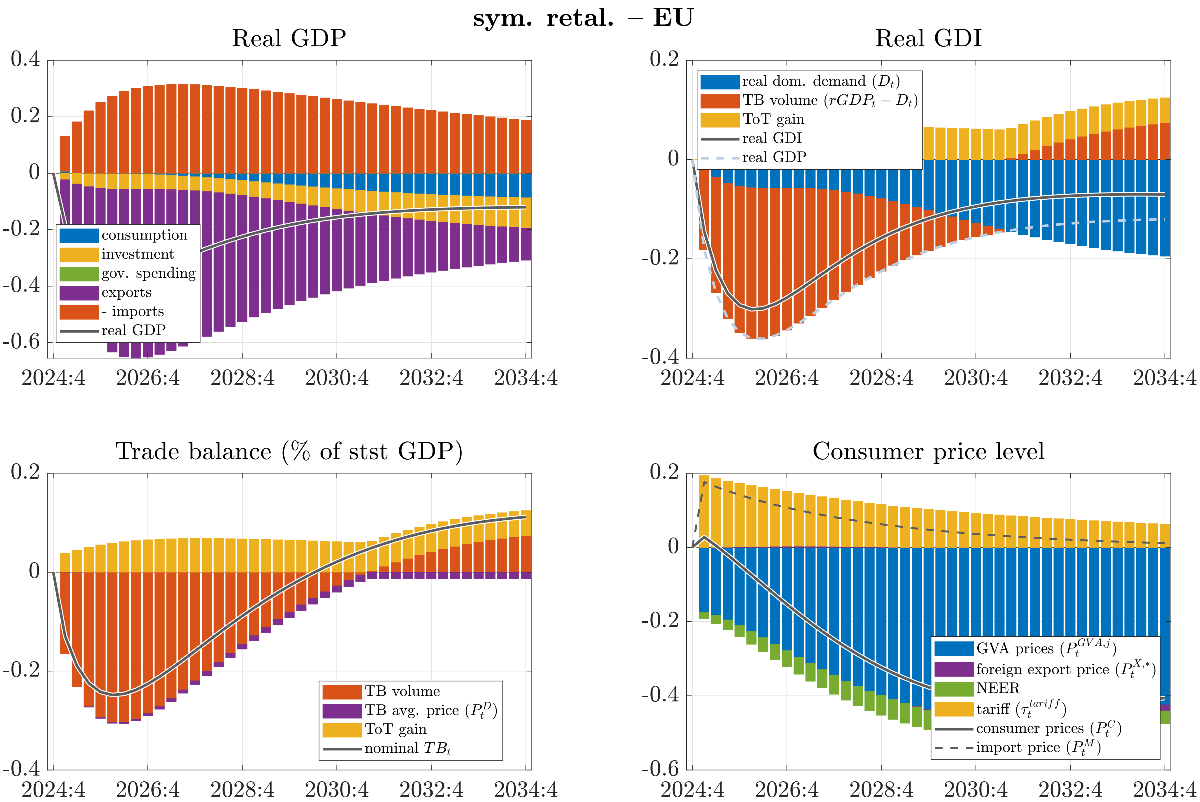
C.3. TIT-FOR-TAT RETALIATIONS

Figure C.6: Macroeconomic effects in the US – tit-for-tat retaliations



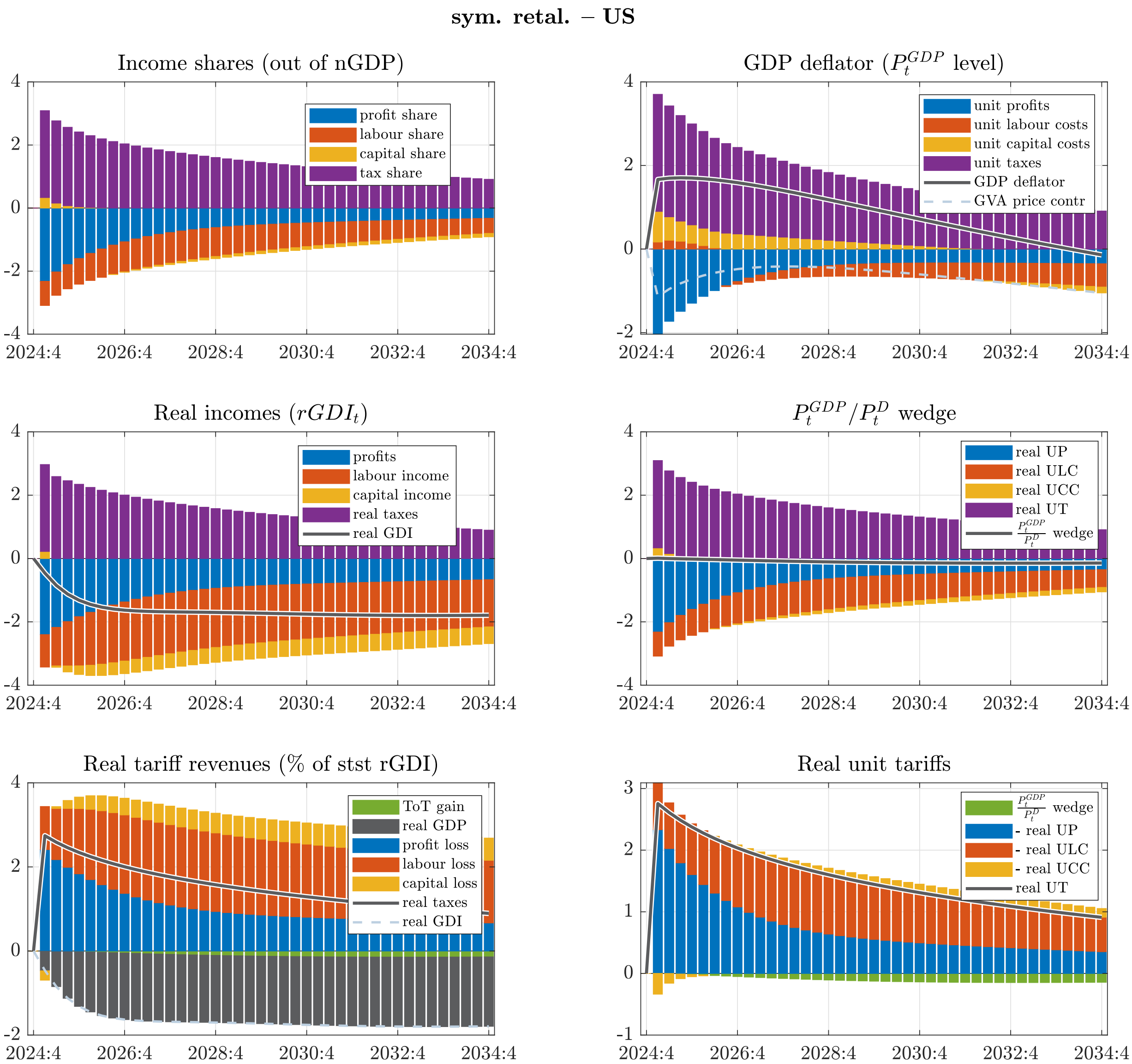
Source: European Commission services.

Figure C.7: Macroeconomic effects in the EU-27 – tit-for-tat retaliations



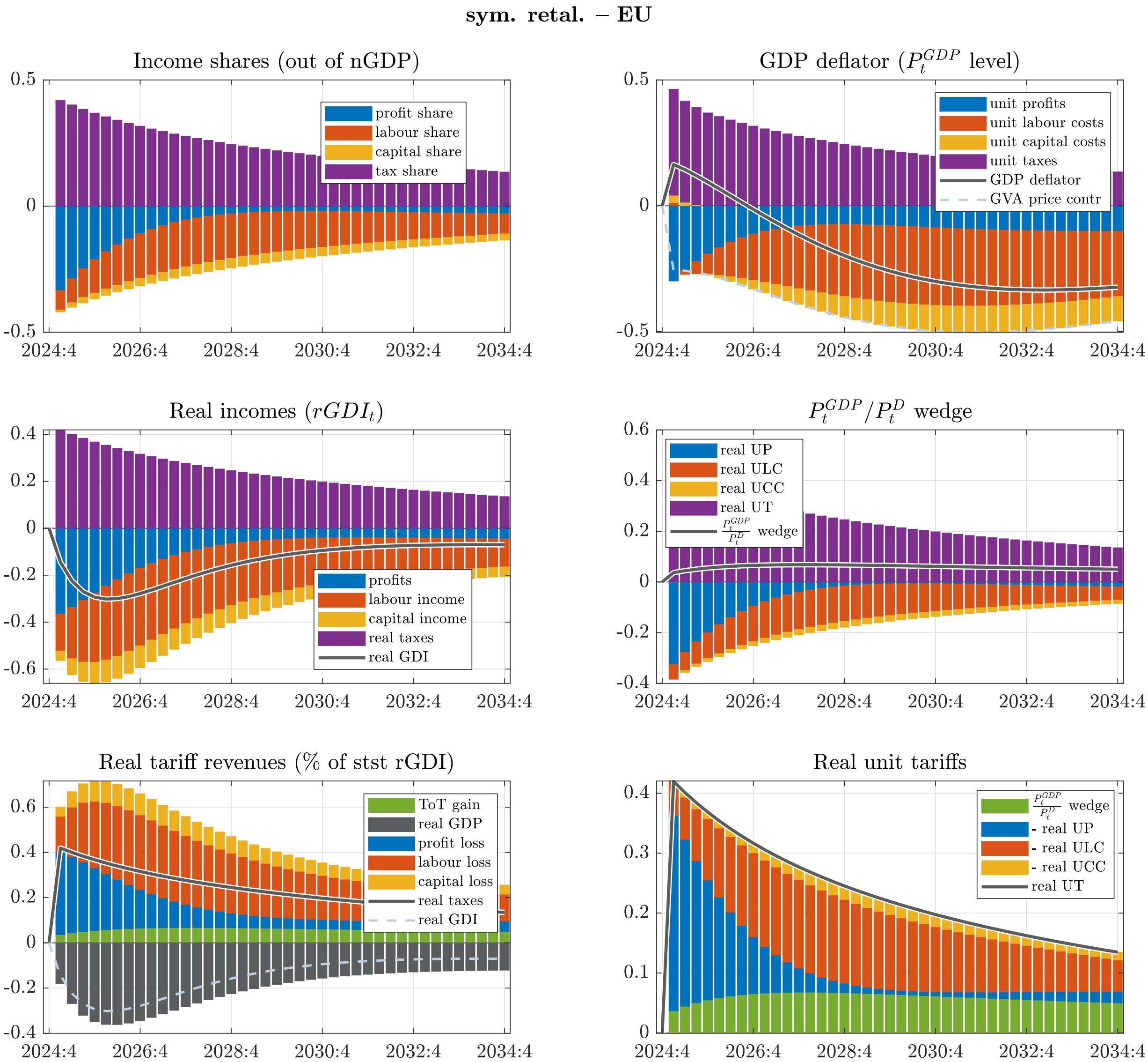
Source: European Commission services.

Figure C.8: US real income decomposition – tit-for-tat retaliations



Note: Panels show the decomposition of income between production factors and product taxes. Left columns focus on income *levels* (except for income *shares*), while right columns adjust for production volumes (dividing through by $rGDP_t$) to display *unit prices*. The first row looks at GDP, while the second row looks at real GDI (deflated by P_t^D). The third row is just a rearranged version of the second row such that bars add up to tariffs, showing how those are "paid for". In the bottom left panel, real GDI is split to real GDP and the terms-of-trade gain. Results are in terms of percentage point deviation from baseline. Underlying equations are (A.28), (A.34), (A.29), (A.32), (A.31), (A.33). Tariffs raise the GDP-deflator above the contribution by GVA prices. *Source:* European Commission services.

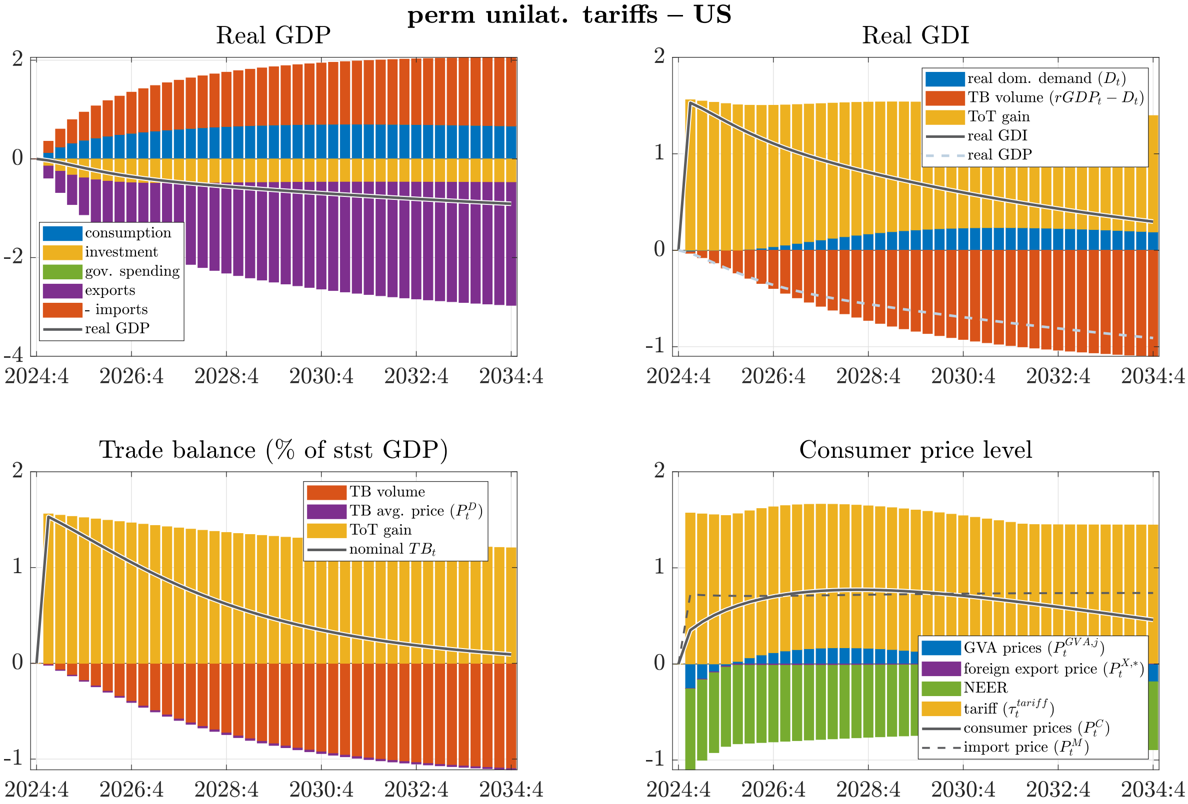
Figure C.9: EU-27 real income decomposition – tit-for-tat retaliations



Note: Panels show the decomposition of income between production factors and product taxes. Left columns focus on income *levels* (except for income *shares*), while right columns adjust for production volumes (dividing through by *rGDP$_t$*) to display *unit prices*. The first row looks at GDP, while the second row looks at real GDI (deflated by P_t^D). The third row is just a rearranged version of the second row such that bars add up to tariffs, showing how those are "paid for". In the bottom left panel, real GDI is split to real GDP and the terms-of-trade gain. Results are in terms of percentage point deviation from baseline. Underlying equations are (A.28), (A.34), (A.29), (A.32), (A.31), (A.33). *Source:* European Commission services.

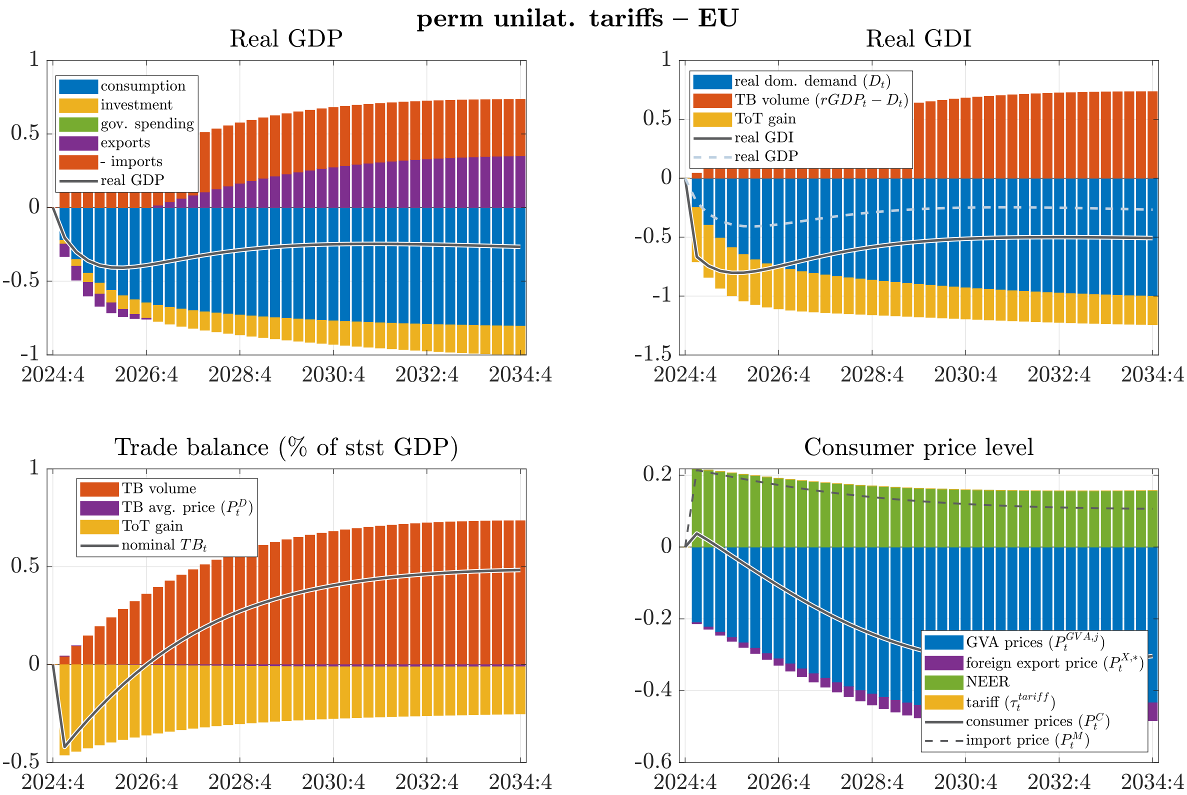
C.4. PERMANENT TARIFFS

Figure C.10: Macroeconomic effects in the US – **permanent** unilateral tariffs



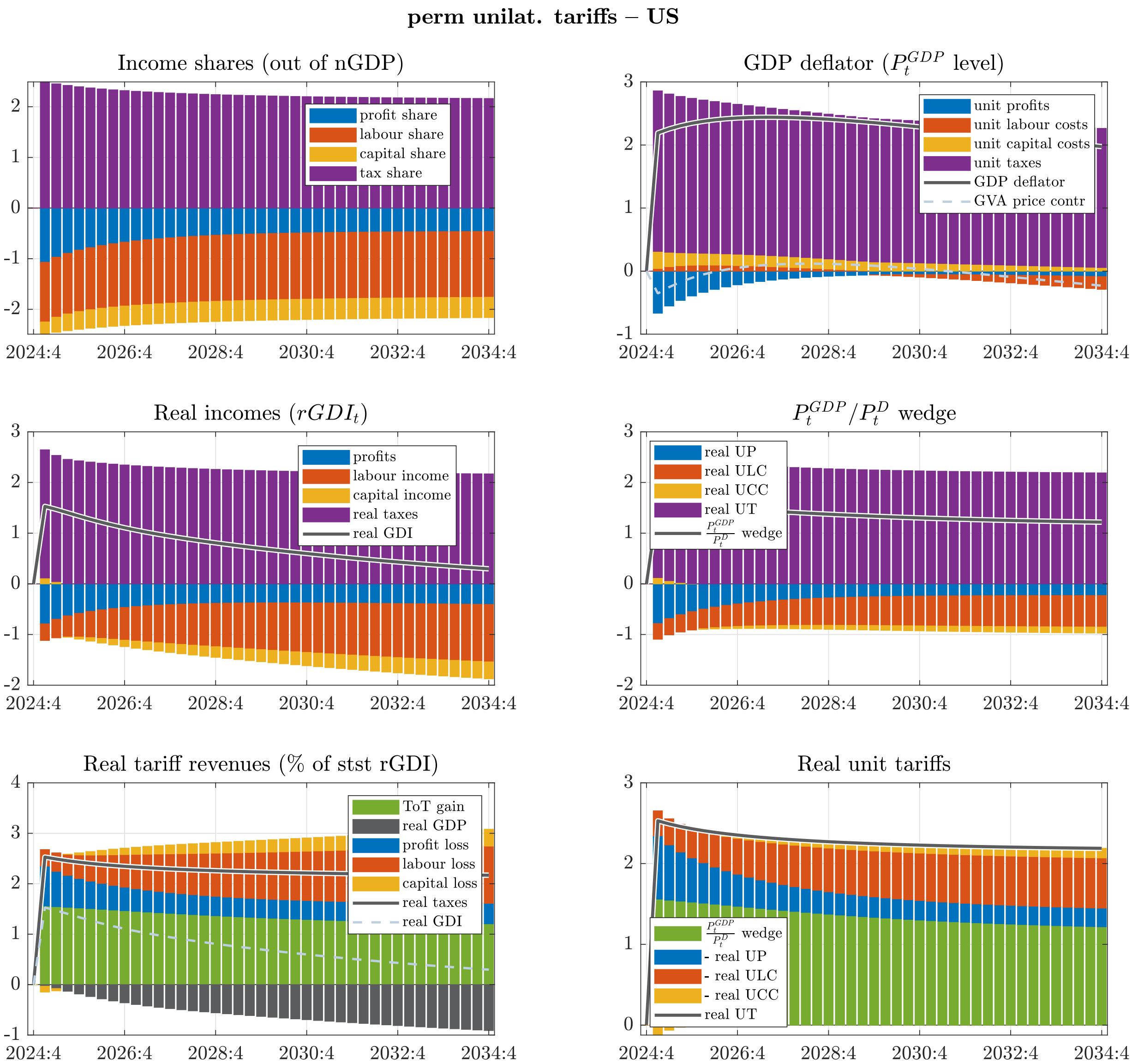
Source: European Commission services.

Figure C.11: Macroeconomic effects in the EU-27 – **permanent** unilateral tariffs



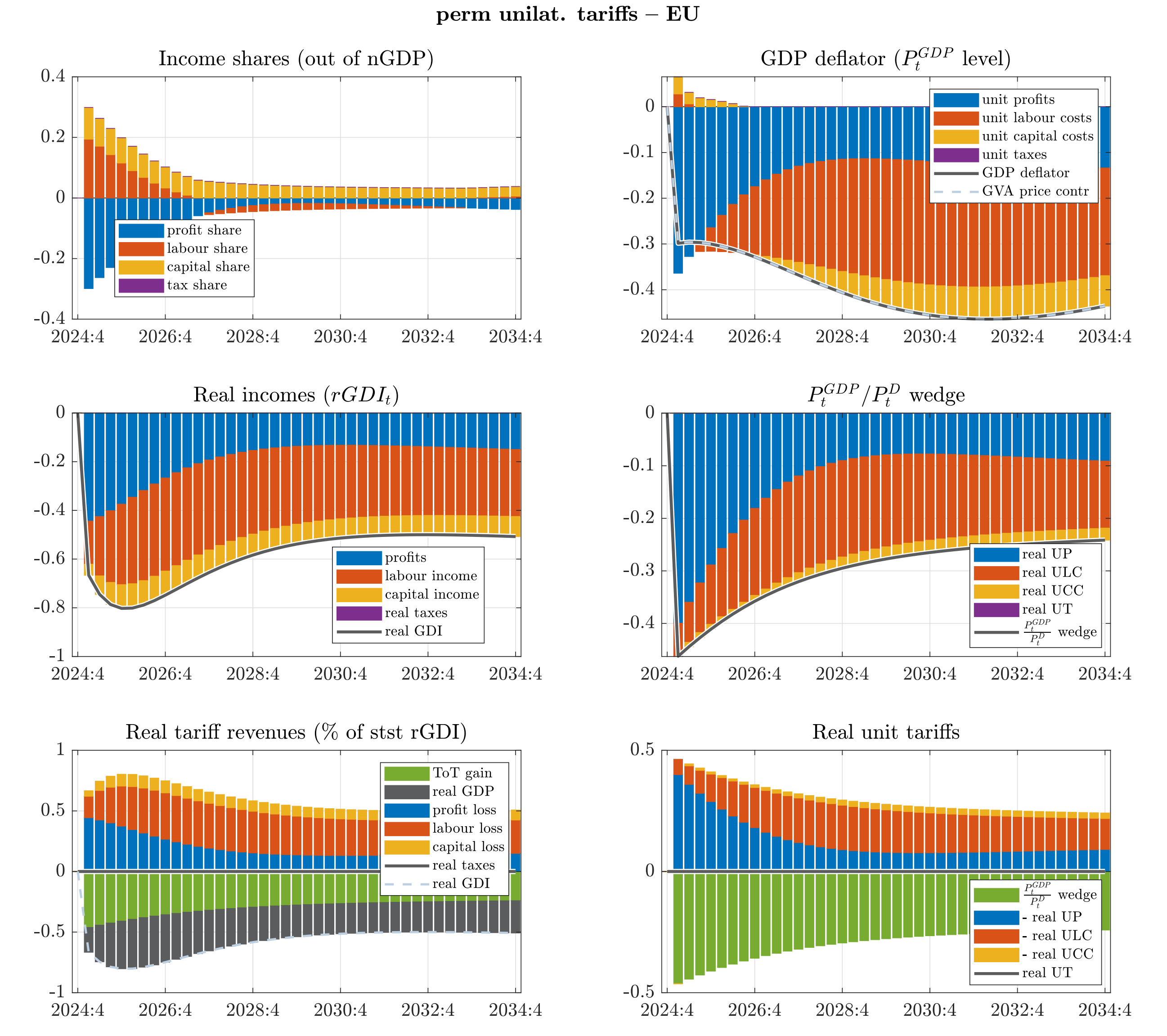
Source: European Commission services.

Figure C.12: US real income decomposition – **permanent** unilateral tariffs



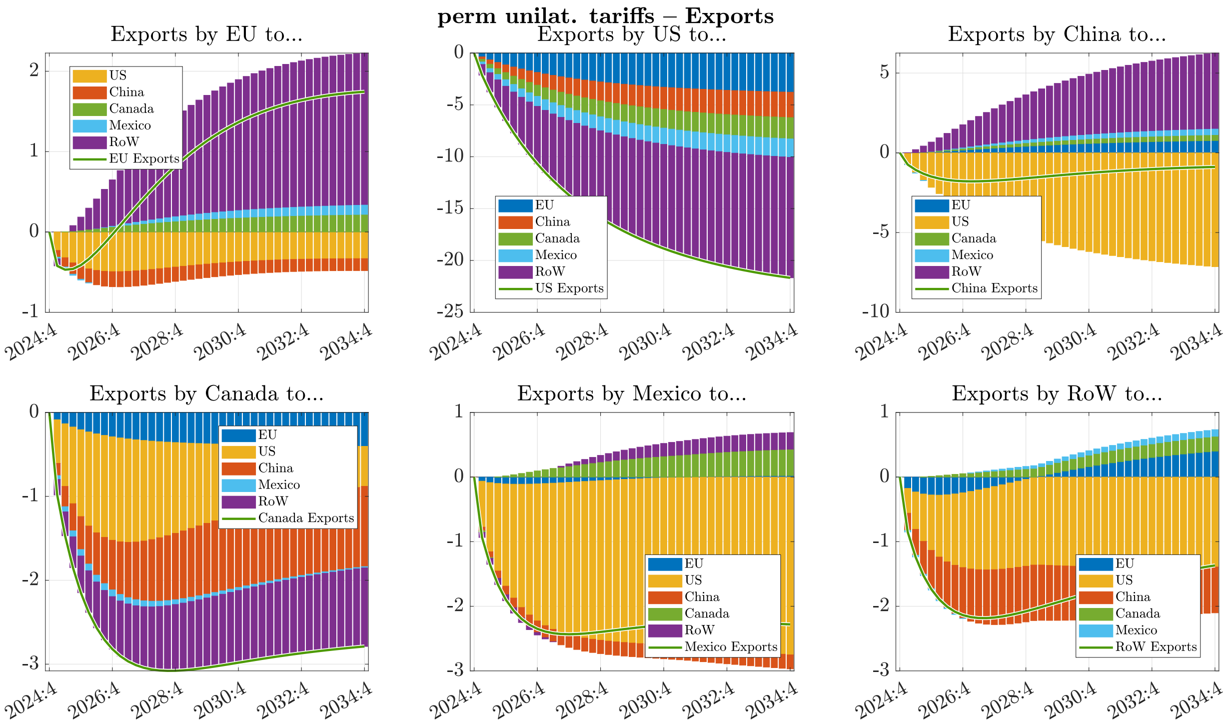
Note: Panels show the decomposition of income between production factors and product taxes. Left columns focus on income *levels* (except for income *shares*), while right columns adjust for production volumes (dividing through by $rGDP_t$) to display *unit prices*. The first row looks at GDP, while the second row looks at real GDI (deflated by P_t^D). The third row is just a rearranged version of the second row such that bars add up to tariffs, showing how those are "paid for". In the bottom left panel, real GDI is split to real GDP and the terms-of-trade gain. Results are in terms of percentage point deviation from baseline. Underlying equations are (A.28), (A.34), (A.29), (A.32), (A.31), (A.33). Tariffs raise the GDP-deflator above the contribution by GVA prices. *Source:* European Commission services.

Figure C.13: EU-27 real income decomposition – **permanent** unilateral tariffs



Note: Panels show the decomposition of income between production factors and product taxes. Left columns focus on income *levels* (except for income *shares*), while right columns adjust for production volumes (dividing through by $rGDP_t$) to display *unit prices*. The first row looks at GDP, while the second row looks at real GDI (deflated by P_t^D). The third row is just a rearranged version of the second row such that bars add up to tariffs, showing how those are "paid for". In the bottom left panel, real GDI is split to real GDP and the terms-of-trade gain. Results are in terms of percentage point deviation from baseline. Underlying equations are (A.28), (A.34), (A.29), (A.32), (A.31), (A.33). *Source:* European Commission services.

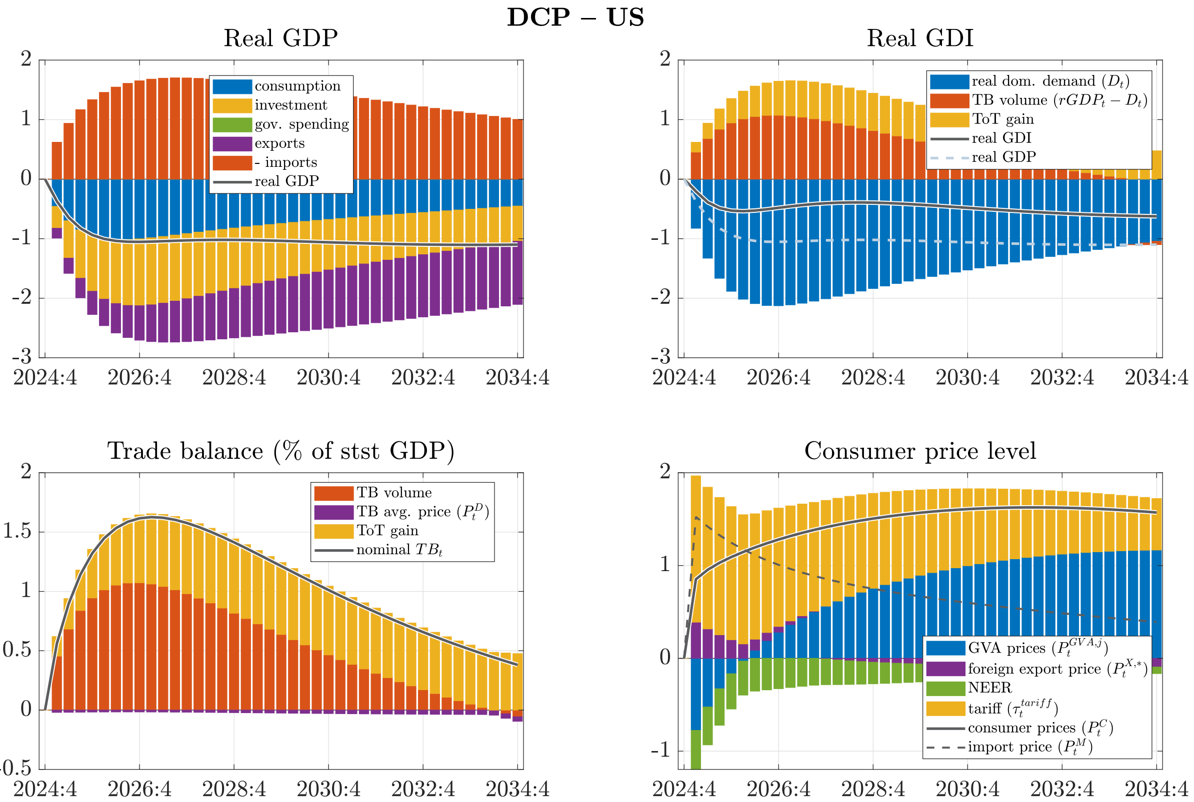
Figure C.14: Trade diversion: export decomposition by destination markets – **permanent** unilateral tariffs



Note: Solid lines report export volumes in %-deviation from baseline, with the *contribution* of various destination markets (EU, US, China, Canada, Mexico, Rest of the World) represented by bars, based on (2.24). *Source:* European Commission services.

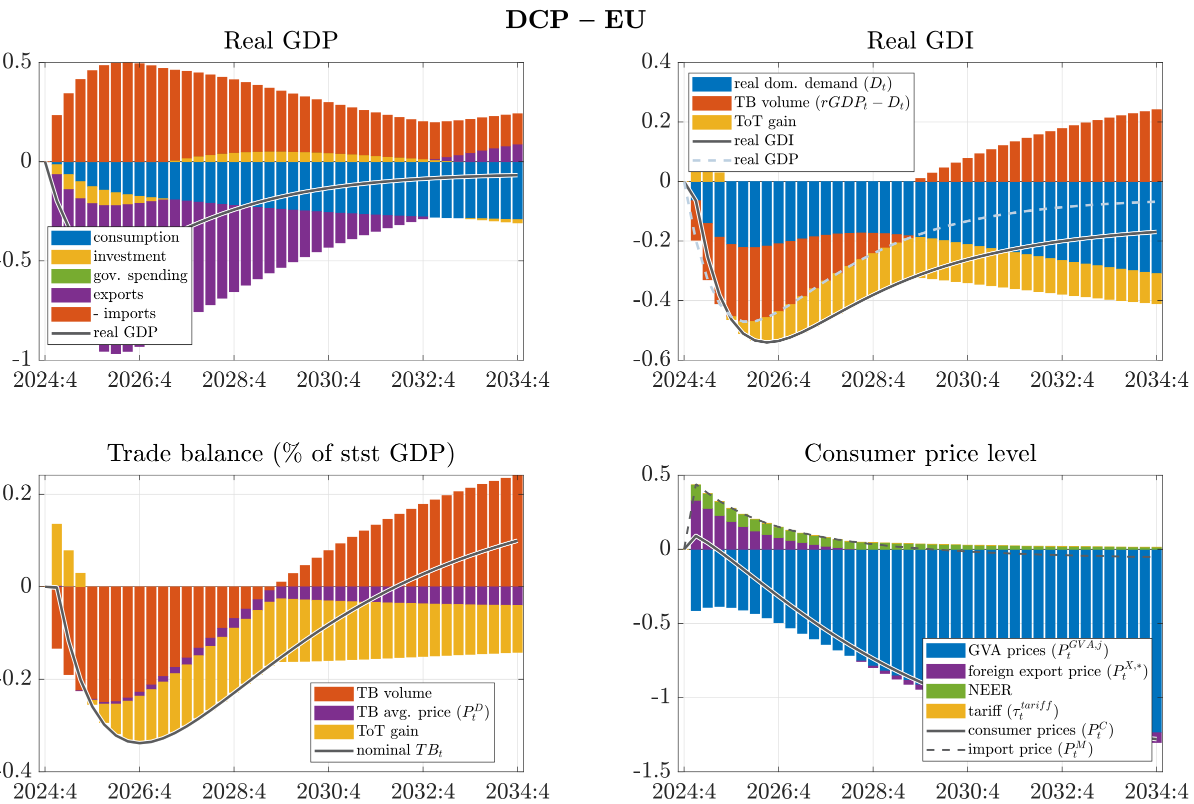
C.5. DOMINANT CURRENCY PRICING

Figure C.15: Macroeconomic effects in the US – unilateral tariffs, DCP



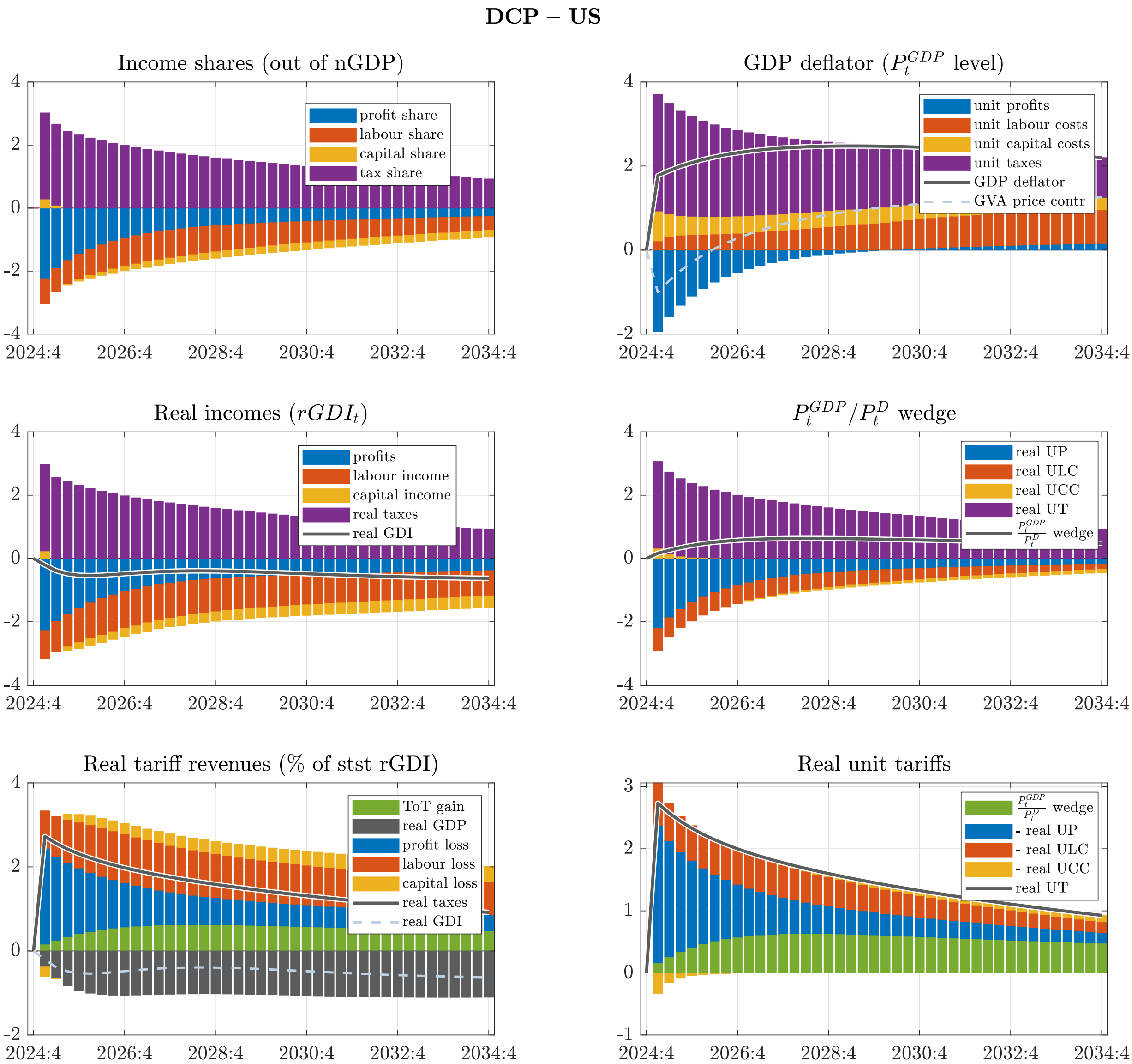
Source: European Commission services.

Figure C.16: Macroeconomic effects in the EU-27 – unilateral tariffs, DCP



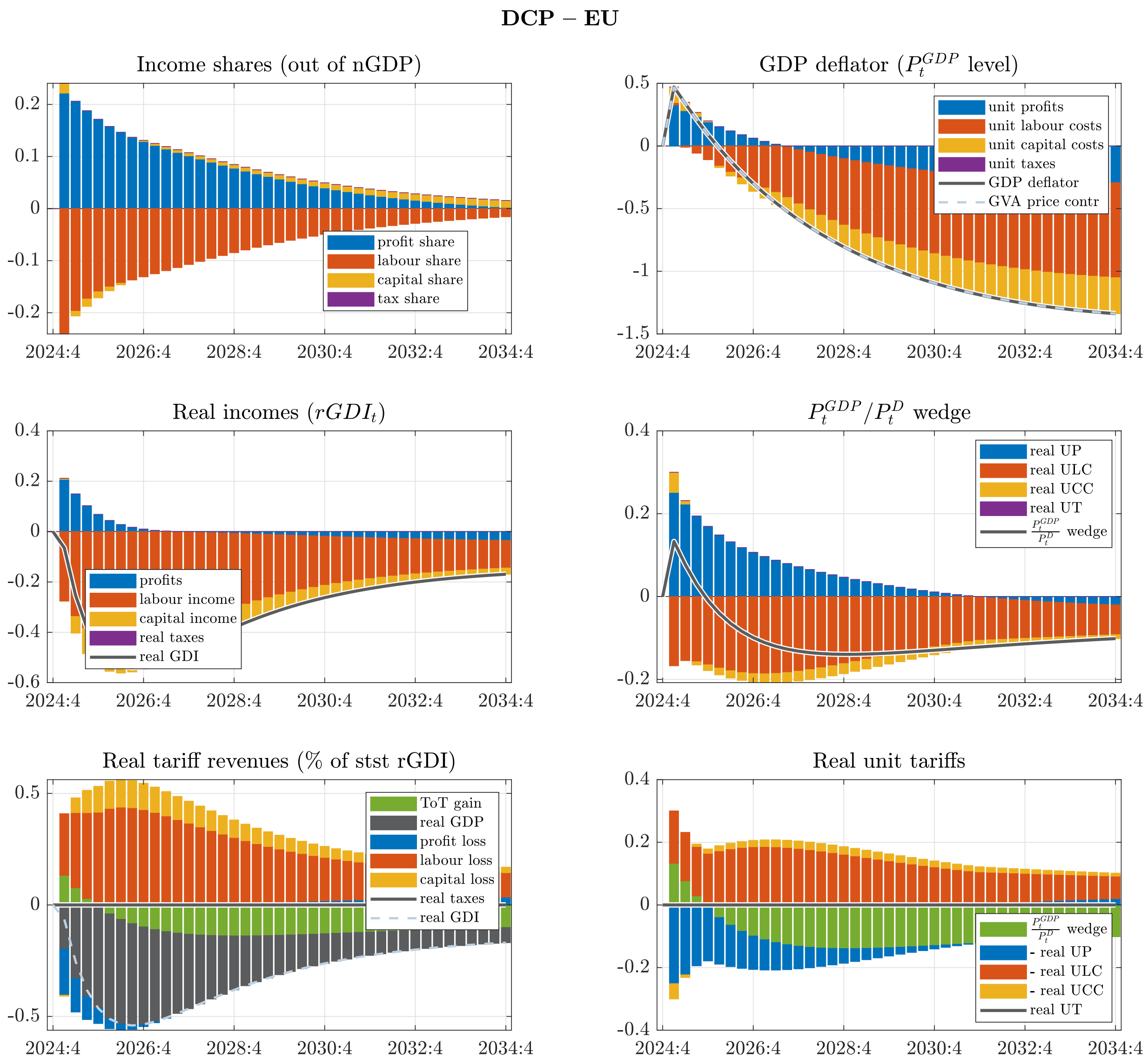
Source: European Commission services.

Figure C.17: US real income decomposition – unilateral tariffs, DCP



Note: Panels show the decomposition of income between production factors and product taxes. Left columns focus on income *levels* (except for income *shares*), while right columns adjust for production volumes (dividing through by $rGDP_t$) to display *unit prices*. The first row looks at GDP, while the second row looks at real GDI (deflated by P_t^D). The third row is just a rearranged version of the second row such that bars add up to tariffs, showing how those are "paid for". In the bottom left panel, real GDI is split to real GDP and the terms-of-trade gain. Results are in terms of percentage point deviation from baseline. Underlying equations are (A.28), (A.34), (A.29), (A.32), (A.31), (A.33). Tariffs raise the GDP-deflator above the contribution by GVA prices. *Source:* European Commission services.

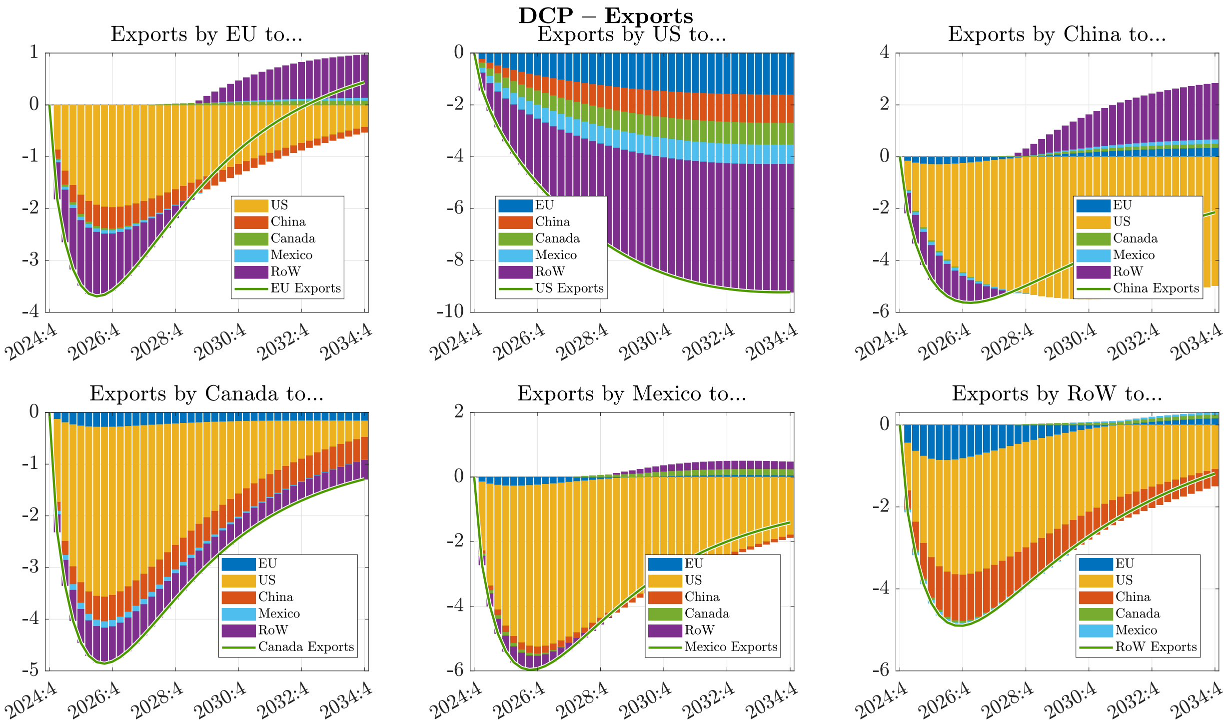
Figure C.18: EU-27 real income decomposition – unilateral tariffs, DCP



Note: Panels show the decomposition of income between production factors and product taxes. Left columns focus on income *levels* (except for income *shares*), while right columns adjust for production volumes (dividing through by $rGDP_t$) to display *unit prices*. The first row looks at GDP, while the second row looks at real GDI (deflated by P_t^D). The third row is just a rearranged version of the second row such that bars add up to tariffs, showing how those are "paid for". In the bottom left panel, real GDI is split to real GDP and the terms-of-trade gain. Results are in terms of percentage point deviation from baseline. Underlying equations are (A.28), (A.34), (A.29), (A.32), (A.31), (A.33). *Source:* European Commission services.

Despite Europe's $ToT_t = P_t^X/P_t^{M1}$ unambigously depreciating (Figure 5.2), Europe's ToT-gain (that is driven by P_t^{GDP}/P_t^D) is briefly positive (Figure C.18). The reason for this is twofold. First, instead of some kind of average between P_t^X and P_t^{M1}, our ToT-gain definition in (A.6),(A.9) uses P_t^D as the average price level, so that the ToT-gain coincides with changes in the purchasing power of domestic production. If there is a generalised increase in both P_t^X and P_t^{M1} (and with DCP export prices are more flexible in local currency), this would leave ToT_t unaffected, but could raise P_t^{GDP}/P_t^D if the import share in domestic expenditures is low, and the export share in GDP is high enough. Second, a trade surplus in steady state (like for the EU) means that an equal rise in P_t^X and P_t^{M1} (i.e. no change in ToT) generates a purchasing power gain.

Figure C.19: Trade diversion: export decomposition by destination markets – unilateral tariffs, DCP



Note: Solid lines report export volumes in %-deviation from baseline, with the *contribution* of various destination markets (EU, US, China, Canada, Mexico, Rest of the World) represented by bars, based on (2.24). *Source:* European Commission services.

D APR 2 "LIBERATION DAY" TARIFF ASSUMPTIONS

The effective bilateral tariff rates used in our model simulations are based on the new Trump administration's announcements ***up to and including 2 April 2025.*** These assumptions underlie the simulations presented in the European Commission's Spring Economic Forecast (Motyovszki, 2025).

These numbers are approximations, using USITC and BEA data to calculate the weights of various product categories (at HTS-2 level) within the total imports of the United States. The values represent *changes* in the effective tariff rates relative to the pre-Trump situation (as opposed to the actual *levels* of the tariff rates).

The assumed increase in bilateral tariffs levied on US imports in our 6-region model is based on:

- 25 percentage points (pps) on goods imported from Mexico and Canada ("border security" tariffs), with only 10pp on oil imports from Canada, and with USMCA-compliant imports (assumed to be 40% of total imports) being fully exempted;
- 54 pps on goods imports from China (20 pps Section 301 "fentanyl" tariffs + 34 pps IEEPA "reciprocal" tariffs);
- 20 pps on goods from the EU (IEEPA "reciprocal" tariffs);
- 23 pps on average from the rest of the world (IEEPA "reciprocal" tariffs).
- These country-specific headline rates are fine-tuned by Section 232 "national security" sectoral tariff hikes of 25 pps in the case of aluminium, steel and automobile imports.
- For other sectors that were temporarily exempted from IEEPA tariffs (e.g. pharmaceuticals and semiconductors) we assumed the country-specific headline rate would eventually apply.
- Trade in services is taxed at 0%.

As a result, the effective tariff rate on total US imports (including services) goes up by around 20 pps, which is the same order of magnitude as estimated by The Budget Lab Yale (2025*b*) at the time.

Table D.1: Rise in bilateral effective US tariff rates (2 Apr 2025 vs 2024)

Effective US tariff rates per origin country and sector (Apr 2)

	EU-27	CN	CA	MX	RoW (in G6 model)	TOTAL
goods	20.70%	52.01%	14.30%	18.00%	23.28%	24.69%
services						0.00%
TOTAL	15.68%	49.67%	12.69%	16.48%	16.95%	19.90%
						19.90%

Note: Numbers refer to percentage point increase in tariffs relative to the 2024 tariff landscape. "TOTAL" refers to the weighted average on goods and services (across rows) and/or trading partners (across columns). Trade weights based on USITC 2023 data, and HTS-2 level product categories.

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